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Health Condition Prediction Using Machine Learning



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1. Task Assignment

Name	Algorithm chosen	Individual Tasks performed
Hanna Yousef Haji	Support Vector Machine (SVM)	1- Introduction (SVM) section. 2- Technical Description (SVM) section. 3- Experimental Results and Analysis (SVM) section. 4- Appendix B.
Khadijha Abduallah Baaqil	Neural Networks (NN)	1- Introduction (NN) section. 2- Technical Description (NN) section. 3- Experimental Results and Analysis (NN) section. 4- Appendix C.
Shaimaa Esam Al_Aqeely	Naïve Bayes	1- Introduction (Naïve Bayes) section. 2- Technical Description (Naïve Bayes) section. 3- Experimental Results and Analysis (Naïve Bayes) section. 4- Appendix D.
Layal Mohammed Abo Khadijah	Logistic Regression	1- Introduction (Logistic Regression) section. 2- Technical Description (Logistic Regression) section. 3- Experimental Results and Analysis (Logistic Regression) section. 4- Appendix E.

Table 1: Task Assignment

We all contributed equally for the rest of the tasks (Introduction - Technical Description - Conclusion – Appendix A) , we also helped each other and reviewed each other's work.

2. Introduction

2.1 Project aim

Machine learning is one of the main branches of artificial intelligence, and it allows computers to learn patterns from data. The aim of this project is to apply different classification algorithms to a health dataset and see how each algorithm affects the model's performance.

2.2 Problem Solved Explanation

The problem we are trying to solve in this project is predicting a patient's medical condition based on their health and lifestyle data.

The dataset has many features like age, blood pressure, glucose, and habits like smoking or exercise. Some features are noisy or not important, which can make prediction harder. Our goal is to see which machine learning algorithm can best classify patients into their correct health condition and how tuning improve the results.

2.3 Selected Dataset

The Health Dataset we selected contains detailed information about patients and their health and lifestyle factors. Each record represents a single patient and includes demographic data such as age and gender. It also tracks clinical measurements like glucose level, blood pressure, BMI, oxygen saturation, cholesterol, triglycerides, and HbA1c.

The dataset further includes lifestyle-related information such as smoking and alcohol habits, physical activity, diet quality, family medical history, stress levels, and average sleep hours. Some columns contain additional notes or noise values that are not relevant for prediction.

This dataset is useful for analyzing patterns in patients' health, understanding how different factors relate to medical conditions, and predicting patient outcomes. It is suitable for applying machine learning algorithms to classify patients' medical conditions and explore how preprocessing and parameter tuning affect prediction performance.

2.4 Chosen Machine learning Algorithms

In this study, we used four machine learning algorithms: **Logistic Regression, Neural Networks, Naïve Bayes, and Support Vector Machine (SVM)**. These algorithms were chosen because they are reliable and work well for predicting outcomes.

Logistic Regression is easy to understand and good for classification problems. **Neural Networks** can learn complex patterns in the data, which helps with complicated relationships. **SVMs** are strong when dealing with many features and work well if the classes are clearly separated. **Naïve Bayes** is fast, works well with large datasets, and assumes features are independent, which makes calculations simple.

We applied these algorithms to predict patients' medical conditions and to compare which one gives the best results after preprocessing and fine-tuning.

3. Technical Description

3.1 Dataset Attributes & Types Description

- **Age** : Patient's age in years (Numerical)
- **Gender** : Patient's gender (Categorical: Male, Female)
- **Glucose** : Blood glucose level (Numerical)
- **Blood Pressure** : Blood pressure measurement (Numerical)
- **BMI** : Body Mass Index (Numerical)
- **Oxygen Saturation** : Blood oxygen saturation level (Numerical)
- **LengthOfStay** : Hospital length of stay in days (Numerical)
- **Cholesterol** : Cholesterol level (Numerical)
- **Triglycerides** : Triglyceride level (Numerical)
- **HbA1c** : Hemoglobin A1c (glycated hemoglobin) (Numerical)
- **Smoking** : Smoking status (Binary: 0 = Non-smoker, 1 = Smoker)
- **Alcohol** : Alcohol consumption (Binary: 0 = No, 1 = Yes)
- **Physical Activity** : Hours of physical activity per week (Numerical)
- **Diet Score** : Diet quality score (Numerical)
- **Family History** : Family medical history (Binary: 0 = No, 1 = Yes)
- **Stress Level** : Stress level on a numeric scale (Numerical)
- **Sleep Hours** : Average hours of sleep per day (Numerical)

- **Random_notes** : Random notes, not related to prediction (Text / String)
- **Noise_col** : Unrelated random values, noisy column (Numerical / Irrelevant)

Target variable:

- **Medical Condition**: Patient's health status (Categorical: Diabetes, Hypertension, Asthma, Obesity, Arthritis, Cancer, Healthy)

3.2 Chosen machine learning algorithms Description and Parameters

Support Vector Machine(SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification tasks. It works by finding the best decision boundary (hyperplane) that separates different classes in the dataset. SVM aims to maximize the margin between classes, which helps improve the model's ability to generalize to new data. It can handle both linear and non-linear data using kernel functions. In this project, SVM is applied to predict patients' medical conditions based on their health and lifestyle features.

Key Parameters:

- **Kernel:** Type of function used to transform data for non-linear separation (e.g., linear, RBF, polynomial).
- **C (Regularization parameter):** Controls the trade-off between maximizing the margin and minimizing classification errors.
- **Gamma:** Determines the influence of individual training samples (used with RBF and polynomial kernels).

Neural Network

Neural Network is a sophisticated machine learning model inspired by the structure and function of the human brain. It consists of interconnected layers of nodes (neurons) that process information to recognize complex patterns and non-linear relationships within the data. It uses a technique called backpropagation to adjust the weights of connections during training to minimize prediction errors.

Since the project involves predicting multiple medical conditions the Neural Network is particularly effective because it can naturally handle multi-class classification problems. The output layer contains a specific node for each class, allowing the model to output a probability distribution across all possible diseases for each patient.

Key Parameters:

- **Hidden Layers:** Defines the architecture of the network (the number of layers and neurons between input and output). A higher number of neurons allows the model to learn more complex features.
- **Learning Rate:** Controls the step size at which the model updates its weights during training. A smaller rate (e.g., 0.1) ensures more precise convergence but takes longer.
- **Training Time:** The number of times the entire dataset is passed forward and backward through the neural network to learn the patterns.
- **Momentum:** A parameter that helps the backpropagation algorithm navigate the error surface more efficiently to avoid getting stuck in local minima.

Naïve Bayes :

Naïve Bayes is a supervised machine learning algorithm based on Bayes' Theorem and commonly used in tasks such as text and medical classification. It is considered a generative model because it estimates the probability distribution of the features within each class, then uses these probabilities to predict the most likely class label for new data. The algorithm assumes that all features are conditionally independent, which makes it fast, simple, and effective for high-dimensional datasets. Unlike discriminative models such as logistic regression, Naïve Bayes does not focus on identifying the most influential features for class separation, but instead relies on calculating class likelihoods based on feature distributions.

Key Parameters:

- **Laplace Correction:** Prevents zero probabilities when certain feature values do not appear in the training data, improving model stability.
- **Kernel Density Estimation (KDE):** Uses a smooth probability density function instead of assuming a normal distribution, helping the model capture more complex patterns in continuous data.
- **Model Type (Distribution Type):** Specifies how the data distribution is modeled, such as Gaussian for continuous features or Multinomial/Bernoulli for text and binary features.

Logistic Regression

Logistic Regression is a basic model in machine learning and statistics used for classification tasks. Instead of predicting a direct numerical value, it estimates the probability that a data point belongs to a certain class, using the Sigmoid function to convert linear outputs into probabilities between 0 and 1.

Since our project deals with multiple classes (like diabetes, obesity, cancer), I adapted Logistic Regression using a Polynomial by Binomial Classification wrapper. This approach turns the multi-class problem into several binary classification tasks (one model per class vs all others), allowing Logistic Regression to handle multi-class data efficiently.

Key Parameters:

- **Max Iterations :** Number of times the model goes through the training data.
- **Regularization:** Prevents overfitting by penalizing large coefficients..
- **Tolerance :** Early stopping criterion; stops training when improvement is very small.
- **Solver:** The optimization algorithm used to compute the model weights (AUTO, LBFGS, etc.).

3.3 Data Preprocessing

The Data Preprocessing methods we used are :

1 - **Removing Irrelevant Columns:**

Columns that do not contribute to predicting the target were removed. (random_notes and noise_col)

2- **Removing Duplicate Instances:**

Eliminating duplicate records to prevent bias and ensure that the model learns from unique examples.

3- **Standardization/Normalization:**

Standardization is Scaling numeric features so that they all have the same range, which helps most algorithms perform better. For Neural Networks, **normalization** is used instead, which scales values between 0 and 1.

4- **String to Binary:**

Converting the string columns to binary values (0,1) , we did it to the gender column .

In the appendix A there is the steps and the filters we used to apply this in Weka and rapid Miner.

4. Experiments and Results

4.1 Support Vector Machine (SVM)

4.1.1 Default SVM Experimental Results

The default parameters in Weka are:

- **Kernel Type:** Polynomial
- **Degree:** 1
- **Gamma:** 0.1
- **Coef0:** 1
- **Cost (C):** 1
- **Batch Size:** 100
- **Epsilon (eps):** 0.001
- **Shrinking:** true

The default parameters in RapidMiner:

- **Kernel Type:** Polynomial
- **Degree:** 3
- **Gamma:** 0.1
- **Coef0:** 1
- **Cost (C):** 1
- **Batch Size:** 80
- **Epsilon (eps):** 0.001
- **Shrinking:** true

The results:

	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	91.3216%	91.97%
	10	91.4564%	92.03%
	15	91.4788%	92.01%
Split Validation	60%	91.3703%	91.82%
	70%	90.639%	91.44%
	80%	90.7151%	91.76%

Table 2: SVM results before fine tuning

4.1.2 Fine-Tuning SVM Parameters and Experimental Results

I tried different kernels and parameter values, and the settings below gave the best accuracy, so they were chosen for fine tuning.

The new values of the SVM parameters after fine tuning in both rapidminer and weka :

- **Kernel Type:** Rbf
- **Gamma:** 0.1
- **Cost (C):** 1
- **Batch Size:** 80
- **Epsilon (eps):** 0.001
- **Shrinking:** true

The results:

	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	91.374	92.52%
	10	91.3441	92.59%
	15	91.3665	92.58%
Split Validation	60%	90.7525	92.25%
	70%	90.4144	92.31%
	80%	90.6776	92.10%

Table 3: SVM results after fine tuning

4.1.3 Results Analysis & Comparison

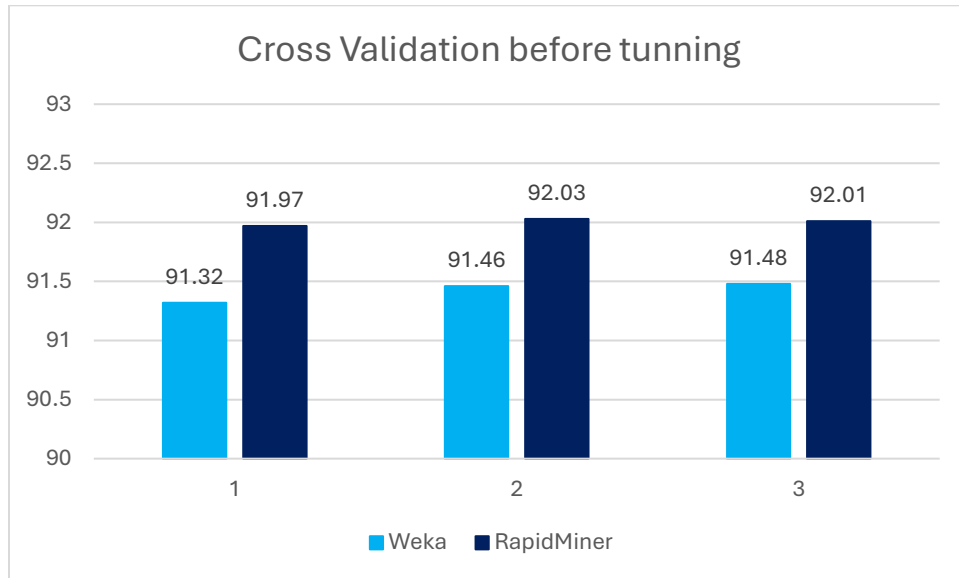


Figure 1: Cross Validation before Fine Tunning

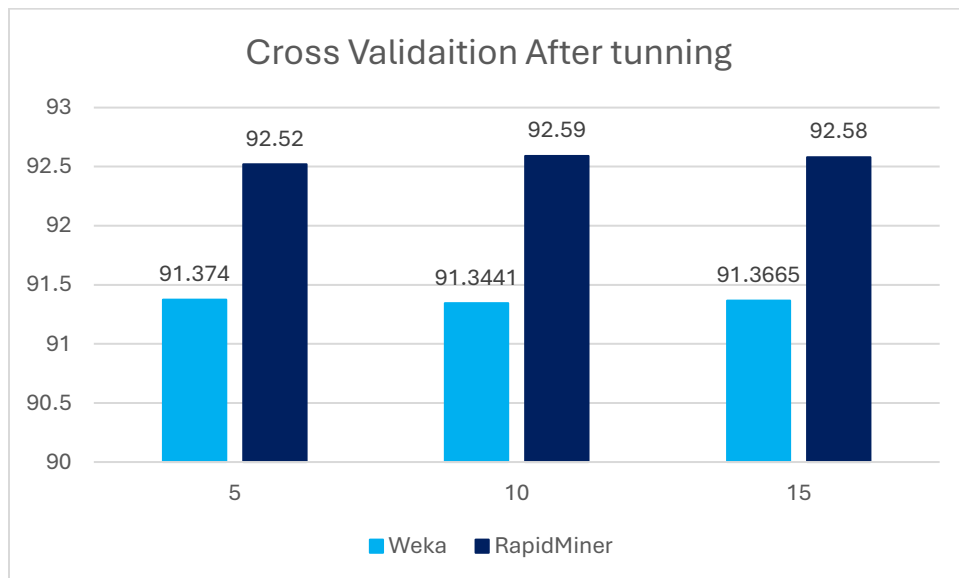


Figure 2: Cross Validation After Fine tuning

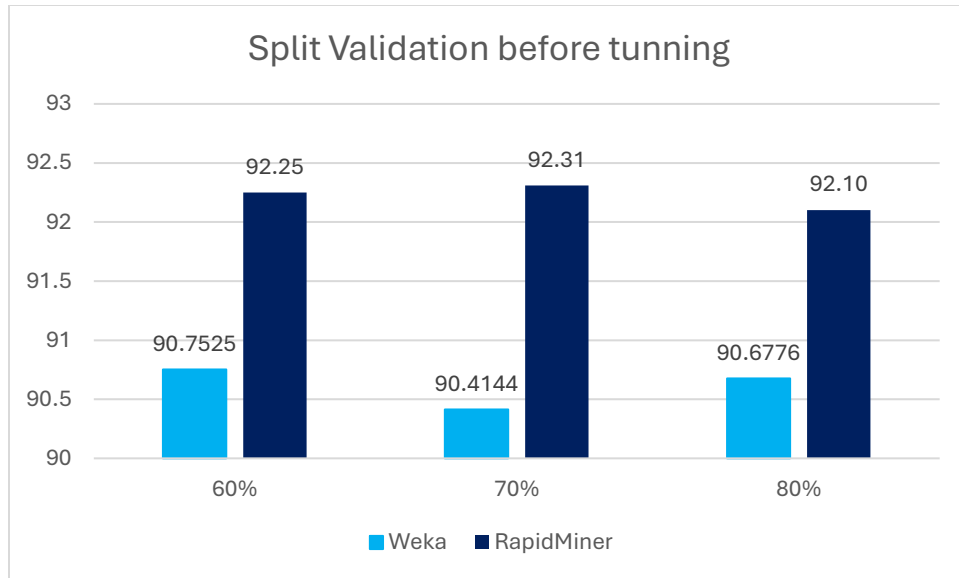


Figure 3: Split Validation Before Fine Tunning

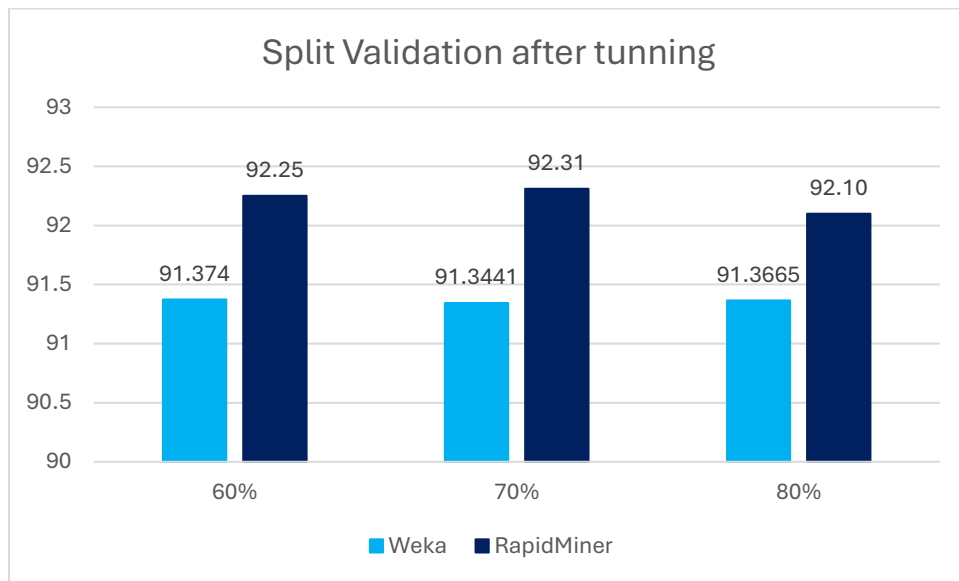


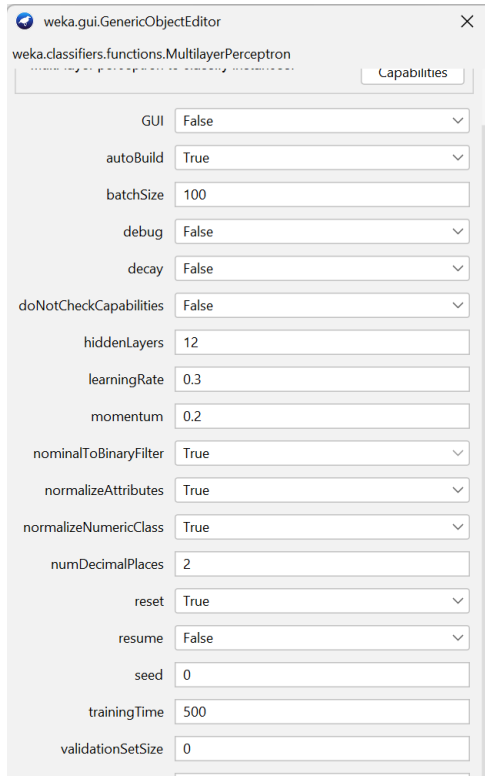
Figure 4: Split Validation After fine Tunning

As we can see the figures above, RapidMiner consistently achieved slightly higher accuracy than Weka across all validation setups. After fine-tuning the SVM parameters, RapidMiner showed a clear improvement, while Weka's accuracy only increased slightly. Cross-validation produced more stable and reliable results than split validation, and the highest accuracy was observed with 10-fold cross-validation in RapidMiner. This indicates that the performance of the SVM model depends on both the software platform and how its parameters are fine-tuned.

4.2 Neural Network (NN)

4.2.1 Default NN Experimental Results

The parameters in Weka are:



weka.gui.GenericObjectEditor

weka.classifiers.functions.MultilayerPerceptron

Capabilities

GUI: False

autoBuild: True

batchSize: 100

debug: False

decay: False

doNotCheckCapabilities: False

hiddenLayers: 12

learningRate: 0.3

momentum: 0.2

nominalToBinaryFilter: True

normalizeAttributes: True

normalizeNumericClass: True

numDecimalPlaces: 2

reset: True

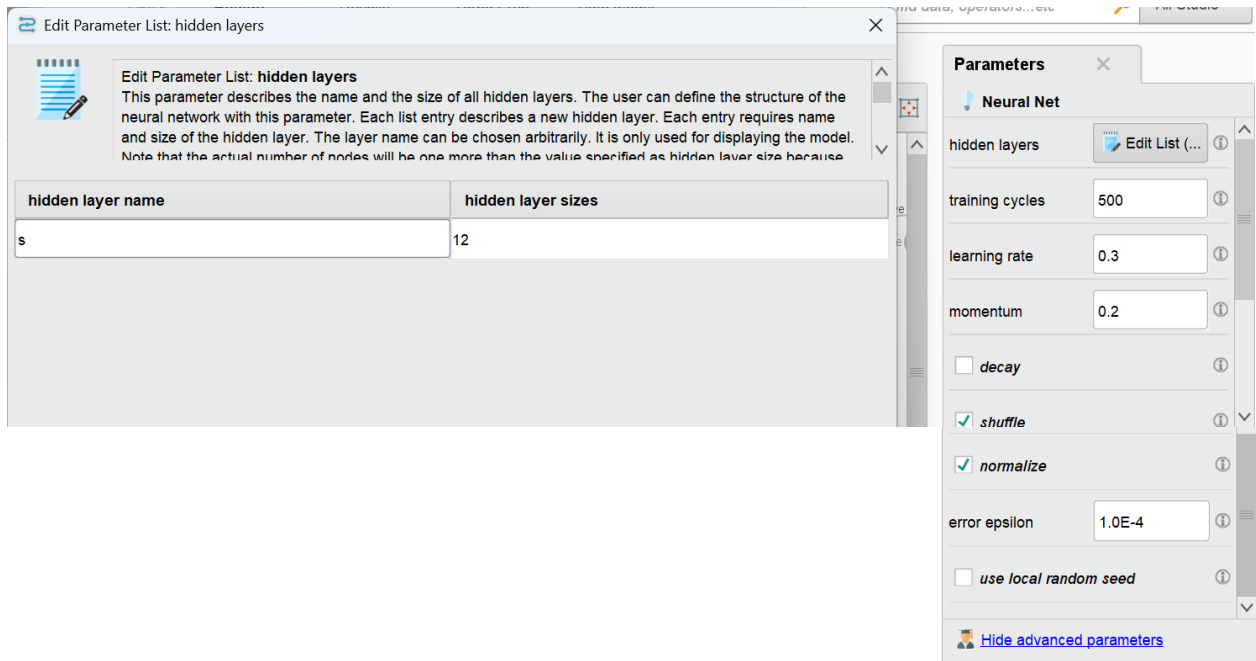
resume: False

seed: 0

trainingTime: 500

validationSetSize: 0

The parameters in RapidMiner:



Edit Parameter List: hidden layers

Edit Parameter List: hidden layers

This parameter describes the name and the size of all hidden layers. The user can define the structure of the neural network with this parameter. Each list entry describes a new hidden layer. Each entry requires name and size of the hidden layer. The layer name can be chosen arbitrarily. It is only used for displaying the model. Note that the actual number of nodes will be one more than the value specified as hidden layer size because

hidden layer name	hidden layer sizes
s	12

Parameters

Neural Net

hidden layers: Edit List (...)

training cycles: 500

learning rate: 0.3

momentum: 0.2

☐ decay

☒ shuffle

☒ normalize

error epsilon: 1.0E-4

☐ use local random seed

[Hide advanced parameters](#)

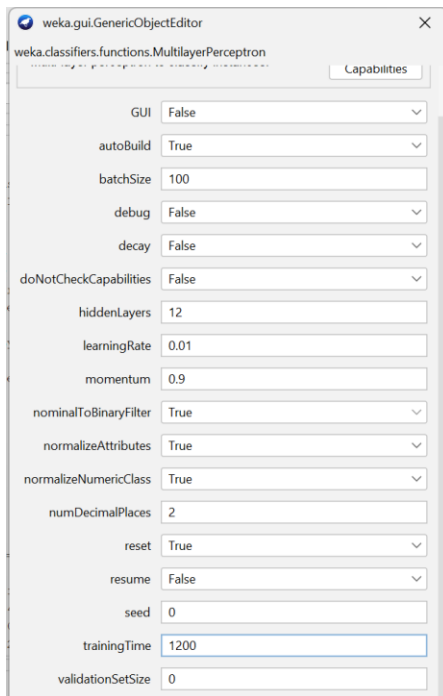
The results:

	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	91.3815 %	91.16%
	10	91.2392 %	91.23%
	15	91.2767 %	91.00%
Split Validation	60%	91.0146 %	90.71%
	70%	90.5642 %	91.01%
	80%	90.453 %	90.56%

Table 4: NN results before fine tuning

4.2.2 Fine-Tuning NN Parameters and Experimental Results

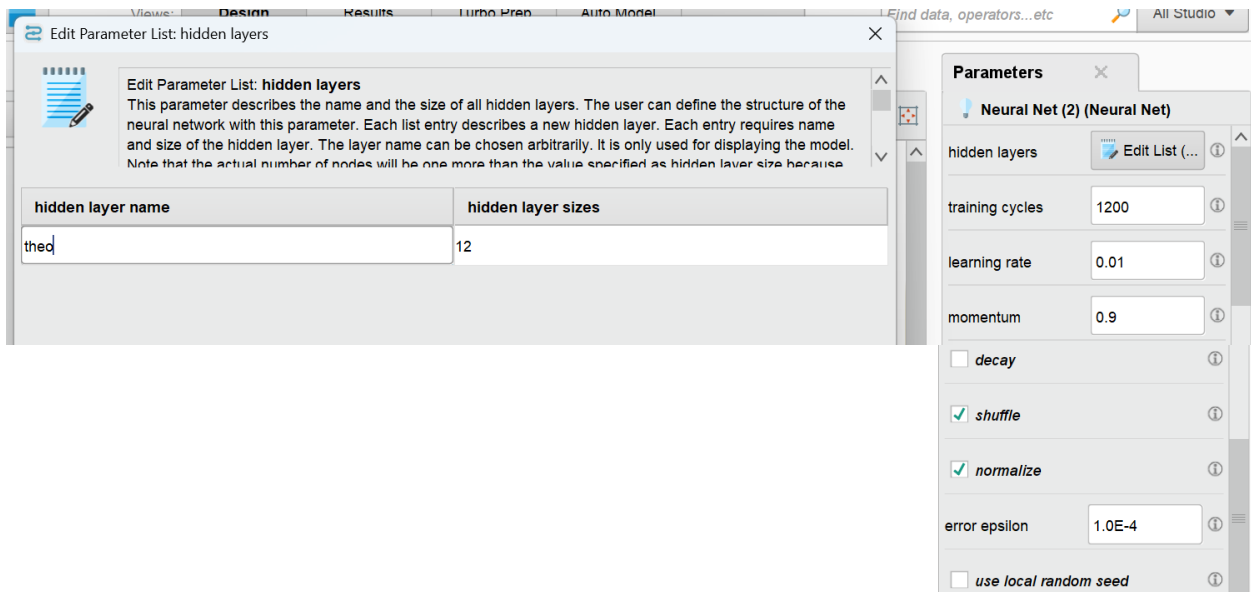
The parameters in Weka are:



The image shows the 'weka.gui.GenericObjectEditor' window for the 'weka.classifiers.functions.MultilayerPerceptron' classifier. It contains a list of parameters with their current values:

Parameter	Value
GUI	False
autoBuild	True
batchSize	100
debug	False
decay	False
doNotCheckCapabilities	False
hiddenLayers	12
learningRate	0.01
momentum	0.9
nominalToBinaryFilter	True
normalizeAttributes	True
normalizeNumericClass	True
numDecimalPlaces	2
reset	True
resume	False
seed	0
trainingTime	1200
validationSetSize	0

The parameters in RapidMiner:



The image shows the 'Edit Parameter List: hidden layers' dialog box in RapidMiner. It contains a table with the following data:

hidden layer name	hidden layer sizes
theq	12

Below the table, there is a description of the parameter and a note. The description states: 'This parameter describes the name and the size of all hidden layers. The user can define the structure of the neural network with this parameter. Each list entry describes a new hidden layer. Each entry requires name and size of the hidden layer. The layer name can be chosen arbitrarily. It is only used for displaying the model. Note that the actual number of nodes will be one more than the value specified as hidden layer size because...'

The 'Parameters' panel on the right shows the following settings for 'Neural Net (2) (Neural Net)':

Parameter	Value
hidden layers	Edit List (...)
training cycles	1200
learning rate	0.01
momentum	0.9
decay	☐
shuffle	☒
normalize	☒
error epsilon	1.0E-4
use local random seed	☐

The results:

	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	91.711 %	91.63%
	10	91.3665 %	91.76%
	15	91.8982 %	91.61%
Split Validation	60%	90.5279 %	91.54%
	70%	91.0384 %	91.24%
	80%	90.79 %	91.24%

Table 5: NN results after fine tuning

4.2.3 Results Analysis & Comparison

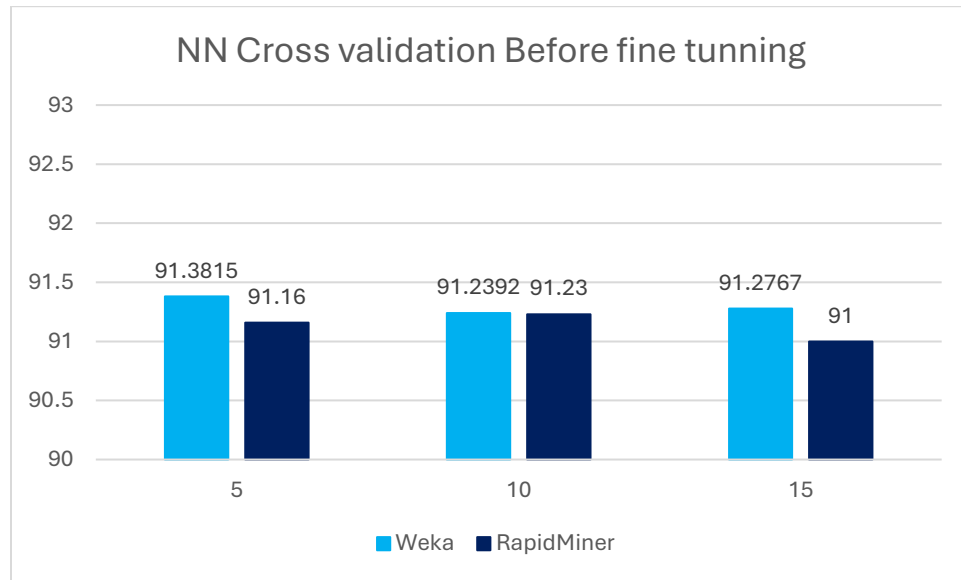


Figure 5: NN cross validation before fine tuning

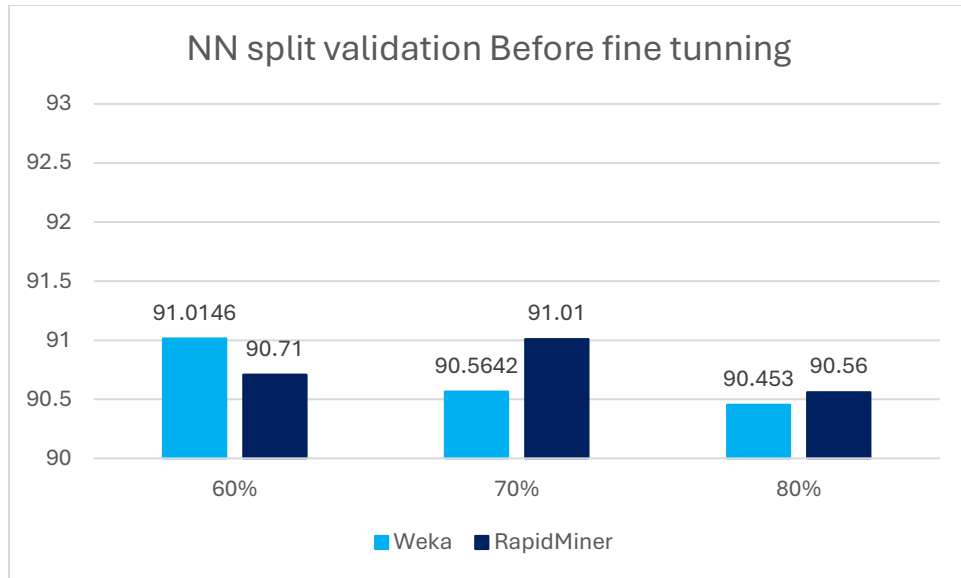


Figure 6: NN split validation before fine tuning

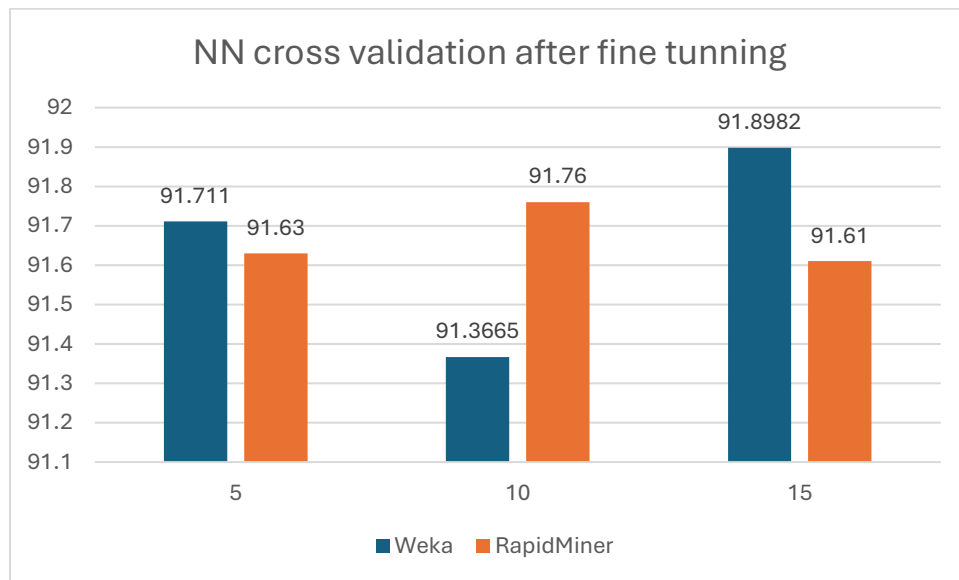


Figure 7: NN cross validation after fine tuning

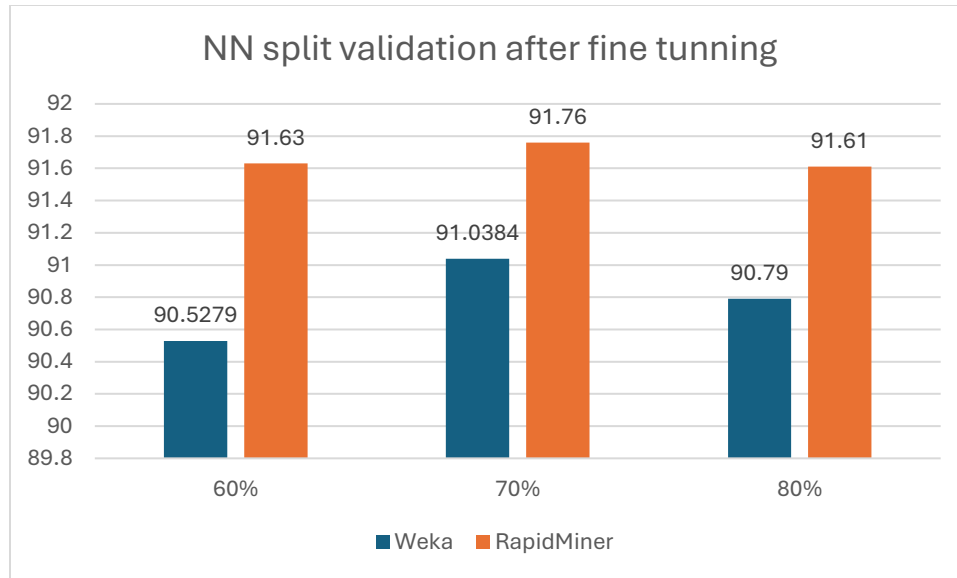


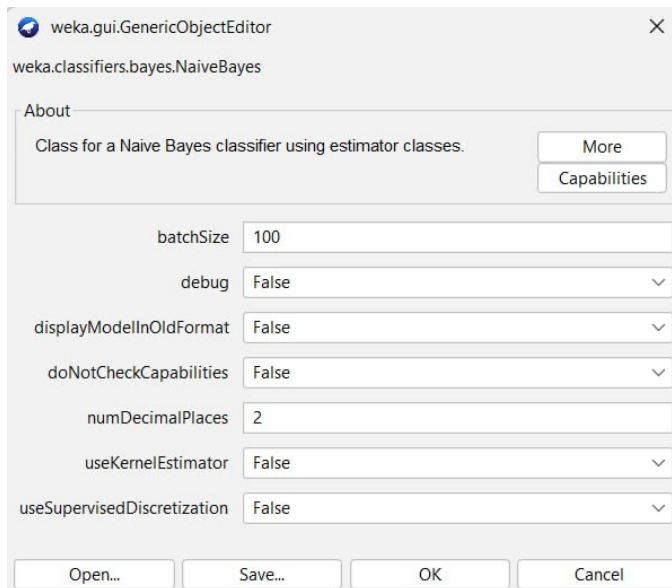
Figure 8: NN split validation after fine tuning

The experimental results demonstrate that the performance leadership fluctuated between the two software platforms depending on the specific validation strategy and hyperparameter configuration. In some test scenarios, RapidMiner demonstrated superior stability and slightly better results, particularly when using robust validation methods. Conversely, Weka outperformed RapidMiner in other instances after fine-tuning the model architecture. Therefore, we can conclude that both platforms are highly capable and comparable, with neither showing a definitive or consistent superiority over the other across all experiments. The observed variations highlight that the optimal tool often depends on the specific tuning and validation approach rather than inherent software differences.

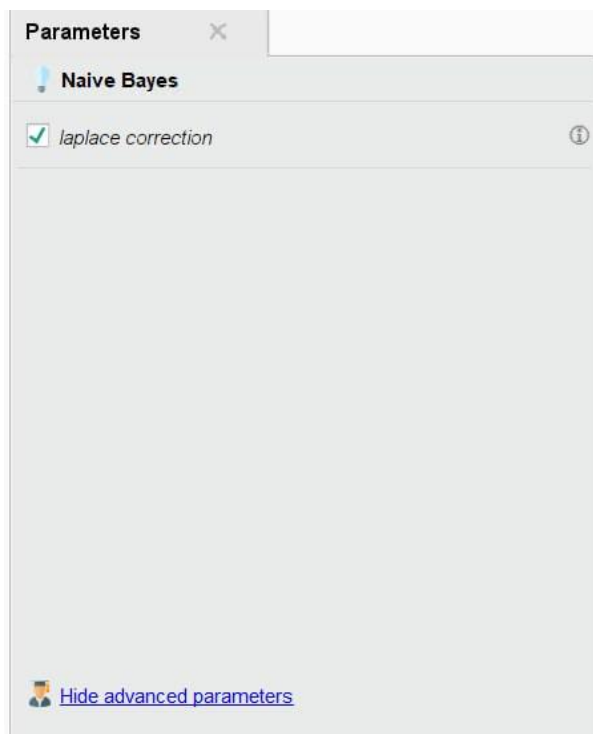
4.3 Naïve Bayes

4.3.1 Default NB Experimental Results

The default parameters in Weka are:



The default parameters in RapidMiner:



The results:

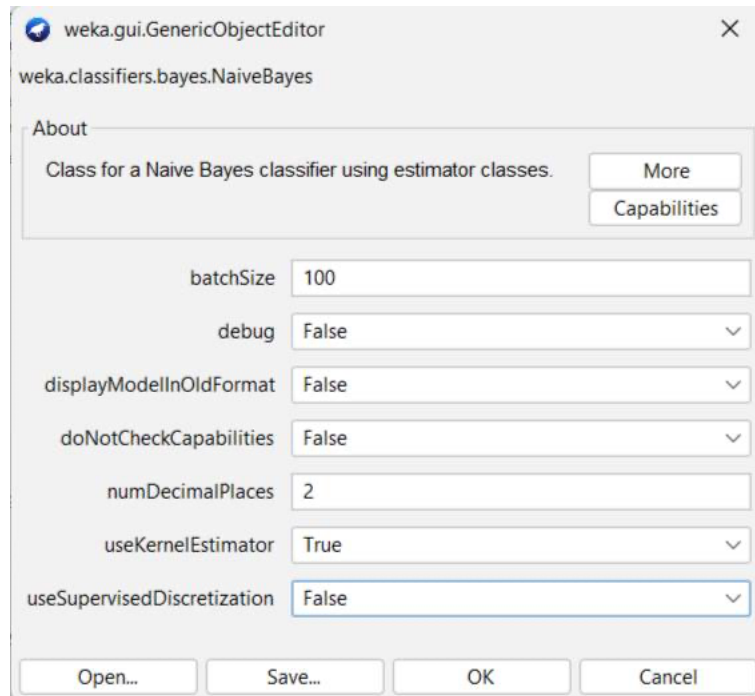
	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	93.3358 %	93.36%
	10	93.4182 %	93.31%
	15	93.4257 %	93.34%
Split Validation	60%	93.4669 %	93.39%
	70%	92.8607 %	93.38%
	80%	92.8491 %	93.33%

Table 6: Naive bayes results before fine tuning

4.3.2 Fine-Tuning NB Parameters and Experimental Results

The new values of the NB parameters after fine tuning in both rapidminer and weka :

Weka:



Rapidminer:

The screenshot shows the 'Parameters' window for the 'Naive Bayes (Kernel)' algorithm. The parameters are as follows:

- laplace correction:** ☐ (unchecked)
- estimation mode:** **greedy** (selected from a dropdown menu)
- minimum bandwidth:** **0.3** (text input)
- number of kernels:** **2** (text input)
- use application grid:** ☐ (unchecked)

The results:

	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	94.5189 %	94.17%
	10	94.5414 %	94.15%
	15	94.5788 %	94.11%
Split Validation	60%	94.4028 %	93.93%
	70%	93.7843 %	94.03%
	80%	93.7851 %	93.82%

Table 7: Naive bayes results after fine tuning

4.3.3 Results Analysis & Comparison

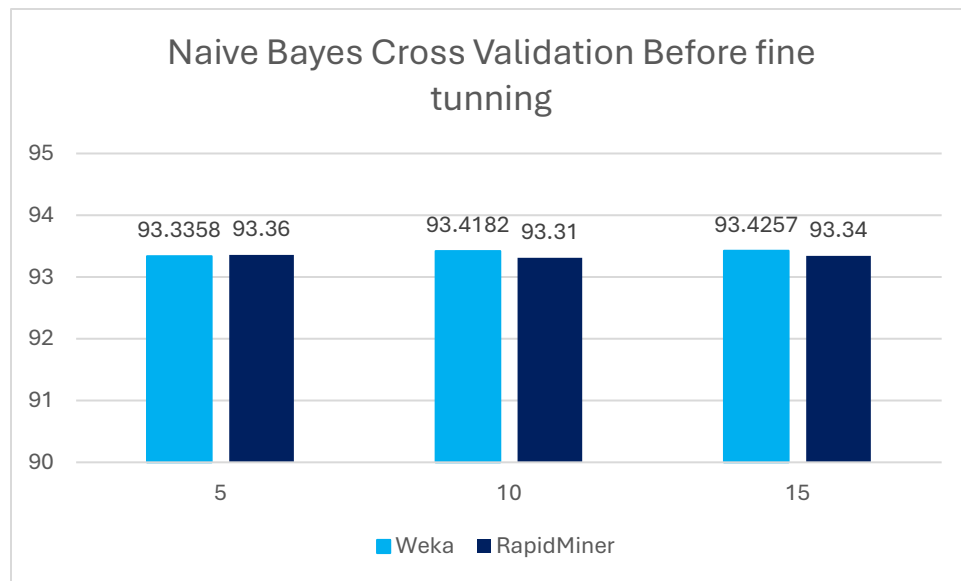


Figure 9: Naive bayes Cross Validation before fine tuning

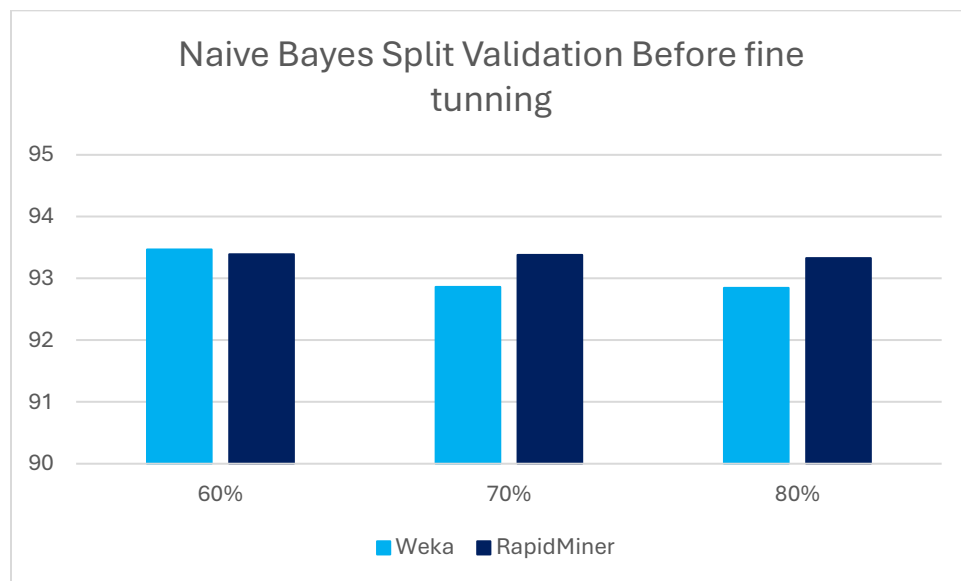


Figure 10: Naive Bayes Split Validation before fine tuning

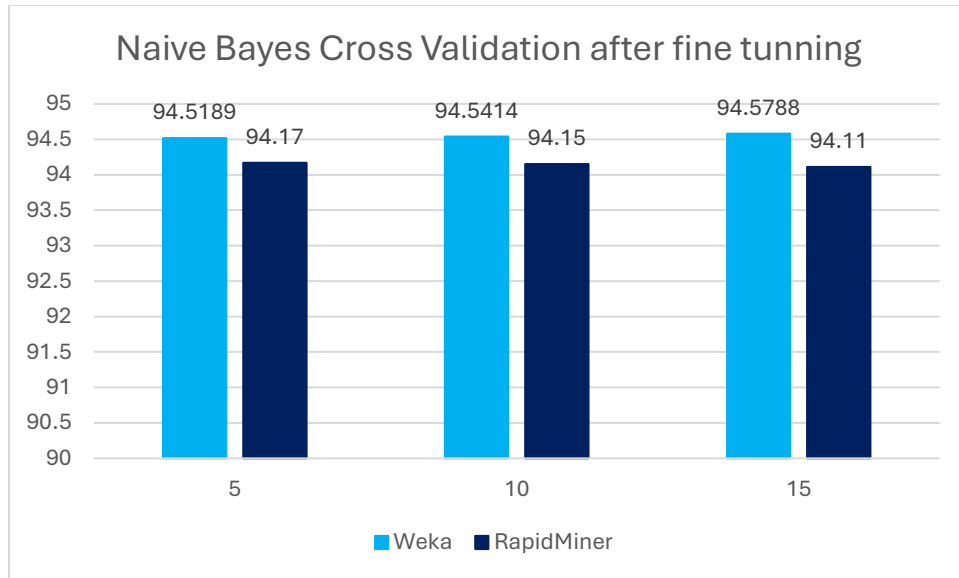


Figure 11: Naive Bayes Cross Validation after fine tuning

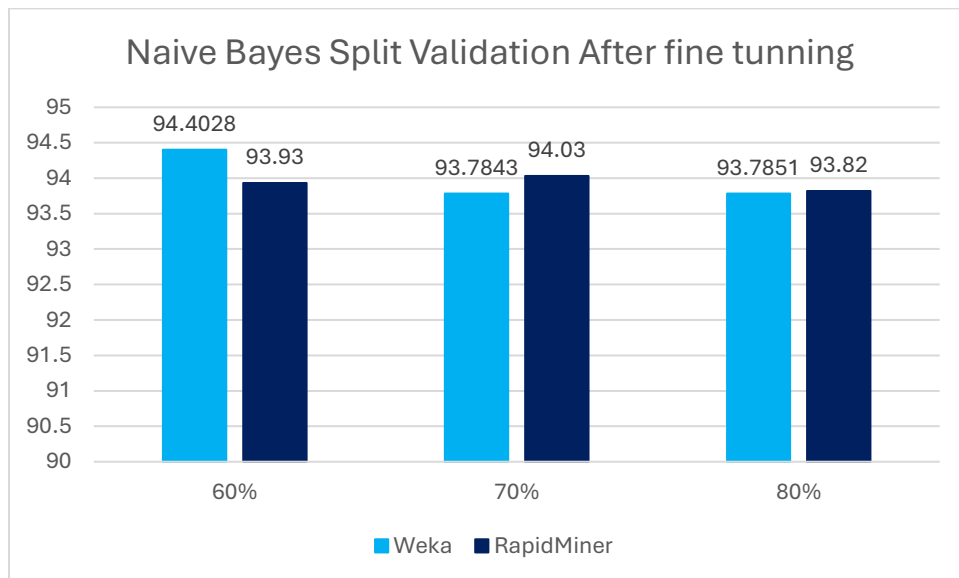


Figure 12: Naive Bayes Split validation after fine tuning

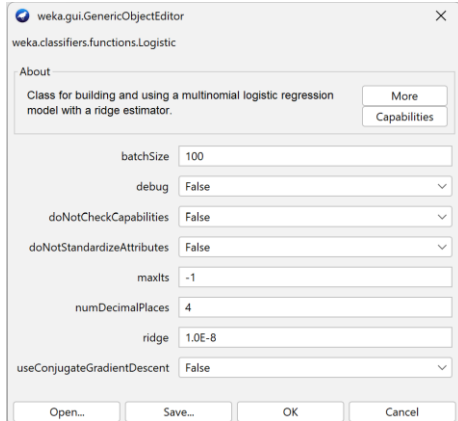
The results show that both RapidMiner and Weka achieved high accuracy using the default Naïve Bayes model, with RapidMiner being slightly more stable across different validation settings. After enabling Kernel Density Estimation during tuning, the accuracy improved in both platforms. The improvement was more noticeable in Weka, while RapidMiner showed a smaller but consistent increase.

Cross-validation provided more stable results than split validation, and the highest accuracy was achieved with 10-fold cross-validation after tuning. Overall, the performance of Naïve Bayes clearly improved after parameter tuning, especially when using kernel estimation.

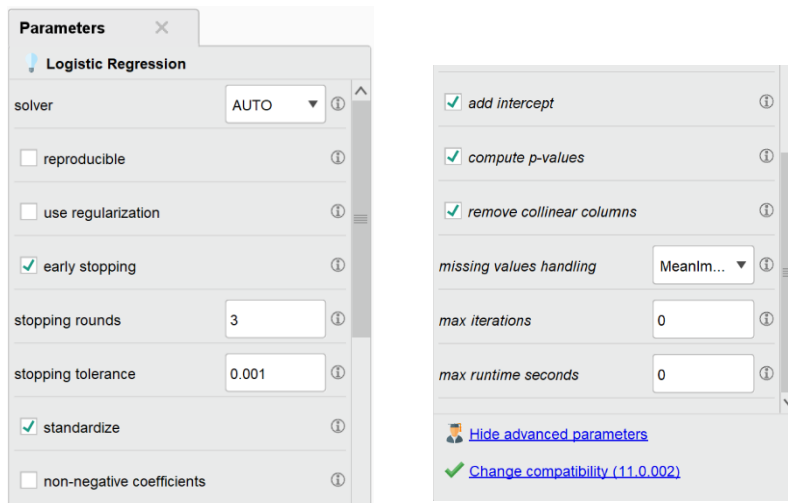
4.4 Logistic Regression

4.4.1 Default LR Experimental Results

The default parameters in Weka are:



The default parameters in RapidMiner:



The results:

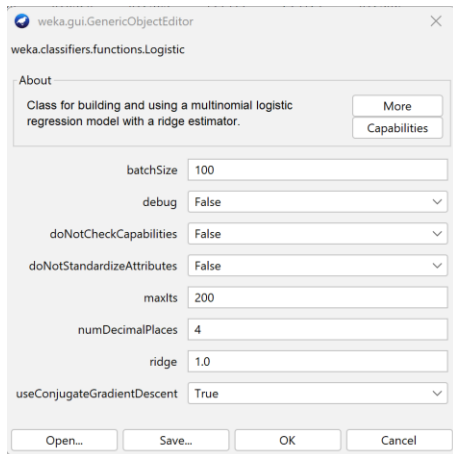
	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	91.6511 %	90.36%
	10	91.6586 %	90.37%
	15	91.6885 %	90.38%
Split Validation	60%	91.2954 %	90.54%
	70%	90.689 %	90.39%
	80%	90.9023 %	90.00%

Table 8: Logistic regression results before fine tuning

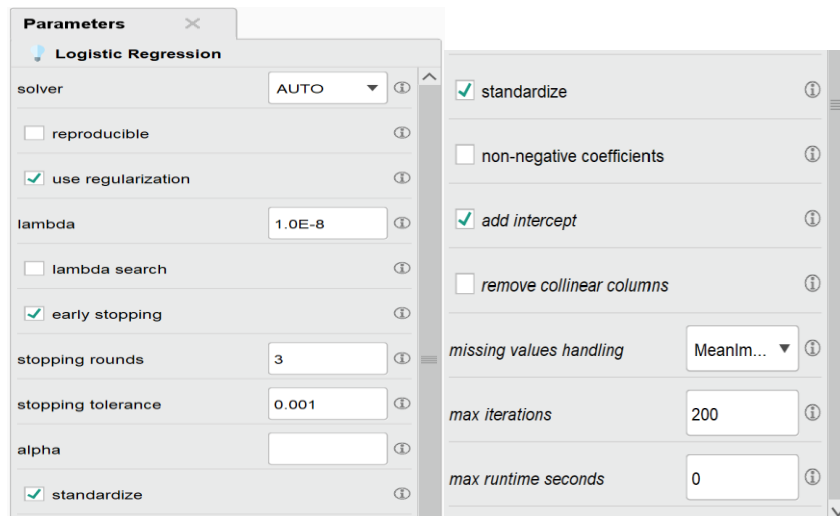
4.4.2 Fine-Tuning LR Parameters and Experimental Results

The new values of the LR parameters after fine tuning in both rapidminer and weka :

Weka:



RapidMiner:



The results:

	Percentage / folds	Weka accuracy	RapidMiner accuracy
Cross Validation	5	91.681 %	90.36%
	10	91.6286 %	90.37%
	15	91.6735 %	90.38%
Split Validation	60%	91.3328 %	90.54%
	70%	90.7888 %	90.39%
	80%	91.052 %	90.00%

Table 9: logistic regression results after fine tuning

4.3.3 Results Analysis & Comparison

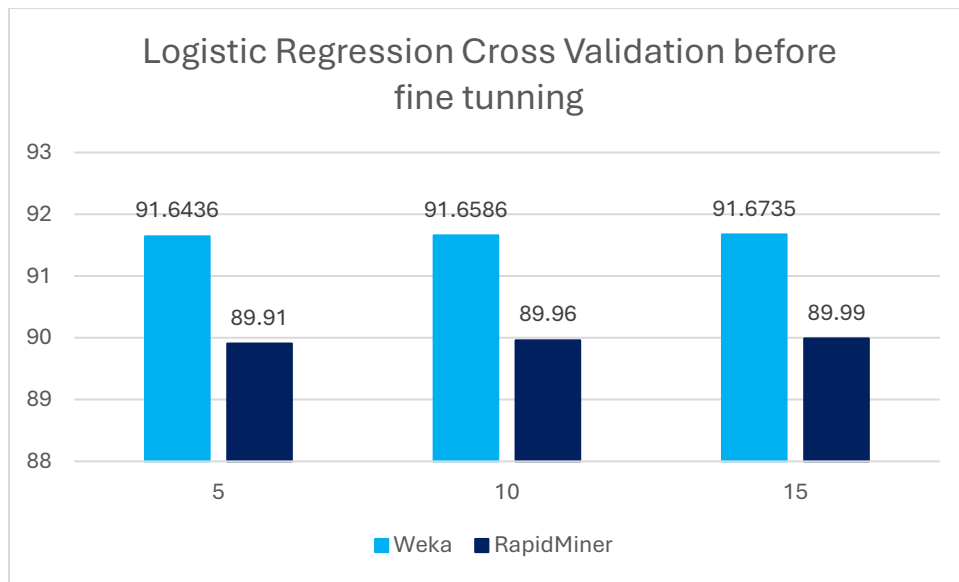


Figure 13: LR cross validation before fine tuning

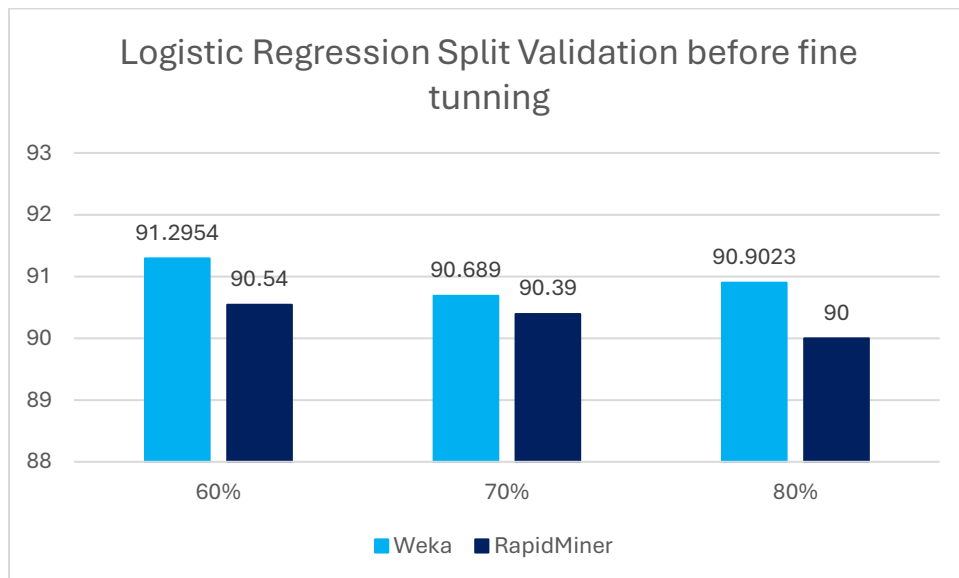
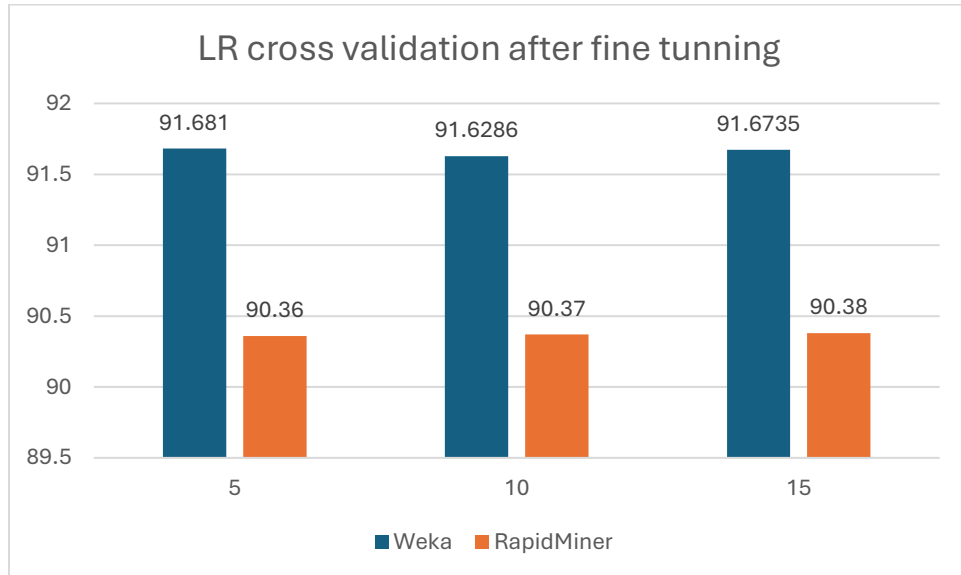
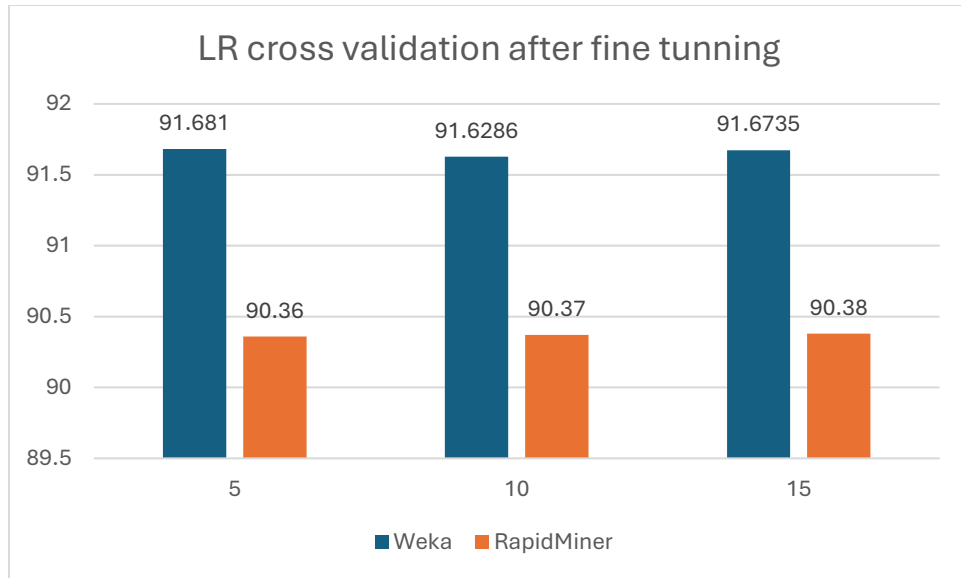


Figure 14: LR split validation before fine tuning



The experimental results clearly showed that Weka consistently outperformed RapidMiner when using the Logistic Regression (LR) algorithm across all six evaluation scenarios, including both Cross-Validation and Split Validation. Weka reached a maximum accuracy of 91.6885%, while RapidMiner's highest was 90.54%, indicating that Weka's default setup was naturally more robust and reliable for this dataset.

Even after fine-tuning, Weka's performance stayed very stable, around 91.6%, suggesting that its default settings were already close to optimal for the task. On the other hand, changing parameters in RapidMiner didn't bring any noticeable improvement; accuracy remained almost the same in both default and tuned runs, showing that it was less sensitive to the regularization settings or had already reached its performance limit. Overall, Weka provided a stronger baseline and greater stability compared to RapidMiner for this classification task.

5. Conclusion

In this project, we tested four machine learning algorithms using both Weka and RapidMiner. We noticed that different algorithms and software platforms gave different results. Fine-tuning the parameters had a noticeable effect, especially for SVM and Naïve Bayes, which reached the highest accuracy after adjusting their settings. Neural Networks improved slightly after tuning, while Logistic Regression remained relatively stable.

When comparing software, RapidMiner often produced more consistent results, particularly for SVM and Neural Networks. On the other hand, Weka sometimes outperformed RapidMiner, especially for SVM and Logistic Regression. This shows that the choice of software can affect performance, but it is not the only factor, the algorithm and parameter tuning matter more.

Overall, we can conclude that SVM and Naïve Bayes were the best performing algorithms for this dataset. Fine-tuning played an important role in improving results, while Logistic Regression was already quite stable. Both Weka and RapidMiner are capable tools, and the best results depend on which algorithm is used, how it is tuned, and the validation method applied. Exploring these combinations is essential to achieve the highest predictive performance.

6. References

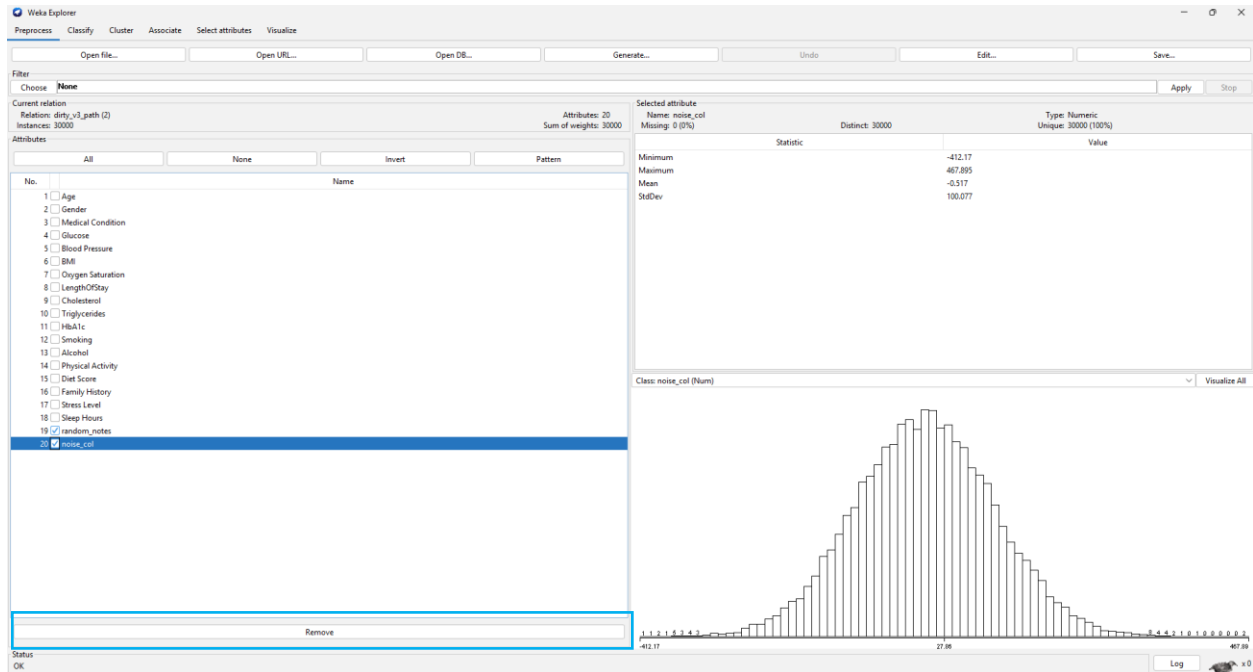
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<https://www.ibm.com/think/topics/logistic-regression#:~:text=In%20logistic%20regression%2C%20the%20model,value%20between%200%20and%201>

7. Appendices

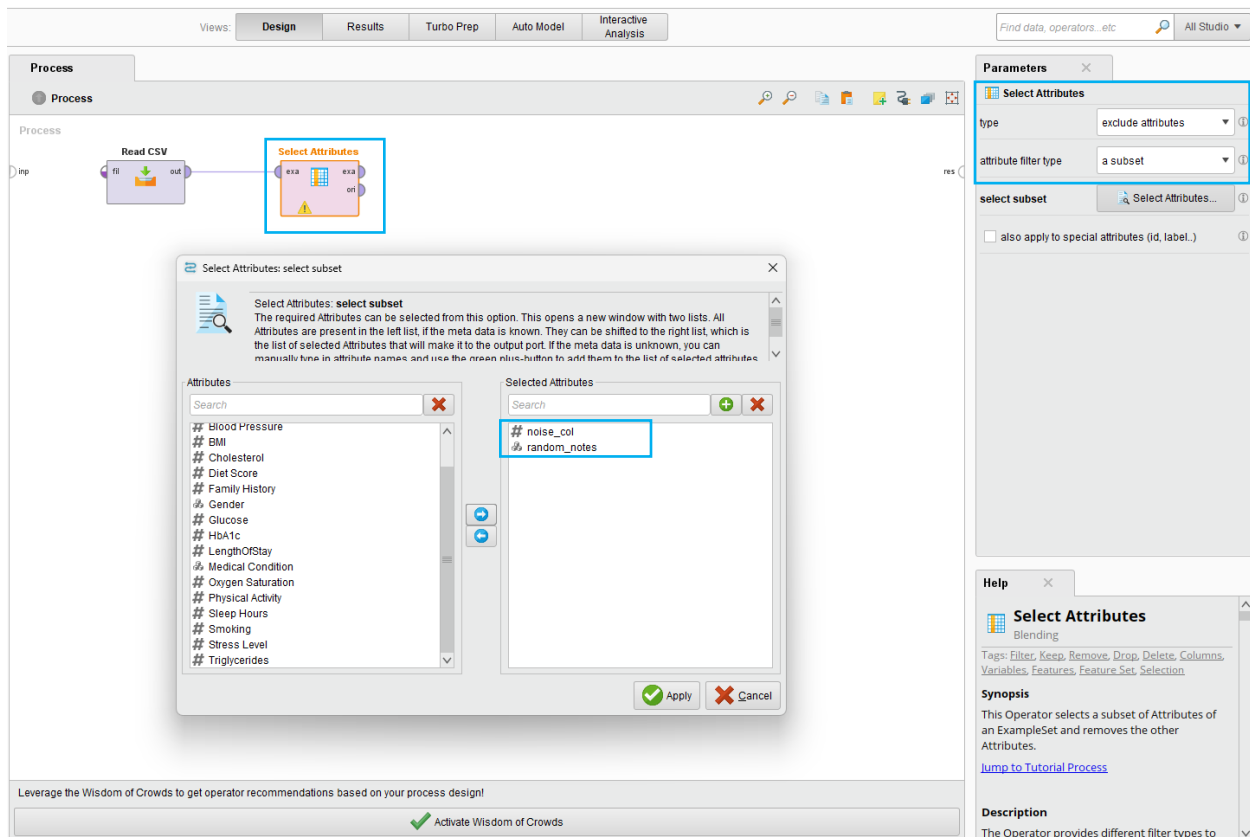
Appendix A

Data preprocessing:

Removing irrelevant columns in weka :



Removing irrelevant columns in rapidMiner



Removing missing rows in weka :

Missing values were handled manually in Weka by sorting the dataset in ascending order for each of the five affected attributes (Age, Gender, Medical Condition, Glucose, and Blood Pressure) to easily identify and delete the rows containing null values

Viewer

Relation: dirty_v3_path (1)

No.	1: Age Numeric	2: Gender Nominal	3: Medical Condition Nominal	4: Glucose Numeric	5: Blood Pressure Numeric	6: BMI Numeric	7: Oxygen Saturation Numeric	8: LengthOfStay Numeric	9: Cholesterol Numeric	10: Triglycerides Numeric	11: HbA1c Numeric	12: Sm Num
25...	89.0	Male	Arthritis	80.35	129.94	27.0	94.99	2.0	172.1	118.74	5.71	
25...		Female	Diabetes	219.73	157.39	29.21	95.93	5.0	213.5	219.72	7.06	
25...		Male	Diabetes	125.74	142.31	34.43	89.32	3.0	227.27	135.65	8.63	
25...			Diabetes		158.48	22.86	98.39	4.0	192.4	157.67	8.99	
25...			Obesity	114.98	106.82	39.13	94.75	3.0	225.25	202.22	6.12	
25...		Male	Healthy	93.31	109.76	29.84	95.67	2.0	211.2	127.6	5.11	
25...		Female	Arthritis	63.96	155.46	22.41	97.97	2.0	200.86	123.24	6.44	
25...			Arthritis	91.61	162.24	21.09	96.42	3.0	184.25	172.21	5.07	
25...		Female	Healthy	86.29		23.78	100.07	1.0	187.67	154.32	5.04	
25...		Female	Hypertension	107.51	186.72	22.78	95.83	5.0	330.53	167.99	6.13	
25...		Female	Obesity	87.55	130.36	29.99	94.37	3.0	240.12	194.68	6.86	
25...		Female	Diabetes	142.29	146.6	24.72	95.78	3.0	200.4	220.51	6.56	
25...		Male	Diabetes	215.76	155.44	35.14			97.97	148.25	5.52	
25...		Male	Diabetes	222.06		29.99			78.24	159.23	9.02	
25...		Male	Hypertension	93.24	158.17	38.21			268.6	227.43	4.72	
25...		Female	Hypertension	112.22	153.7	32.34			20.98	159.78	6.06	
25...		Female	Hypertension	121.61	156.54	28.46			74.95	264.9	6.63	
25...		Female	Healthy	94.64		22.86			95.99	131.49	4.93	
25...		Male	Hypertension	102.64	167.2	24.63			40.23	155.61	4.79	
25...		Male	Hypertension	119.25		17.02			55.13	128.85	4.81	
25...		Female	Cancer	90.49		22.86			227.4	66.66	5.63	
25...		Female	Hypertension	113.06	129.53	23.47			55.89	231.8	6.35	
25...		Male	Obesity	98.14		30.12			95.86	157.12	5.89	

Undo

Copy

Search...

Clear search

Delete selected instance

Delete ALL selected instances

Insert new instance

Set instance weight

Add instance

Undo

OK

Cancel

Viewer												
Relation: dirty_v3_path (1)-weka.filters.unsupervised.attribute.Remove-R19-20												
1: Age	2: Gender	3: Medical Condition	4: Glucose	5: Blood Pressure	6: BMI	7: Oxygen Saturation	8: LengthOfStay	9: Cholesterol	10: Triglycerides	11: HbA1c	12: Smoking	
Numeric	Nominal	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	
47.0	Male	Diabetes	318.51	154.54	35.55	98.05	6.0	250.15	231.71	5.63	0.0	
44.0	Male	Asthma		76.98	23.24	94.12	3.0	169.14	161.91	6.15	0.0	
31.0	Female	Asthma		81.32	30.62	98.31	3.0	233.63	153.95	5.8	1.0	
52.0	Female	Asthma		87.65	27.63	84.12	2.0	179.95	120.5	5.11	0.0	
30.0		Healthy		88.51	30.88	91.56	1.0	177.67	157.0	4.87	0.0	
46.0	Female	Asthma		88.67	28.57	86.74	2.0	197.38	178.07	5.13	1.0	
34.0	Female	Obesity		89.78	30.84	98.18	2.0	246.81	208.91	5.87	0.0	
68.0	Male	Cancer		91.6	36.65	94.94	8.0	168.2	205.72	6.24	0.0	
53.0	Male	Diabetes		91.67	31.76	99.86	3.0	191.35	197.56	9.79	0.0	
46.0				92.03	26.2	85.65	2.0	210.92	154.85	5.19	1.0	
30.0	Female	Asthma		92.36	23.05	88.06	2.0	212.22	105.28	4.87	1.0	
41.0	Male	Cancer		93.44	26.05	99.59	15.0	208.71	181.46	4.95	0.0	
71.0	Female			94.82	26.98	95.38	3.0	255.71	138.29	8.07	0.0	
28.0		Obesity		94.86	28.26	98.36	3.0	176.09	224.23	5.42	1.0	
70.0	Female	Diabetes		94.95	29.95	95.9	3.0	214.59	158.18	8.79	0.0	
40.0	Female	Obesity		95.2	29.76	101.2	4.0	214.14	145.23	5.75	0.0	
51.0	Male	Arthritis		95.2	36.02	96.15	4.0	246.45	129.56	5.95	0.0	
32.0	Female	Asthma		95.32	23.79	102.3	3.0	182.74	179.91	4.14	1.0	
49.0	Female	Obesity		95.42	45.22	94.4	4.0	180.96				
67.0	Female	Hypertension		95.98	27.32	90.82	4.0	213.89	216.73	5.77	1.0	
70.0	Male			96.72	27.84	96.07	17.0	189.0	200.15	5.24	1.0	
39.0		Healthy		96.92	26.15	94.75	1.0	182.24	135.42	5.27	0.0	
41.0	Female			97.12	23.64	94.73	3.0	188.75	187.42	8.75	0.0	
Right click (or left+alt) for context menu												
Add instance Undo OK Cancel												
Relation: dirty_v3_path												
No.	1: Age	2: Gender	3: Medical Condition	4: Glucose	5: Blood Pressure	6: BMI	7: Oxygen Saturation	8: LengthOfStay	9: Cholesterol	10: Triglycerides	11: HbA1c	12: Sm Num
	Numeric	Nominal	Nominal	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	
18...	62.0	Male		293.6	116.12	27.75	90.44	4.0	253.88	161.81	9.31	
18...	61.0	Male		294.07	128.99	29.18	96.94	3.0	183.99	126.67	8.49	
18...	59.0	Male		312.69	146.35	34.19	98.68	3.0	177.71	122.24	10.34	
18...	52.0		Diabetes	67.6	157.64	30.07	98.15	8.0	204.55	208.99	6.08	
18...	54.0		Diabetes	76.27		27.29	93.75	3.0	204.53	194.46	9.32	
18...	57.0		Diabetes	79.93	143.73	25.12	93.71	8.0	214.21	127.61	7.73	
18...	55.0		Diabetes	81.14	156.4	23.94	92.74	5.0	174.46	257.71	8.73	
18...	48.0		Diabetes	83.67	107.09	36.42	92.3	5.0	223.15	207.88	7.93	
18...	43.0		Diabetes	84.93	163.84	25.57	94.97	7.0	148.38	128.35	6.38	
18...	55.0		Diabetes	85.61	156.08	28.7	89.03	7.0	188.49	106.54	5.94	
18...	61.0		Diabetes	86.25		24.9	95.03	5.0	208.61	138.28	6.36	
18...	63.0		Diabetes	86.8	148.1	27.32	94.12	7.0	186.25	149.48	7.75	
18...	41.0		Diabetes	87.09	154.36	29.21	92.75	6.0	242.25	214.27	8.71	
18...	47.0		Diabetes	89.83	160.51	25.41	97.05	7.0	216.86	187.04	7.56	
18...	65.0		Diabetes	93.24	136.48	22.88	94.12	4.0	243.35	203.65	9.47	
18...	40.0		Diabetes	93.29	134.99	35.69	97.53	4.0	203.48	171.78	9.54	
18...	57.0		Diabetes			33.75	98.61	8.0	178.13	180.18	10.45	
18...	50.0		Diabetes	97.26	129.59	33.63	95.85	4.0	236.11	172.36	7.25	
18...	47.0		Diabetes	97.4	129.63	25.62	95.48	8.0	221.19	162.32	7.98	
18...	48.0		Diabetes	97.62	141.45	24.19	95.74	4.0	213.93	166.58	6.63	
18...	71.0		Diabetes	98.26	139.34	35.91	96.7	5.0	235.0	117.22	9.02	
18...	46.0		Diabetes	98.63	157.17	35.71	96.31	8.0	198.2	188.05	7.67	
18...	52.0		Diabetes	99.02	145.98	37.68	91.14	8.0	193.49	198.0	8.09	
Right click (or left+alt) for context menu												

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Viewer

Relation: dirty_v3_path

No.	1: Age Numeric	2: Gender Nominal	3: Medical Condition Nominal	4: Glucose Numeric	5: Blood Pressure Numeric	6: BMI Numeric	7: Oxygen Saturation Numeric	8: LengthOfStay Numeric	9: Cholesterol Numeric	10: Triglycerides Numeric	11: HbA1c Numeric	12: Sm Num
15...	59.0	Male	Arthritis	125.51		20.68	97.68	4.0	194.8	197.23	5.96	
15...	73.0	Female		116.28	90.71	30.66	95.77	2.0	242.56	172.89	5.44	
15...	17.0	Female		102.57	92.29	25.17	93.22	4.0	245.08	148.05	5.06	
15...	54.0	Female		63.78	92.63	25.37	95.35	2.0	202.88	164.51	5.05	
15...	15.0	Female		102.93	93.99	27.83	85.82	2.0	165.85	162.35	4.41	
15...	35.0	Female		105.75	95.44	30.06	83.95	4.0	190.53	195.19	6.15	
15...	64.0	Female		120.5	97.37	45.74	92.59	3.0	272.9	184.44	5.78	
15...	40.0	Female		89.95	97.45	24.95	96.11	3.0	209.74	163.67	5.76	
15...	63.0	Female		96.29	97.99	24.08	96.03	1.0	202.31	106.54	4.7	
15...	46.0	Female		110.03	99.38	19.4	98.96	1.0	207.83	138.27	4.79	
15...	55.0	Female		97.94	100.41	26.34	89.28	3.0	220.28	133.39	4.8	
15...	34.0	Female		95.73	101.07	26.93	94.18	1.0	137.89	145.84	4.57	
15...	57.0	Female		291.9	101.09	32.91	95.28	7.0	171.52	112.69	9.39	
15...	59.0	Female		102.98	101.15	29.34	96.13	1.0	205.83	152.35	4.97	
15...	33.0	Female		89.74	101.29	22.74	85.63	6.0	174.81	196.92	5.21	
15...	51.0	Female		81.12	101.43	22.87	94.36	2.0	185.91	188.21	4.31	
15...	38.0	Female		106.18	101.68	31.97	99.54	3.0	192.65	237.11	4.77	
15...	33.0	Female		98.25	101.94	30.01	87.05	4.0	219.66	129.87	4.72	
15...	66.0	Female		102.06	102.19	28.78	96.8	2.0	186.72	121.75	5.05	
15...	61.0	Female		56.04	102.25	20.07	100.4	1.0	163.94	135.69	4.99	
15...	21.0	Female		83.89	102.48	18.86	97.72	1.0	213.31	134.18	4.77	
15...	57.0	Female		195.54	102.53	27.58	93.24	4.0	185.56	171.85	7.88	
15...	69.0	Female		210.81	103.03	34.69	95.43	6.0	225.9	175.93	7.93	

Add instance Undo OK Cancel

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Viewer

Relation: dirty_v3_path

No.	1: Age Numeric	2: Gender Nominal	3: Medical Condition Nominal	4: Glucose Numeric	5: Blood Pressure Numeric	6: BMI Numeric	7: Oxygen Saturation Numeric	8: LengthOfStay Numeric	9: Cholesterol Numeric	10: Triglycerides Numeric	11: HbA1c Numeric	12: Sm Num
13...	50.0	Male	Hypertension	131.83	222.72	29.31	96.74	2.0	286.69	223.96	6.22	
13...	76.0	Female	Hypertension	85.45	223.1	33.01	96.76	6.0	272.76	218.66	6.89	
13...	55.0	Female	Hypertension	127.45	224.25	35.8	96.47	5.0	201.87	220.16	5.22	
13...	65.0	Female	Hypertension	44.7		31.69	95.87	6.0	162.77	81.08	4.86	
13...	64.0	Male	Obesity	53.7		31.61	96.75	4.0	251.87	271.09	7.02	
13...	47.0	Female	Healthy	58.5		19.92	96.38	2.0	163.26	108.45	4.92	
13...	75.0	Male	Hypertension	59.79		32.06	95.83	6.0	237.53	169.78	6.3	
13...	43.0	Male	Healthy	60.87		23.16	99.64	2.0	177.74	131.86	4.91	
13...	23.0	Male	Healthy	61.34		28.81	99.73	2.0	210.99	116.14	5.36	
13...	55.0	Female	Hypertension	63.35		24.12	98.8	3.0	184.53	191.94	5.55	
13...	47.0	Male	Hypertension	63.44		22.93	96.49	5.0	216.04	244.62	4.38	
13...	36.0	Female	Healthy	63.63		22.69	96.02	1.0	185.23	150.55	5.22	
13...	34.0	Male	Healthy	64.43		23.5	98.18	2.0	226.43	141.08	5.57	
13...	32.0	Male	Healthy	64.45		23.34	95.38	1.0	160.41	127.34	5.07	
13...	28.0	Male	Healthy	64.5		26.58	94.81	1.0	179.26	119.97	5.06	
13...	47.0	Female	Healthy	64.64		22.4	95.72	2.0	203.61	139.52	4.51	
13...	57.0	Male	Hypertension	65.32		27.17	96.61	4.0	257.67	142.23	5.74	
13...	74.0	Male	Hypertension	65.65		24.61	95.74	5.0	226.13	173.1	5.06	
13...	61.0	Female	Hypertension	66.07		27.92	94.71	4.0	230.77	249.11	7.23	
13...	63.0	Female	Healthy	66.27		28.77	96.47	1.0	157.73	158.22	5.14	
13...	68.0	Female	Cancer	66.51		23.99	99.48	13.0	226.57	141.91	5.86	
13...	59.0	Female	Hypertension	67.44		27.11	92.1	6.0	227.11	233.32	4.73	
13...	52.0	Male	Hypertension	67.85		27.62	100.38	6.0	250.26	264.57	4.94	

Add instance Undo OK Cancel

Removing missing rows in rapidMiner:

The screenshot displays the rapidMiner software interface. The main process flow includes 'Read CSV', 'Select Attributes', and 'Filter Examples'. The 'Filter Examples' operator is highlighted with a blue box. A dialog box titled 'Create Filters: filters' is open, showing a list of attributes (Age, Gender, Medical Condition, Glucose, Blood Pressure) with the condition 'is missing' selected for each. The 'Match any' radio button is selected at the bottom of the dialog. On the right, the 'Parameters' panel for 'Filter Examples' shows the 'condition class' set to 'custom filters' and the 'invert filter' checkbox checked. Below the parameters, there is a 'Help' section for 'Filter Examples' with a synopsis and a link to the tutorial process.

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators...etc All Studio

Process

Process

Read CSV

Select Attributes

Filter Examples

Create Filters: filters

Create Filters: filters

This is the default parameter for defining filter conditions via 'Add Filters...' dialog window. It is also available when the 'custom_filters' condition class is selected. This option allows the definition of a custom filter condition. A condition consists of an Attribute, a comparison function and a value to match. More conditions can be added by the 'Add Entry' button. Several filters can be joined either by 'Match all' or 'Match any'.

Attribute	Comparison	Value	Action
Age	is missing		✖
Gender	is missing		✖
Medical Condition	is missing		✖
Glucose	is missing		✖
Blood Pressure	is missing		✖

☐ Match all ☒ Match any ☒ Preselect comparators

Add Entry OK Cancel

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

Parameters

Filter Examples

filters

Add Filters...

condition class

custom filters

☒ invert filter

Hide advanced parameters

Change compatibility (12.0.002)

Help

Filter Examples

AI Studio Core

Tags: Select, Keep, Remove, Drop, Delete, Rows, Cases, Instances, Lines, Observations, Filter, Missing, Filter.

Synopsis

This Operator selects which Examples of an ExampleSet are kept and which Examples are removed.

Jump to Tutorial Process

Description

Removing Duplicate rows in weka :

Filter

Choose **RemoveDuplicates**

Current relation

Relation: dirty_v3_path (2)-weka.filters.unsupervised.attribute.Remove-R19-20

Instances: 30000

Attributes: 18

Sum of weights: 30000

Attributes

All None Invert Pattern

No.	Name
1	☐ Age
2	☐ Gender
3	☐ Medical Condition
4	☐ Glucose
5	☐ Blood Pressure
6	☐ BMI
7	☐ Oxygen Saturation
8	☐ LengthOfStay
9	☐ Cholesterol
10	☐ Triglycerides
11	☐ HbA1c
12	☐ Smoking
13	☐ Alcohol
14	☐ Physical Activity
15	☐ Diet Score
16	☐ Family History
17	☐ Stress Level
18	☐ Sleep Hours

Removing Duplicate rows in rapidMiner:

The screenshot displays the RapidMiner software interface. The main workspace shows a process flow with four operators: Read CSV, Select Attributes, Filter Examples, and Remove Duplicates. The Remove Duplicates operator is highlighted with a blue box. To the right, the Parameters panel for the Remove Duplicates operator is visible, showing the attribute filter type set to 'all' and several unchecked options: invert selection, include special attributes, and treat missing values as duplicates. Below the Parameters panel, the Help panel provides a synopsis of the operator, stating it removes duplicate examples from an ExampleSet by comparing all examples with each other on the basis of the specified attributes. Two examples are considered duplicate if the selected attributes have the same values in them. A link to the tutorial process is also provided.

Process: A connected set of Operators that help you to transform and analyze your data. Also known as: flow, program, pipeline, diagram

Process:

Read CSV → Select Attributes → Filter Examples → Remove Duplicates

Parameters

Remove Duplicates

attribute filter type: all

☐ invert selection

☐ include special attributes

☐ treat missing values as duplicates

Help

Remove Duplicates

AI Studio Core

Tags: Deduplication, Matches, Matching, Replicates, Copies, Cleansing, Quality, Distinct, Equal, Filter, Duplicates

Synopsis

This operator removes duplicate examples from an ExampleSet by comparing all examples with each other on the basis of the specified attributes. Two examples are considered duplicate if the selected attributes have the same values in them.

[Jump to Tutorial Process](#)

String to binary in weka :

The screenshot shows the Weka GUI. The Filter tab is selected, and the NominalToBinary filter is chosen. The current relation is 'dirty_v3_path (2)-weka.filters.unsupervised.attribute.Remove-R19-20' with 30000 instances. The attributes section shows a list of 18 attributes, all of which are currently unchecked. The filter is set to 'All'.

Filter

Choose **NominalToBinary** -R first-last

Current relation

Relation: dirty_v3_path (2)-weka.filters.unsupervised.attribute.Remove-R19-20

Instances: 30000

Attributes

Attributes: 18
Sum of weights: 30000

All None Invert Pattern

No.	Name
1	☐ Age
2	☐ Gender
3	☐ Medical Condition
4	☐ Glucose
5	☐ Blood Pressure
6	☐ BMI
7	☐ Oxygen Saturation
8	☐ LengthOfStay
9	☐ Cholesterol
10	☐ Triglycerides
11	☐ HbA1c
12	☐ Smoking
13	☐ Alcohol
14	☐ Physical Activity
15	☐ Diet Score
16	☐ Family History
17	☐ Stress Level
18	☐ Sleep Hours

String to binary in rapidMiner :

The screenshot displays the rapidMiner software interface. The main workspace shows a process flow with five operators: Read CSV, Select Attributes, Filter Examples, Remove Duplicates, and One-Hot Encoding. The One-Hot Encoding operator is highlighted with a blue box. A dialog box titled "Select Attributes: attributes" is open, showing a list of attributes with "Gender" selected. The "Parameters" panel on the right shows the configuration for the One-Hot Encoding operator, including the attribute filter type set to "subset" and the "perform encoding" checkbox checked. The "Help" panel on the right provides a synopsis of the operator, stating it can remove nominal attributes with too many values and perform an encoding on the remaining nominal attributes.

Process

Process

Read CSV

Select Attributes

Filter Examples

Remove Duplicates

One-Hot Encoding

Select Attributes: attributes

Select Attributes: attributes

The attribute which should be chosen.

Attributes

Search

Medical Condition

Selected Attributes

Search

Gender

Apply

Cancel

Parameters

One-Hot Encoding

Model Simulator

Tags: Nominal, Numerical, Dummy, Coding, One-Hot, Types

Synopsis

This operator can remove nominal attributes with too many values and perform an encoding on the remaining nominal attributes which will transform them into new numerical ones.

[Jump to Tutorial Process](#)

Description

Hide advanced parameters

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

Standardization in weka :

Filter

Choose **Standardize**

Current relation

Relation: dirty_v3_path (2)-weka.filters.unsupervised.attribute.Remove-R19-20

Instances: 30000

Attributes: 18

Sum of weights: 30000

Attributes

All None Invert Pattern

No.	Name
1	☐ Age
2	☐ Gender
3	☐ Medical Condition
4	☐ Glucose
5	☐ Blood Pressure
6	☐ BMI
7	☐ Oxygen Saturation
8	☐ LengthOfStay
9	☐ Cholesterol
10	☐ Triglycerides
11	☐ HbA1c
12	☐ Smoking
13	☐ Alcohol
14	☐ Physical Activity
15	☐ Diet Score
16	☐ Family History
17	☐ Stress Level
18	☐ Sleep Hours

Standardization in rapidMiner :

The screenshot displays the RapidMiner software interface. At the top, a workflow is visible in the 'Process' tab, consisting of several operators: 'Read CSV', 'Select Attributes', 'Filter Examples', 'Remove Duplicates', 'Normalize' (highlighted with a blue box), and 'One-Hot Encoding'. Below the workflow, a 'Select Attributes: attributes' dialog box is open, showing a list of attributes on the left and a 'Selected Attributes' list on the right. The attributes listed include Age, Blood Pressure, BMI, Cholesterol, Diet Score, Glucose, HbA1c, LengthOfStay, Oxygen Saturation, Physical Activity, Sleep Hours, Stress Level, and Triglycerides. The 'Selected Attributes' list currently contains Alcohol, Family History, and Smoking. To the right of the main interface, the 'Parameters' panel for the 'Normalize' operator is visible. It shows the 'attribute filter type' set to 'subset', the 'attributes' field with a 'Select Attributes...' button, the 'invert selection' checkbox checked, the 'include special attributes' checkbox unchecked, and the 'method' set to 'Z-transformation'. Below the parameters panel, a 'Help' panel for the 'Normalize' operator is also visible, providing a synopsis and description of the operator.

Process

Process

inp

Read CSV

Select Attributes

Filter Examples

Remove Duplicates

Normalize

One-Hot Encoding

res

res

Select Attributes: attributes

Select Attributes: attributes

The attribute which should be chosen.

Attributes

Search

Age

Blood Pressure

BMI

Cholesterol

Diet Score

Glucose

HbA1c

LengthOfStay

Oxygen Saturation

Physical Activity

Sleep Hours

Stress Level

Triglycerides

Selected Attributes

Search

Alcohol

Family History

Smoking

Apply

Cancel

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

Parameters

Normalize

attribute filter type

subset

attributes

Select Attributes...

invert selection

include special attributes

method

Z-transformation

Hide advanced parameters

Change compatibility (12.0.002)

Help

Normalize

AI Studio Core

Tags: Normalization, Standardize, Z-Transformation, Scaling, Features, Attributes, Variables, Columns

Synopsis

This Operator normalizes the values of the selected Attributes.

[Jump to Tutorial Process](#)

Description

Normalization is used to scale values so they fit

Normalization in weka :

Filter

Choose **Normalize -S 1.0 -T 0.0**

Current relation

Relation: dirty_v3_path (2)-weka.filters.unsupervised.attribute.Remove-R19-20

Instances: 30000

Attributes: 18

Sum of weights: 30000

Attributes

All None Invert Pattern

No.	Name
1	☒ Age
2	☐ Gender
3	☐ Medical Condition
4	☐ Glucose
5	☐ Blood Pressure
6	☐ BMI
7	☐ Oxygen Saturation
8	☐ LengthOfStay
9	☐ Cholesterol
10	☐ Triglycerides
11	☐ HbA1c
12	☐ Smoking
13	☐ Alcohol
14	☐ Physical Activity
15	☐ Diet Score
16	☐ Family History
17	☐ Stress Level
18	☐ Sleep Hours

Normalization in rapidMiner :

The screenshot displays the rapidMiner software interface for data preprocessing. The main workspace shows a sequence of operators: Read CSV, Select Attributes, Filter Examples, Remove Duplicates, Normalize, and One-Hot Encoding. The Normalize operator is currently selected and highlighted with a blue border.

On the right side, the **Parameters** panel for the Normalize operator is visible. It includes the following settings:

- attribute filter type: subset
- attributes: Select Attributes...
- ☒ invert selection
- ☐ include special attributes
- method: range transformation
- min: 0.0
- max: 1.0

Below the parameters, there are links for [Hide advanced parameters](#) and [Change compatibility \(12.0.002\)](#).

In the center, a dialog box titled "Select Attributes: attributes" is open. It prompts the user to "Select Attributes: attributes" and "The attribute which should be chosen." The dialog contains two lists:

- Attributes:** A list of attributes including Age, Blood Pressure, BMI, Cholesterol, Diet Score, Glucose, HbA1c, LengthOfStay, Oxygen Saturation, Physical Activity, Sleep Hours, Stress Level, and Triglycerides.
- Selected Attributes:** A list of attributes including Alcohol, Family History, and Smoking.

At the bottom of the dialog, there are "Apply" and "Cancel" buttons.

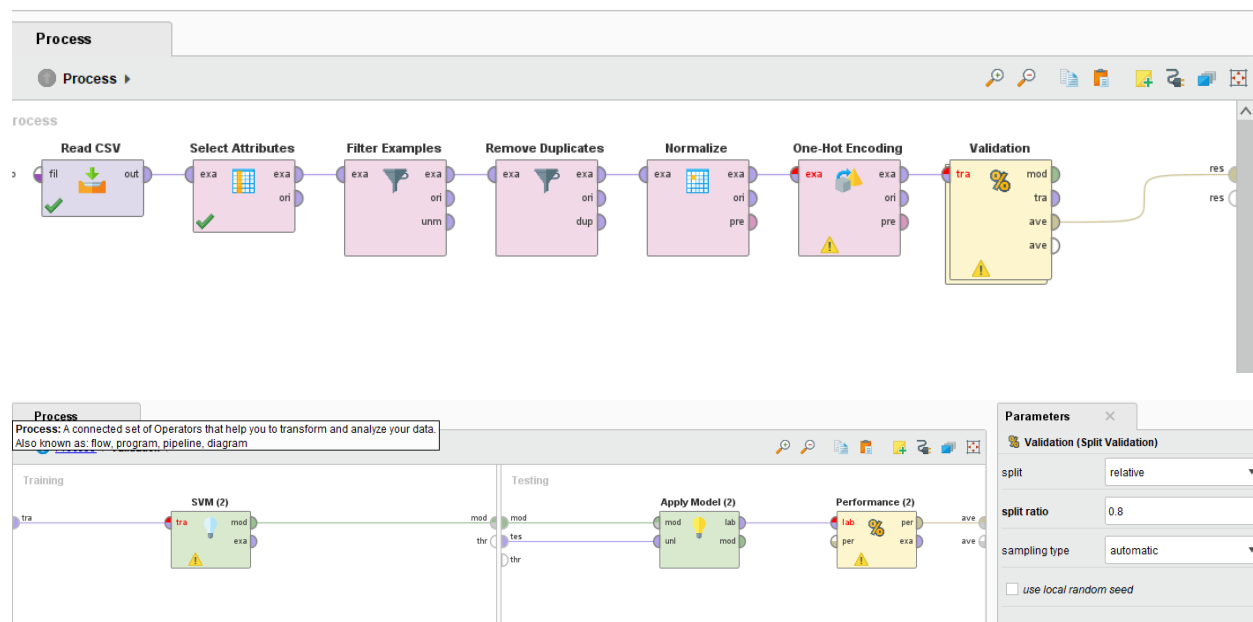
At the bottom of the main interface, there is a status bar with the text "Leverage the Wisdom of Crowds to get operator recommendations based on your process design!" and a checkbox labeled "Activate Wisdom of Crowds" which is currently checked.

Appendix B

Support Vector Machine SVM

Before Fine tuning :

In RapidMiner :



Split 60 :

accuracy: 91.82%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	397	1	10	5	2	3	16	91.47%
pred. Cancer	0	232	0	0	3	0	1	98.31%
pred. Obesity	7	0	719	0	18	29	34	89.10%
pred. Healthy	7	0	0	634	0	1	0	98.75%
pred. Diabetes	0	5	18	0	1294	5	14	96.86%
pred. Arthritis	7	0	30	2	7	263	69	69.58%
pred. Hypertension	6	6	30	0	34	67	1365	90.52%
class recall	93.63%	95.08%	89.10%	98.91%	95.29%	71.47%	91.06%	

Split 70 :

accuracy: 91.44%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	299	1	9	0	2	3	10	92.28%
pred. Cancer	0	175	0	0	1	0	0	99.43%
pred. Obesity	4	0	527	0	16	28	28	87.40%
pred. Healthy	6	0	0	479	0	0	0	98.76%
pred. Diabetes	0	3	16	0	969	3	14	96.42%
pred. Arthritis	6	0	25	2	3	191	49	69.20%
pred. Hypertension	3	4	28	0	28	51	1023	89.97%
class recall	94.03%	95.63%	87.11%	99.58%	95.09%	69.20%	91.01%	

Split 80 :

accuracy: 91.76%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	196	0	5	0	0	3	4	94.23%
pred. Cancer	0	119	0	0	0	0	0	100.00%
pred. Obesity	4	0	356	0	10	18	16	88.12%
pred. Healthy	3	0	0	319	0	0	1	98.76%
pred. Diabetes	0	3	9	0	650	1	12	96.30%
pred. Arthritis	5	0	14	1	5	130	37	67.71%
pred. Hypertension	4	0	19	0	14	32	680	90.79%
class recall	92.45%	97.54%	88.34%	99.69%	95.73%	70.65%	90.67%	

After fine tuning :

Cross validation : fold 15

accuracy: 92.58% +/- 1.03% (micro average: 92.58%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	979	10	11	2	0	14	21	94.41%
pred. Cancer	0	569	0	0	9	0	0	98.44%
pred. Obesity	18	0	1839	0	43	73	74	89.84%
pred. Healthy	16	0	0	1598	0	0	1	98.95%
pred. Diabetes	5	13	19	0	3266	6	37	97.61%
pred. Arthritis	13	0	44	2	9	624	126	76.28%
pred. Hypertension	30	18	104	0	69	204	3489	89.14%
class recall	92.27%	93.28%	91.18%	99.75%	96.17%	67.75%	93.09%	

Fold 10 :

accuracy: 92.59% +/- 0.65% (micro average: 92.59%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	982	11	11	2	1	12	21	94.42%
pred. Cancer	0	571	0	0	10	0	1	98.11%
pred. Obesity	15	0	1838	0	47	75	66	90.05%
pred. Healthy	17	0	0	1596	0	0	1	98.88%
pred. Diabetes	4	12	21	0	3261	5	36	97.66%
pred. Arthritis	12	1	46	4	9	623	129	75.61%
pred. Hypertension	31	15	101	0	68	206	3494	89.25%
class recall	92.55%	93.61%	91.13%	99.63%	96.02%	67.64%	93.22%	

Fold 5 :

accuracy: 92.52% +/- 0.53% (micro average: 92.52%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	976	9	12	3	2	15	24	93.76%
pred. Cancer	0	570	0	0	7	0	1	98.62%
pred. Obesity	16	0	1837	0	48	71	69	90.00%
pred. Healthy	18	0	0	1595	0	0	1	98.82%
pred. Diabetes	3	12	21	0	3262	4	33	97.81%
pred. Arthritis	13	1	42	4	8	627	131	75.91%
pred. Hypertension	35	18	105	0	69	204	3489	89.01%
class recall	91.99%	93.44%	91.08%	99.56%	96.05%	68.08%	93.09%	

Split validation 60 :

accuracy: 92.25%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	393	1	4	2	0	4	11	94.70%
pred. Cancer	0	227	0	0	0	0	0	100.00%
pred. Obesity	5	0	725	0	19	29	26	90.17%
pred. Healthy	10	0	0	636	0	0	0	98.45%
pred. Diabetes	1	6	16	0	1299	1	12	97.30%
pred. Arthritis	5	0	26	3	7	257	60	71.79%
pred. Hypertension	10	10	36	0	33	77	1390	89.33%
class recall	92.69%	93.03%	89.84%	99.22%	95.66%	69.84%	92.73%	

split validation 70 :

accuracy: 92.31%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	296	3	4	2	0	3	7	93.97%
pred. Cancer	0	170	0	0	0	0	0	100.00%
pred. Obesity	5	0	541	0	14	27	20	89.13%
pred. Healthy	7	0	0	476	0	0	0	98.55%
pred. Diabetes	0	5	12	0	975	2	10	97.11%
pred. Arthritis	5	0	21	3	5	197	44	71.64%
pred. Hypertension	5	5	27	0	25	47	1043	90.54%
class recall	93.08%	92.90%	89.42%	98.96%	95.68%	71.38%	92.79%	

split validation 80 :

accuracy: 92.10%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypertension	class precision
pred. Asthma	195	1	4	1	0	3	3	94.20%
pred. Cancer	0	116	0	0	0	0	0	100.00%
pred. Obesity	4	0	365	0	11	14	14	89.46%
pred. Healthy	3	0	0	317	0	0	0	99.06%
pred. Diabetes	0	3	6	0	648	1	11	96.86%
pred. Arthritis	4	0	10	2	4	131	35	70.43%
pred. Hypertension	6	2	18	0	16	35	687	89.92%
class recall	91.98%	95.08%	90.57%	99.06%	95.43%	71.20%	91.60%	

Weka :

Before tuning :

Weka 5 folds:

```
Time taken to build model: 1.68 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      12196           91.3216 %
Incorrectly Classified Instances    1159           8.6784 %
Kappa statistic                    0.8918
Mean absolute error                 0.2056
Root mean squared error            0.3037
Relative absolute error            89.2734 %
Root relative squared error        89.4922 %
Total Number of Instances         13355

=== Detailed Accuracy By Class ===
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.896	0.005	0.941	0.896	0.918	0.911	0.982	0.876	Asthma
	0.905	0.001	0.977	0.905	0.940	0.938	0.995	0.923	Cancer
	0.913	0.021	0.884	0.913	0.899	0.880	0.974	0.846	Obesity
	0.996	0.002	0.984	0.996	0.990	0.988	0.999	0.983	Healthy
	0.955	0.009	0.973	0.955	0.964	0.952	0.987	0.956	Diabetes
	0.592	0.015	0.747	0.592	0.660	0.643	0.922	0.524	Arthritis
	0.925	0.057	0.864	0.925	0.894	0.851	0.950	0.833	Hypertension
Weighted Avg.	0.913	0.023	0.912	0.913	0.912	0.892	0.972	0.871	

```
=== Confusion Matrix ===

  a   b   c   d   e   f   g  <-- classified as
951   1   24   23   2   12   48 |  a = Asthma
11 552   0   0   14   1   32 |  b = Cancer
12   0 1842   0   28   28  107 |  c = Obesity
 2   0   0 1595   0   5   0 |  d = Healthy
 3   12   47   0 3244   8   82 |  e = Diabetes
 8   0   85   2   5  545  276 |  f = Arthritis
24   0   85   1  40  131 3467 |  g = Hypertension
```

10 folds:

```
Time taken to build model: 1.7 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      12214          91.4564 %
Incorrectly Classified Instances    1141          8.5436 %
Kappa statistic                    0.8936
Mean absolute error                 0.2056
Root mean squared error            0.3037
Relative absolute error             89.2649 %
Root relative squared error        89.4825 %
Total Number of Instances         13355

=== Detailed Accuracy By Class ===
```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.903	0.005	0.939	0.903	0.921	0.914	0.983	0.880	Asthma
	0.910	0.001	0.977	0.910	0.942	0.940	0.995	0.925	Cancer
	0.914	0.021	0.887	0.914	0.900	0.882	0.975	0.848	Obesity
	0.995	0.002	0.985	0.995	0.990	0.988	0.999	0.984	Healthy
	0.955	0.009	0.973	0.955	0.964	0.952	0.987	0.956	Diabetes
	0.604	0.015	0.747	0.604	0.668	0.650	0.924	0.531	Arthritis
	0.924	0.055	0.868	0.924	0.895	0.853	0.951	0.836	Hypertension
Weighted Avg.	0.915	0.023	0.914	0.915	0.913	0.894	0.972	0.873	

```
=== Confusion Matrix ===

  a   b   c   d   e   f   g  <-- classified as
958   1   23   22   2   11   44 |   a = Asthma
 10 555    0    0   15    0   30 |   b = Cancer
 12   0 1843    0   29   29  104 |   c = Obesity
  4   0    0 1594    0    4    0 |   d = Healthy
  3  12   47    0 3243    7   84 |   e = Diabetes
  9   0   82    2    5  556  267 |   f = Arthritis
 24   0   83    1   38  137 3465 |   g = Hypertension
```

15 folds :

Time taken to build model: 1.63 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	12217	91.4788 %
Incorrectly Classified Instances	1138	8.5212 %
Kappa statistic	0.8938	
Mean absolute error	0.2056	
Root mean squared error	0.3037	
Relative absolute error	89.2615 %	
Root relative squared error	89.4798 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.901	0.005	0.943	0.901	0.921	0.915	0.983	0.882	Asthma
	0.920	0.001	0.977	0.920	0.948	0.946	0.995	0.932	Cancer
	0.914	0.021	0.884	0.914	0.899	0.881	0.974	0.846	Obesity
	0.996	0.002	0.986	0.996	0.991	0.989	0.999	0.985	Healthy
	0.955	0.009	0.974	0.955	0.964	0.952	0.987	0.956	Diabetes
	0.598	0.015	0.746	0.598	0.664	0.646	0.924	0.528	Arthritis
	0.925	0.055	0.868	0.925	0.896	0.854	0.952	0.837	Hypertension
Weighted Avg.	0.915	0.023	0.914	0.915	0.913	0.894	0.972	0.873	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
956	0	25	21	2	12	45		a = Asthma
10	561	0	0	13	0	26		b = Cancer
12	0	1844	0	28	28	105		c = Obesity
3	0	0	1595	0	4	0		d = Healthy
3	13	50	0	3242	7	81		e = Diabetes
10	0	84	1	5	551	270		f = Arthritis
20	0	82	1	40	137	3468		g = Hypertension

60 split :

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances	4881	91.3703 %
Incorrectly Classified Instances	461	8.6297 %
Kappa statistic	0.8923	
Mean absolute error	0.2057	
Root mean squared error	0.3037	
Relative absolute error	89.289 %	
Root relative squared error	89.5276 %	
Total Number of Instances	5342	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.871	0.004	0.951	0.871	0.909	0.903	0.979	0.866	Asthma
	0.932	0.000	0.995	0.932	0.962	0.961	0.997	0.956	Cancer
	0.917	0.021	0.888	0.917	0.902	0.884	0.976	0.851	Obesity
	0.998	0.003	0.976	0.998	0.987	0.986	0.998	0.976	Healthy
	0.956	0.010	0.971	0.956	0.963	0.951	0.986	0.953	Diabetes
	0.599	0.014	0.754	0.599	0.668	0.651	0.922	0.531	Arthritis
	0.925	0.057	0.864	0.925	0.893	0.851	0.950	0.832	Hypertension
Weighted Avg.	0.914	0.023	0.913	0.914	0.912	0.892	0.971	0.870	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
371	0	11	13	3	5	23		a = Asthma
1	218	0	0	4	0	11		b = Cancer
5	0	760	0	12	10	42		c = Obesity
1	0	0	622	0	0	0		d = Healthy
2	1	19	0	1303	2	36		e = Diabetes
3	0	36	1	1	221	107		f = Arthritis
7	0	30	1	19	55	1386		g = Hypertension

70 split :

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances	3631	90.639 %
Incorrectly Classified Instances	375	9.361 %
Kappa statistic	0.8831	
Mean absolute error	0.2058	
Root mean squared error	0.3039	
Relative absolute error	89.372 %	
Root relative squared error	89.6299 %	
Total Number of Instances	4006	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.850	0.005	0.935	0.850	0.890	0.882	0.976	0.847	Asthma
	0.936	0.000	0.994	0.936	0.964	0.963	0.997	0.957	Cancer
	0.911	0.024	0.874	0.911	0.892	0.872	0.974	0.836	Obesity
	0.998	0.004	0.970	0.998	0.984	0.982	0.998	0.970	Healthy
	0.949	0.007	0.978	0.949	0.963	0.951	0.985	0.958	Diabetes
	0.601	0.016	0.736	0.601	0.661	0.642	0.923	0.524	Arthritis
	0.916	0.061	0.853	0.916	0.884	0.837	0.946	0.819	Hypertension
Weighted Avg.	0.906	0.025	0.906	0.906	0.905	0.884	0.969	0.862	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
272	0	10	12	1	5	20		a = Asthma
1	160	0	0	2	0	8		b = Cancer
6	0	560	0	7	8	34		c = Obesity
1	0	0	456	0	0	0		d = Healthy
3	1	16	0	985	1	32		e = Diabetes
2	0	26	1	1	170	83		f = Arthritis
6	0	29	1	11	47	1028		g = Hypertension

80 split :

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances	2423	90.7151 %
Incorrectly Classified Instances	248	9.2849 %
Kappa statistic	0.8842	
Mean absolute error	0.2058	
Root mean squared error	0.3039	
Relative absolute error	89.3462 %	
Root relative squared error	89.5798 %	
Total Number of Instances	2671	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.869	0.004	0.951	0.869	0.908	0.901	0.977	0.867	Asthma
	0.946	0.000	1.000	0.946	0.972	0.972	0.997	0.968	Cancer
	0.902	0.024	0.871	0.902	0.886	0.865	0.973	0.831	Obesity
	0.997	0.004	0.969	0.997	0.983	0.980	0.998	0.968	Healthy
	0.952	0.009	0.973	0.952	0.962	0.950	0.984	0.953	Diabetes
	0.602	0.016	0.737	0.602	0.663	0.644	0.927	0.527	Arthritis
	0.913	0.060	0.857	0.913	0.884	0.837	0.947	0.823	Hypertension
Weighted Avg.	0.907	0.025	0.907	0.907	0.906	0.884	0.969	0.863	

=== Confusion Matrix ===

a	b	c	d	e	f	g	<-- classified as
193	0	6	9	2	3	9	a = Asthma
0	106	0	0	2	0	4	b = Cancer
3	0	370	0	5	6	26	c = Obesity
1	0	0	310	0	0	0	d = Healthy
0	0	9	0	641	2	21	e = Diabetes
0	0	18	1	0	112	55	f = Arthritis
6	0	22	0	9	29	691	g = Hypertension

Appendix C

Neural network:

Before fine tuning:

Weka:

Cross Validation: 5

The screenshot shows the Weka Explorer interface with the MultilayerPerceptron classifier selected. The 'Test options' section on the left indicates 'Cross-validation' with 'Folds' set to 5. The 'Classifier output' pane on the right displays the following results:

Time taken to build model: 24.1 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	12204	91.3815 %
Incorrectly Classified Instances	1151	8.6185 %
Kappa statistic	0.8929	
Mean absolute error	0.0317	
Root mean squared error	0.1382	
Relative absolute error	13.7467 %	
Root relative squared error	40.7277 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.921	0.006	0.930	0.921	0.925	0.919	0.987	0.955	Asthma
	0.959	0.002	0.964	0.959	0.961	0.960	0.999	0.987	Cancer
	0.904	0.022	0.882	0.904	0.893	0.874	0.989	0.947	Obesity
	0.992	0.002	0.983	0.992	0.988	0.986	1.000	0.997	Healthy
	0.955	0.011	0.968	0.955	0.962	0.949	0.993	0.987	Diabetes
	0.628	0.016	0.739	0.628	0.679	0.660	0.967	0.736	Arthritis
	0.909	0.049	0.878	0.909	0.893	0.851	0.983	0.958	Hypertension
Weighted Avg.	0.914	0.022	0.913	0.914	0.913	0.893	0.988	0.954	

=== Confusion Matrix ===

Cross Validation: 10

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 12**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☒ Cross-validation Folds **10**
☐ Percentage split % **66**
 More options...

(Norm) Medical Condition

Start Stop

Result list (right-click for options):
 00:58:13 - functions.MultilayerPerceptron
 01:02:53 - functions.MultilayerPerceptron
01:21:08 - functions.MultilayerPerceptron

==== Stratified cross-validation ====

==== Summary ====

Correctly Classified Instances	12185	91.2392 %
Incorrectly Classified Instances	1170	8.7608 %
Kappa statistic	0.8911	
Mean absolute error	0.0312	
Root mean squared error	0.1385	
Relative absolute error	13.5366 %	
Root relative squared error	40.8157 %	
Total Number of Instances	13355	

==== Detailed Accuracy By Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.896	0.004	0.948	0.896	0.922	0.915	0.987	0.956	Asthma
	0.967	0.001	0.972	0.967	0.970	0.968	0.999	0.987	Cancer
	0.899	0.019	0.895	0.899	0.897	0.879	0.989	0.946	Obesity
	0.991	0.002	0.983	0.991	0.987	0.985	1.000	0.996	Healthy
	0.953	0.008	0.975	0.953	0.964	0.951	0.992	0.985	Diabetes
	0.630	0.022	0.681	0.630	0.654	0.630	0.965	0.710	Arthritis
	0.915	0.053	0.872	0.915	0.893	0.850	0.982	0.955	Hypertension
Weighted Avg.	0.912	0.022	0.912	0.912	0.912	0.892	0.988	0.951	

==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
951	2	15	21	1	21	50	1	a = Asthma
0	590	0	0	17	0	3	1	b = Cancer

Status: OK Log

Cross Validation: 15

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 12**

Test options:
☐ Use training set
☐ Supplied test set (Set...)
☒ Cross-validation Folds **15**
☐ Percentage split % **66**
 More options...

(Norm) Medical Condition

Start Stop

Result list (right-click for options):
 00:58:13 - functions.MultilayerPerceptron
 01:02:53 - functions.MultilayerPerceptron
 01:21:08 - functions.MultilayerPerceptron
01:30:22 - functions.MultilayerPerceptron

==== Stratified cross-validation ====

==== Summary ====

Correctly Classified Instances	12190	91.2767 %
Incorrectly Classified Instances	1165	8.7233 %
Kappa statistic	0.8916	
Mean absolute error	0.0317	
Root mean squared error	0.1382	
Relative absolute error	13.7655 %	
Root relative squared error	40.7258 %	
Total Number of Instances	13355	

==== Detailed Accuracy By Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.905	0.005	0.935	0.905	0.920	0.913	0.986	0.946	Asthma
	0.957	0.001	0.975	0.957	0.966	0.965	0.999	0.987	Cancer
	0.901	0.020	0.889	0.901	0.895	0.876	0.989	0.946	Obesity
	0.993	0.002	0.985	0.993	0.989	0.987	1.000	0.995	Healthy
	0.952	0.010	0.970	0.952	0.961	0.948	0.992	0.985	Diabetes
	0.654	0.020	0.707	0.654	0.679	0.657	0.970	0.735	Arthritis
	0.908	0.050	0.876	0.908	0.892	0.849	0.981	0.953	Hypertension
Weighted Avg.	0.913	0.022	0.912	0.913	0.912	0.892	0.988	0.951	

==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
960	0	19	14	0	20	48	1	a = Asthma
0	594	0	0	21	0	4	1	b = Cancer

Status: OK Log

Split Validation: 60%

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 12**

Test options

☐ Use training set

☐ Supplied test set

☐ Cross-validation Folds 15

☒ Percentage split % 60

(Nom) Medical Condition

Result list (right-click for options)

00:58:13 - functions.MultilayerPerceptron

01:02:53 - functions.MultilayerPerceptron

01:21:08 - functions.MultilayerPerceptron

01:30:22 - functions.MultilayerPerceptron

01:55:18 - functions.MultilayerPerceptron

Classifier output

Time taken to test model on test split: 0.02 seconds

==== Summary ====

Correctly Classified Instances	4862	91.0146 %
Incorrectly Classified Instances	480	8.9854 %
Kappa statistic	0.8979	
Mean absolute error	0.0312	
Root mean squared error	0.1433	
Relative absolute error	13.5463 %	
Root relative squared error	42.2479 %	
Total Number of Instances	5342	

==== Detailed Accuracy By Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.887	0.006	0.931	0.887	0.909	0.901	0.989	0.936	Asthma
	0.966	0.001	0.987	0.966	0.976	0.975	0.999	0.990	Cancer
	0.887	0.015	0.918	0.887	0.902	0.884	0.987	0.949	Obesity
	0.989	0.002	0.986	0.989	0.987	0.985	1.000	0.999	Healthy
	0.954	0.012	0.964	0.954	0.959	0.945	0.993	0.984	Diabetes
	0.610	0.018	0.712	0.610	0.657	0.636	0.964	0.700	Arthritis
	0.923	0.061	0.855	0.923	0.887	0.842	0.977	0.946	Hypertension
Weighted Avg.	0.910	0.024	0.910	0.910	0.909	0.888	0.986	0.946	

==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
a	378	0	3	7	2	7	29	a = Asthma
b	0	168	0	3	0	0	0	b = Cancer
c	9	0	565	0	4	13	24	c = Obesity

Status OK



Split Validation: 70%

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 12**

Test options

☐ Use training set

☐ Supplied test set

☐ Cross-validation Folds 15

☒ Percentage split % 70

(Nom) Medical Condition

Result list (right-click for options)

00:58:13 - functions.MultilayerPerceptron

01:02:53 - functions.MultilayerPerceptron

01:21:08 - functions.MultilayerPerceptron

01:30:22 - functions.MultilayerPerceptron

01:55:18 - functions.MultilayerPerceptron

01:58:26 - functions.MultilayerPerceptron

Classifier output

==== Summary ====

Correctly Classified Instances	3628	90.5642 %
Incorrectly Classified Instances	378	9.4358 %
Kappa statistic	0.8829	
Mean absolute error	0.0331	
Root mean squared error	0.1451	
Relative absolute error	14.3655 %	
Root relative squared error	42.8065 %	
Total Number of Instances	4006	

==== Detailed Accuracy By Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.900	0.010	0.889	0.900	0.894	0.885	0.990	0.952	Asthma
	0.982	0.002	0.960	0.982	0.971	0.970	0.999	0.988	Cancer
	0.919	0.023	0.877	0.919	0.898	0.879	0.990	0.957	Obesity
	0.982	0.001	0.989	0.982	0.986	0.984	1.000	0.998	Healthy
	0.948	0.009	0.973	0.948	0.960	0.947	0.990	0.984	Diabetes
	0.671	0.026	0.660	0.671	0.665	0.640	0.964	0.693	Arthritis
	0.877	0.044	0.886	0.877	0.882	0.836	0.979	0.948	Hypertension
Weighted Avg.	0.906	0.021	0.907	0.906	0.906	0.884	0.987	0.949	

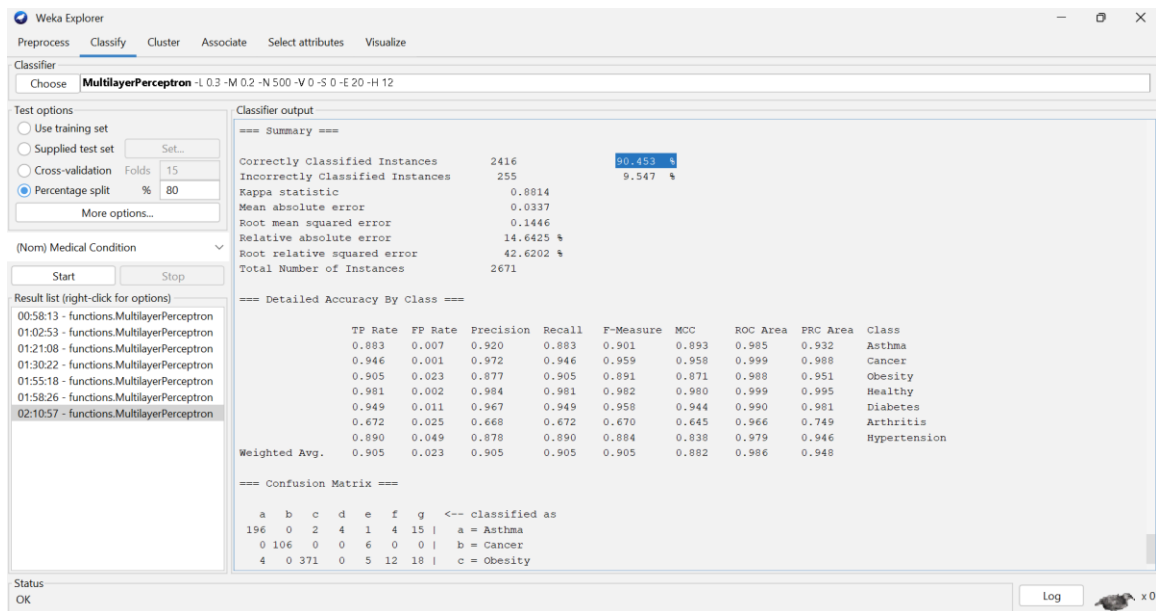
==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
a	288	1	4	4	1	6	16	a = Asthma
b	0	168	0	0	3	0	0	b = Cancer
c	9	0	565	0	4	13	24	c = Obesity

Status OK

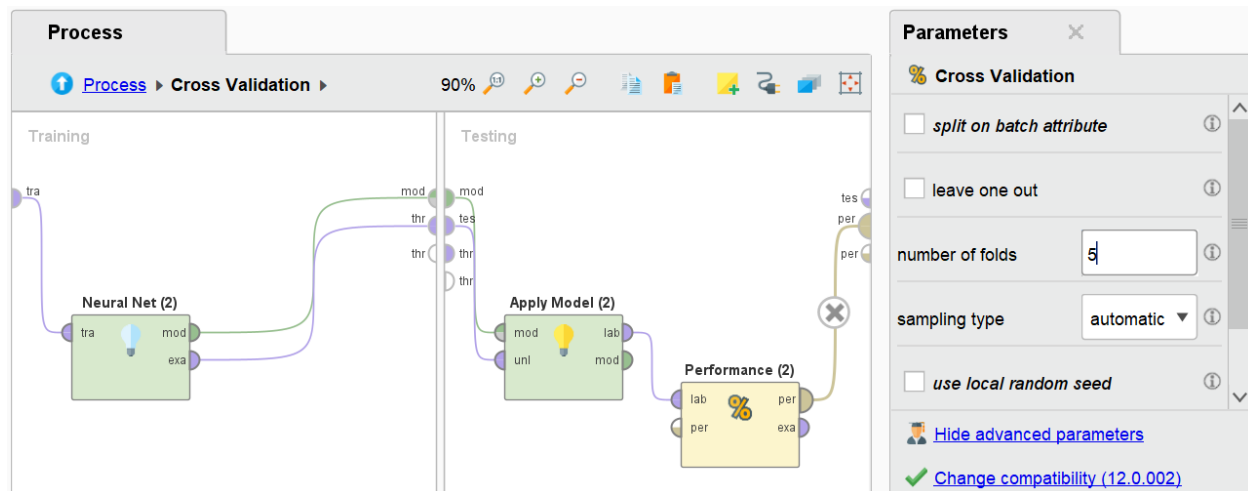


Split Validation: 80%



RapidMiner:

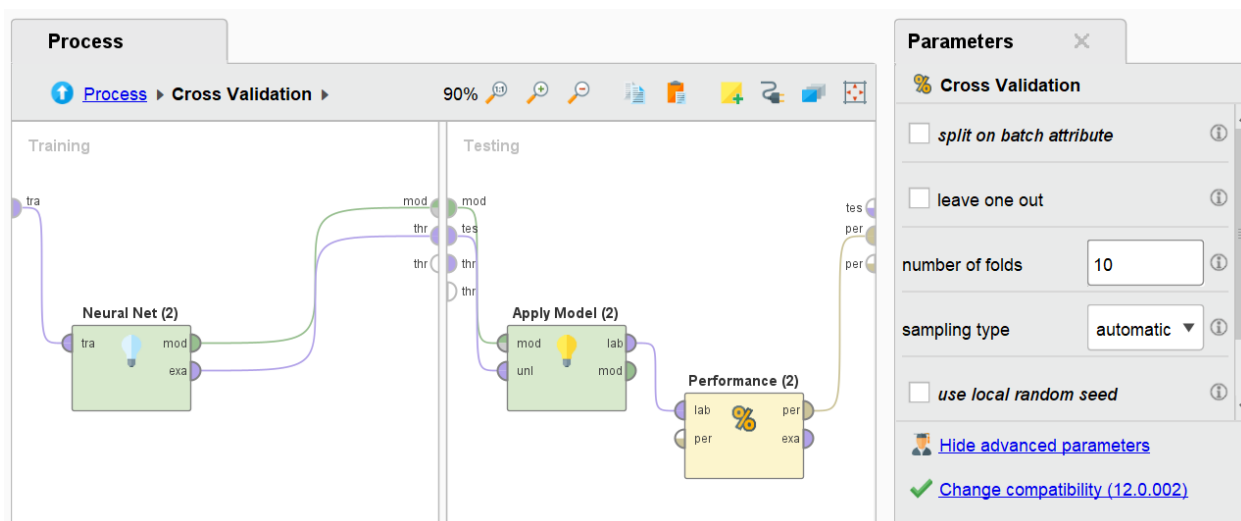
Cross Validation: 5



accuracy: 91.16% +/- 0.94% (micro average: 91.16%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	959	2	14	21	1	10	34	92.12%
pred. Cancer	2	581	0	0	19	0	0	96.51%
pred. Obesity	19	0	1797	0	49	63	61	90.35%
pred. Healthy	11	0	0	1572	0	1	2	99.12%
pred. Diab...	0	22	29	0	3233	4	59	96.59%
pred. Arthritis	21	0	67	6	8	595	155	69.84%
pred. Hype...	49	5	110	3	86	248	3437	87.28%
class recall	90.39%	95.25%	89.09%	98.13%	95.20%	64.60%	91.70%	

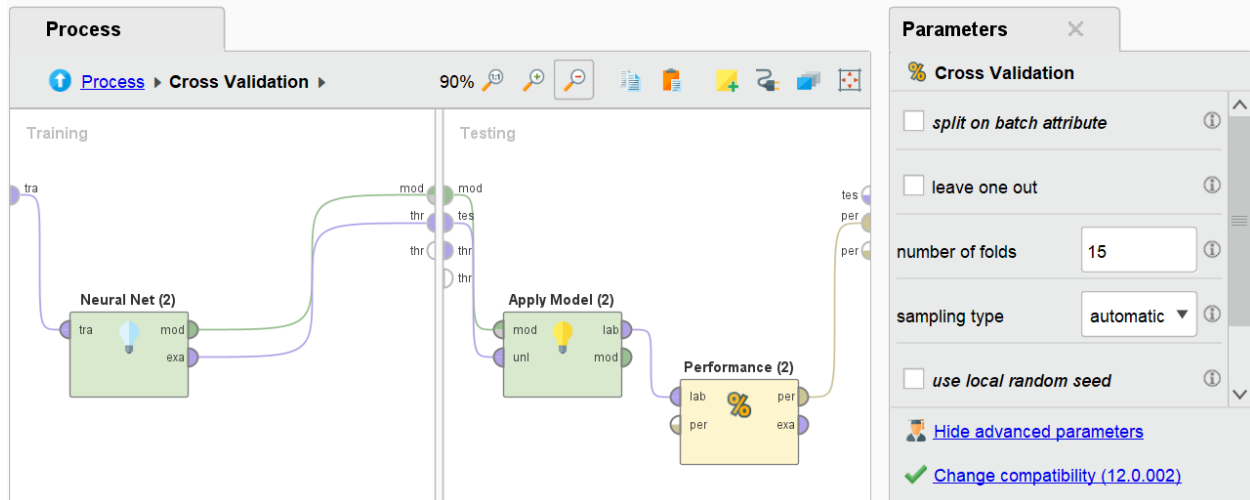
Cross Validation: 10



accuracy: 91.23% +/- 1.15% (micro average: 91.23%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	951	1	20	7	3	12	35	92.42%
pred. Cancer	0	576	0	0	19	0	0	96.81%
pred. Obesity	20	0	1801	0	53	61	78	89.47%
pred. Healthy	22	0	1	1586	0	2	2	98.33%
pred. Diab...	1	23	26	0	3240	1	51	96.95%
pred. Arthritis	15	1	76	8	12	636	188	67.95%
pred. Hype...	52	9	93	1	69	209	3394	88.69%
class recall	89.63%	94.43%	89.29%	99.00%	95.41%	69.06%	90.55%	

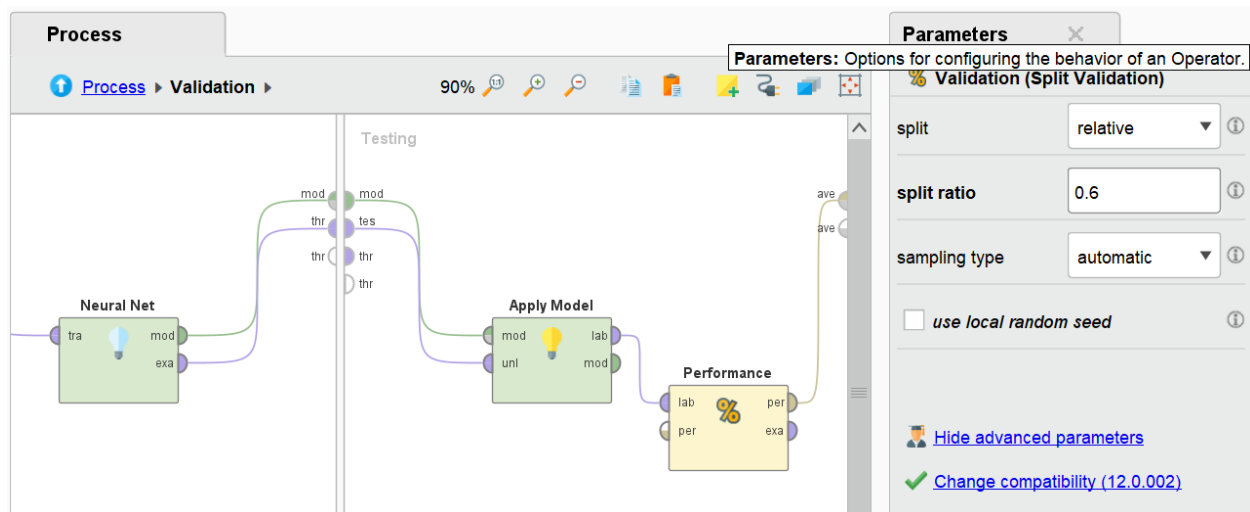
Cross Validation: 15



accuracy: 91.00% +/- 1.08% (micro average: 91.00%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	961	0	17	11	1	12	31	93.03%
pred. Cancer	1	579	0	0	19	0	2	96.34%
pred. Obesity	18	0	1801	0	54	81	91	88.07%
pred. Healthy	18	0	1	1585	0	4	4	98.33%
pred. Diab...	1	24	28	0	3231	3	53	96.74%
pred. Arthritis	17	0	57	5	7	609	180	69.60%
pred. Hype...	45	7	113	1	84	212	3387	88.00%
class recall	90.57%	94.92%	89.29%	98.94%	95.14%	66.12%	90.37%	

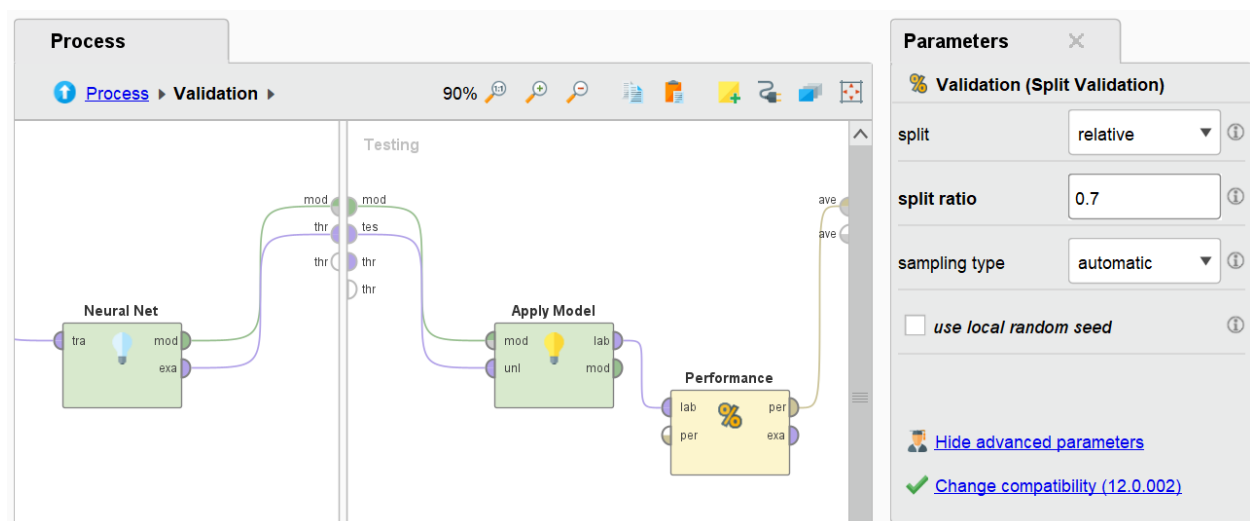
Split Validation: 60%



accuracy: 90.71%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	384	1	5	6	2	5	24	89.93%
pred. Cancer	1	224	0	0	6	0	2	96.14%
pred. Obesity	7	1	725	0	23	27	36	88.52%
pred. Healthy	7	0	1	629	0	0	0	98.74%
pred. Diab...	1	10	24	0	1289	5	26	95.13%
pred. Arthritis	11	0	21	3	8	247	64	69.77%
pred. Hype...	13	8	31	3	30	84	1347	88.85%
class recall	90.57%	91.80%	89.84%	98.13%	94.92%	67.12%	89.86%	

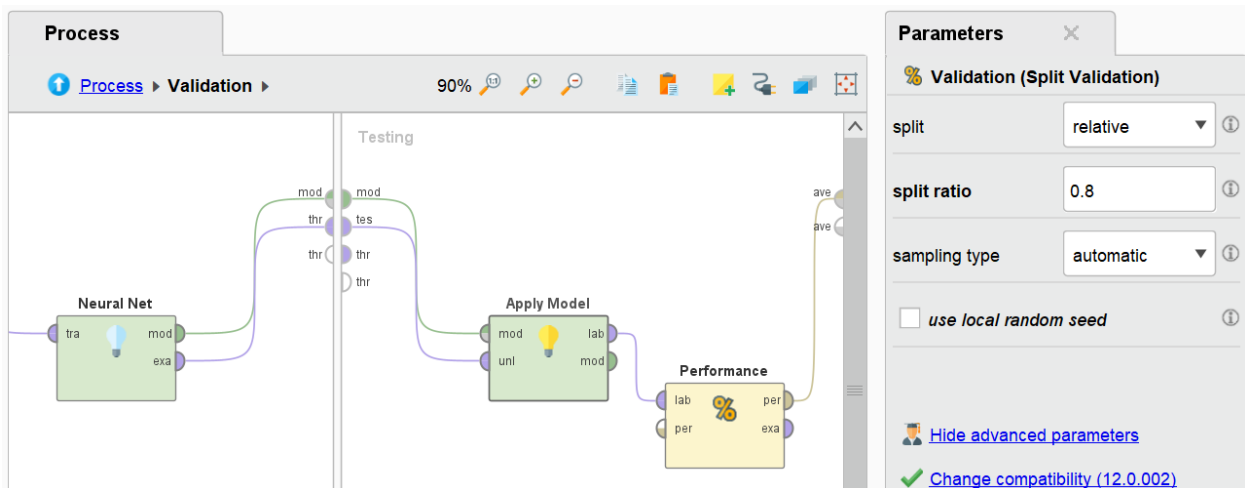
Split Validation: 70%



accuracy: 91.01%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precl...
pred. Asthma	289	0	9	5	1	4	7	91.75%
pred. Cancer	0	177	0	0	6	0	0	96.72%
pred. Obesity	3	0	519	0	8	15	22	91.53%
pred. Healthy	4	0	1	470	0	0	0	98.95%
pred. Diab...	0	6	20	0	971	3	17	95.48%
pred. Arthritis	10	0	26	5	6	187	45	67.03%
pred. Hype...	12	0	30	1	27	67	1033	88.29%
class recall	90.88%	96.72%	85.79%	97.71%	95.29%	67.75%	91.90%	

Split Validation: 80%



accuracy: 90.56%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	170	0	3	1	0	2	1	96.05%
pred. Cancer	1	118	0	0	6	0	0	94.40%
pred. Obesity	5	0	357	0	10	14	9	90.38%
pred. Healthy	8	0	0	317	0	0	2	96.94%
pred. Diab...	0	3	6	0	642	1	10	96.98%
pred. Arthritis	7	0	19	1	3	125	39	64.43%
pred. Hype...	21	1	18	1	18	42	689	87.22%
class recall	80.19%	96.72%	88.59%	99.06%	94.55%	67.93%	91.87%	

After fine tuning:

Weka:

Cross Validation: 5

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron** -L 0.01 -M 0.9 -N 1200 -V 0 -S 0 -E 20 -H 12

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **5**
☐ Percentage split %: 80
 More options...

(Nom) Medical Condition
 Start Stop

Result list (right-click for options):
 00:58:13 - functions.MultilayerPerceptron
 01:02:53 - functions.MultilayerPerceptron
 01:21:08 - functions.MultilayerPerceptron
 01:30:22 - functions.MultilayerPerceptron
 01:55:18 - functions.MultilayerPerceptron
 01:58:26 - functions.MultilayerPerceptron
 02:10:57 - functions.MultilayerPerceptron
02:16:18 - functions.MultilayerPerceptron

Classifier output

==== Summary ====

Correctly Classified Instances	12248	91.711 %
Incorrectly Classified Instances	1107	8.289 %
Kappa statistic	0.8969	
Mean absolute error	0.0318	
Root mean squared error	0.1347	
Relative absolute error	13.809 %	
Root relative squared error	39.6821 %	
Total Number of Instances	13355	

==== Detailed Accuracy By Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.907	0.006	0.931	0.907	0.919	0.912	0.986	0.952	Asthma
	0.959	0.001	0.972	0.959	0.965	0.964	0.999	0.990	Cancer
	0.905	0.019	0.896	0.905	0.900	0.883	0.991	0.957	Obesity
	0.991	0.002	0.982	0.991	0.986	0.984	1.000	0.996	Healthy
	0.958	0.010	0.971	0.958	0.964	0.952	0.994	0.989	Diabetes
	0.646	0.018	0.726	0.646	0.684	0.663	0.968	0.731	Arthritis
	0.918	0.048	0.883	0.918	0.900	0.861	0.983	0.957	Hypertension
Weighted Avg.	0.917	0.021	0.916	0.917	0.916	0.897	0.989	0.955	

==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
962	0	13	23	1	14	48	1	a = Asthma
0	585	0	0	21	0	4	1	b = Cancer
21	0	1825	0	27	50	94	1	c = Obesity

Status: OK Log x0

Cross Validation: 10

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron -L 0.01 -M 0.9 -N 1200 -V 0 -S 0 -E 20 -H 12**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds **10**
- ☐ Percentage split % **80**

(Norm) Medical Condition

Start Stop

Result list (right-click for options):

- 00:58:13 - functions.MultilayerPerceptron
- 01:02:53 - functions.MultilayerPerceptron
- 01:21:08 - functions.MultilayerPerceptron
- 01:30:22 - functions.MultilayerPerceptron
- 01:55:18 - functions.MultilayerPerceptron
- 01:58:26 - functions.MultilayerPerceptron
- 02:10:57 - functions.MultilayerPerceptron
- 02:16:18 - functions.MultilayerPerceptron
- 02:38:17 - functions.MultilayerPerceptron**

Classifier output:

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	12202	91.3665 %
Incorrectly Classified Instances	1153	8.6335 %
Kappa statistic	0.8927	
Mean absolute error	0.0326	
Root mean squared error	0.1347	
Relative absolute error	14.1429 %	
Root relative squared error	39.6876 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.907	0.005	0.945	0.907	0.925	0.919	0.999	0.956	Asthma
	0.951	0.001	0.970	0.951	0.960	0.958	0.999	0.989	Cancer
	0.900	0.020	0.888	0.900	0.894	0.875	0.990	0.961	Obesity
	0.991	0.002	0.986	0.991	0.988	0.987	1.000	0.999	Healthy
	0.956	0.010	0.970	0.956	0.962	0.950	0.994	0.989	Diabetes
	0.648	0.021	0.695	0.648	0.671	0.648	0.973	0.756	Arthritis
	0.911	0.048	0.881	0.911	0.896	0.854	0.984	0.960	Hypertension
Weighted Avg.	0.914	0.021	0.913	0.914	0.913	0.893	0.990	0.959	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
962	0	18	16	1	17	47	1	a = Asthma
0	580	0	0	28	0	2	1	b = Cancer

Status: OK

Log

Cross Validation: 15

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron -L 0.01 -M 0.9 -N 1200 -V 0 -S 0 -E 20 -H 12**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation Folds **15**
- ☐ Percentage split % **80**

(Norm) Medical Condition

Start Stop

Result list (right-click for options):

- 00:58:13 - functions.MultilayerPerceptron
- 01:02:53 - functions.MultilayerPerceptron
- 01:21:08 - functions.MultilayerPerceptron
- 01:30:22 - functions.MultilayerPerceptron
- 01:55:18 - functions.MultilayerPerceptron
- 01:58:26 - functions.MultilayerPerceptron
- 02:10:57 - functions.MultilayerPerceptron
- 02:16:18 - functions.MultilayerPerceptron
- 02:38:17 - functions.MultilayerPerceptron
- 02:56:45 - functions.MultilayerPerceptron**

Classifier output:

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	12273	91.8982 %
Incorrectly Classified Instances	1082	8.1018 %
Kappa statistic	0.8993	
Mean absolute error	0.0314	
Root mean squared error	0.1322	
Relative absolute error	13.6278 %	
Root relative squared error	38.9504 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.906	0.005	0.939	0.906	0.922	0.916	0.993	0.967	Asthma
	0.951	0.001	0.970	0.951	0.960	0.958	0.999	0.987	Cancer
	0.915	0.019	0.894	0.915	0.905	0.888	0.991	0.959	Obesity
	0.995	0.002	0.989	0.995	0.992	0.991	1.000	0.997	Healthy
	0.956	0.009	0.972	0.956	0.964	0.952	0.994	0.989	Diabetes
	0.661	0.020	0.710	0.661	0.685	0.663	0.973	0.755	Arthritis
	0.916	0.044	0.890	0.916	0.903	0.865	0.986	0.965	Hypertension
Weighted Avg.	0.919	0.020	0.919	0.919	0.919	0.900	0.991	0.961	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
961	0	19	15	0	19	47	1	a = Asthma
0	580	0	0	28	0	2	1	b = Cancer

Status: OK

Log

Split Validation: 60%

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron -L 0.01 -M 0.9 -N 1200 -V 0 -S 0 -E 20 -H 12**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds: 10
- ☒ Percentage split %: 60

(Nom) Medical Condition

Result list (right-click for options):

- 22:41:01 - functions.MultilayerPerceptron
- 22:44:33 - functions.MultilayerPerceptron
- 22:47:53 - functions.MultilayerPerceptron
- 22:52:49 - functions.MultilayerPerceptron**

Classifier output:

=== Summary ===

Correctly Classified Instances	4836	90.5279 %
Incorrectly Classified Instances	506	9.4721 %
Kappa statistic	0.8823	
Mean absolute error	0.0331	
Root mean squared error	0.1444	
Relative absolute error	14.3771 %	
Root relative squared error	42.5755 %	
Total Number of Instances	5342	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.894	0.007	0.923	0.894	0.908	0.901	0.991	0.954	Asthma
	0.944	0.002	0.961	0.944	0.953	0.950	0.999	0.989	Cancer
	0.879	0.019	0.893	0.879	0.886	0.866	0.987	0.953	Obesity
	0.994	0.001	0.990	0.994	0.992	0.991	1.000	0.999	Healthy
	0.958	0.015	0.957	0.958	0.957	0.943	0.993	0.985	Diabetes
	0.667	0.026	0.660	0.667	0.663	0.638	0.965	0.706	Arthritis
	0.891	0.048	0.878	0.891	0.884	0.838	0.978	0.948	Hypertension
Weighted Avg.	0.905	0.023	0.906	0.905	0.905	0.883	0.987	0.950	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
381	0	6	5	1	7	26	1	a = Asthma
0	221	0	0	13	0	0	1	b = Cancer

Status: OK

 x0

Split Validation: 70%

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **MultilayerPerceptron -L 0.01 -M 0.9 -N 1200 -V 0 -S 0 -E 20 -H 12**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds: 15
- ☒ Percentage split %: 70

(Nom) Medical Condition

Result list (right-click for options):

- 00:58:13 - functions.MultilayerPerceptron
- 01:02:53 - functions.MultilayerPerceptron
- 01:21:08 - functions.MultilayerPerceptron
- 01:30:22 - functions.MultilayerPerceptron
- 01:55:18 - functions.MultilayerPerceptron
- 01:58:26 - functions.MultilayerPerceptron
- 02:10:57 - functions.MultilayerPerceptron
- 02:16:18 - functions.MultilayerPerceptron
- 02:38:17 - functions.MultilayerPerceptron
- 02:56:45 - functions.MultilayerPerceptron
- 03:30:44 - functions.MultilayerPerceptron
- 03:34:26 - functions.MultilayerPerceptron**

Classifier output:

=== Summary ===

Correctly classified Instances	3647	91.0384 %
Incorrectly classified Instances	359	8.9616 %
Kappa statistic	0.8887	
Mean absolute error	0.0331	
Root mean squared error	0.1415	
Relative absolute error	14.3936 %	
Root relative squared error	41.7224 %	
Total Number of Instances	4006	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.894	0.006	0.929	0.894	0.911	0.903	0.989	0.951	Asthma
	0.977	0.002	0.954	0.977	0.965	0.964	0.999	0.982	Cancer
	0.885	0.017	0.907	0.885	0.895	0.877	0.988	0.951	Obesity
	0.996	0.003	0.974	0.996	0.985	0.983	1.000	0.999	Healthy
	0.948	0.008	0.975	0.948	0.961	0.948	0.991	0.984	Diabetes
	0.714	0.029	0.654	0.714	0.682	0.658	0.969	0.762	Arthritis
	0.899	0.045	0.887	0.899	0.893	0.851	0.981	0.954	Hypertension
Weighted Avg.	0.910	0.020	0.912	0.910	0.911	0.891	0.987	0.954	

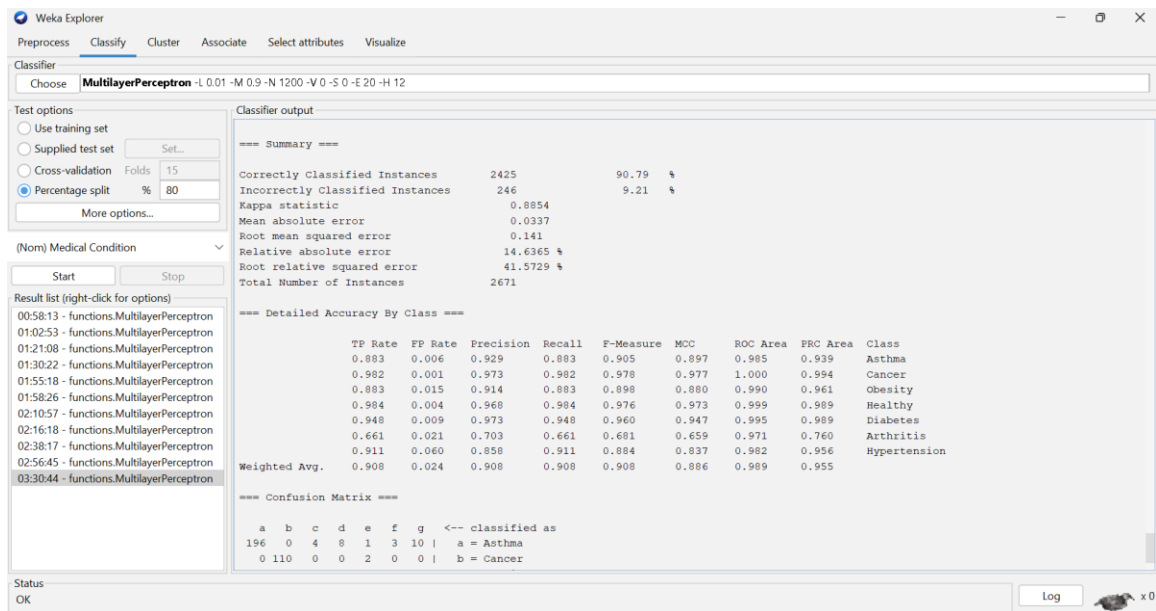
=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
286	0	4	8	0	11	11	1	a = Asthma
0	167	0	0	4	0	0	1	b = Cancer

Status: OK

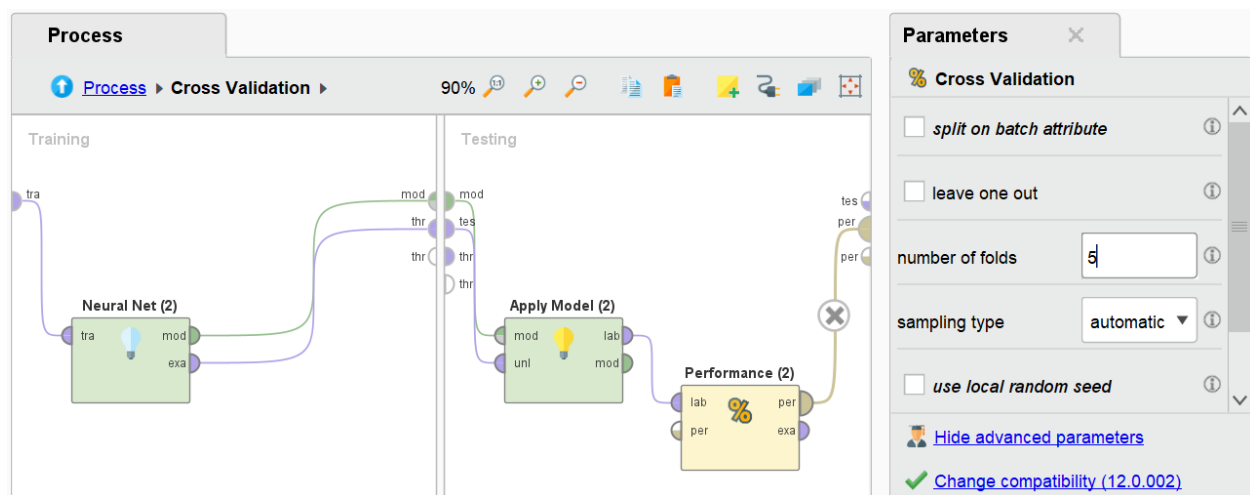
 x0

Split Validation: 80%



RapidMiner:

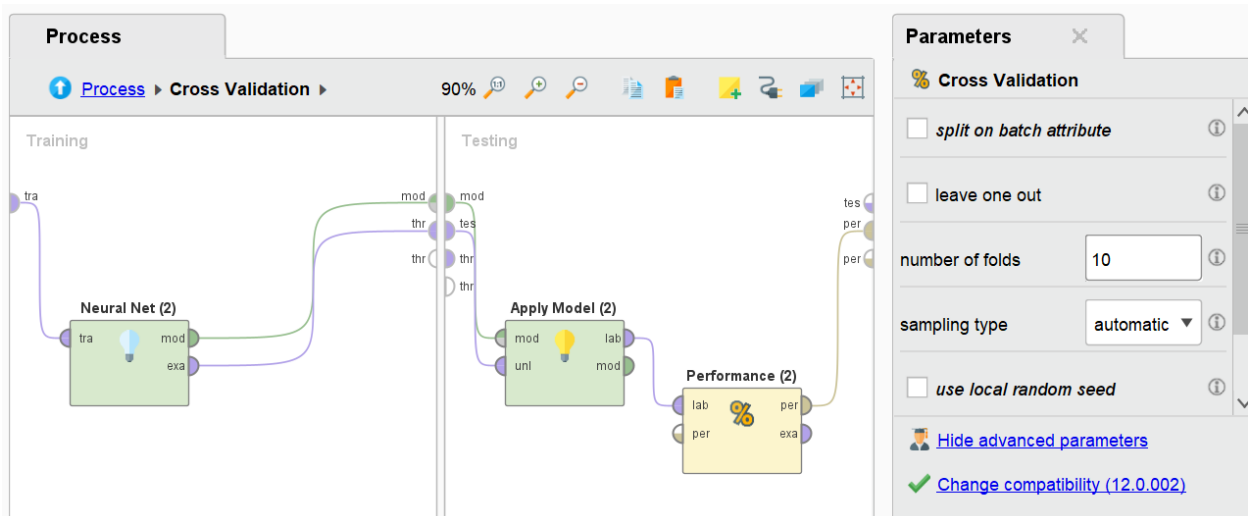
Cross Validation: 5



accuracy: 91.63% +/- 0.35% (micro average: 91.63%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	966	0	18	10	2	11	30	93.15%
pred. Cancer	0	584	0	0	18	0	2	96.69%
pred. Obesity	21	0	1817	0	44	64	76	89.86%
pred. Healthy	15	0	0	1588	0	2	2	98.82%
pred. Diab...	1	21	32	0	3239	4	46	96.89%
pred. Arthritis	24	0	47	4	11	617	166	71.00%
pred. Hype...	34	5	103	0	82	223	3426	88.46%
class recall	91.05%	95.74%	90.08%	99.13%	95.38%	66.99%	91.41%	

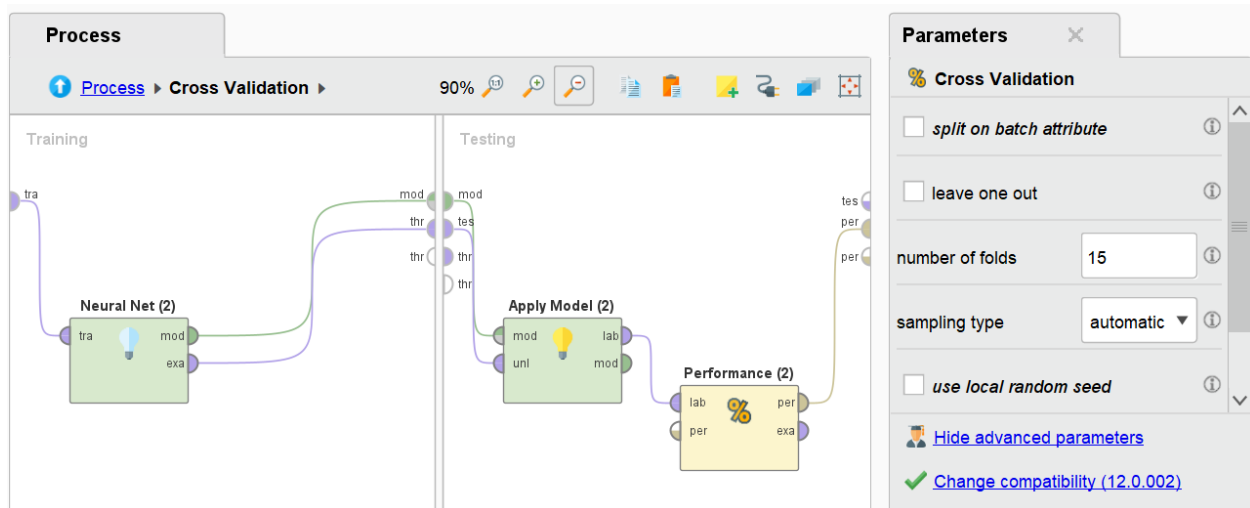
Cross Validation: 10



accuracy: 91.76% +/- 0.89% (micro average: 91.76%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	961	0	14	7	3	10	32	93.57%
pred. Cancer	0	582	0	0	19	0	0	96.84%
pred. Obesity	16	0	1829	0	45	68	71	90.14%
pred. Healthy	15	0	1	1591	0	4	3	98.57%
pred. Diab...	2	24	21	0	3246	3	53	96.92%
pred. Arthritis	18	0	63	3	9	638	181	69.96%
pred. Hype...	49	4	89	1	74	198	3408	89.14%
class recall	90.57%	95.41%	90.68%	99.31%	95.58%	69.27%	90.93%	

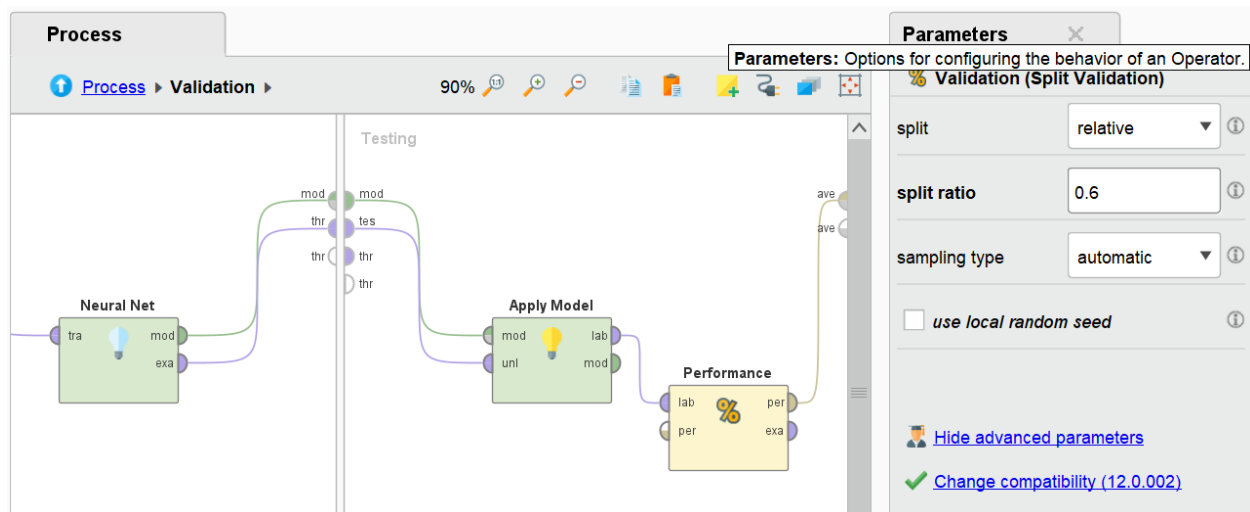
Cross Validation: 15



accuracy: 91.61% +/- 0.94% (micro average: 91.61%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	967	1	13	11	2	13	23	93.88%
pred. Cancer	0	581	0	0	25	0	0	95.87%
pred. Obesity	15	0	1834	0	38	78	96	88.99%
pred. Healthy	17	0	0	1584	0	2	3	98.63%
pred. Diab...	0	22	21	0	3243	4	57	96.89%
pred. Arthritis	22	0	53	5	12	606	149	71.55%
pred. Hype...	40	6	96	2	76	218	3420	88.65%
class recall	91.14%	95.25%	90.93%	98.88%	95.49%	65.80%	91.25%	

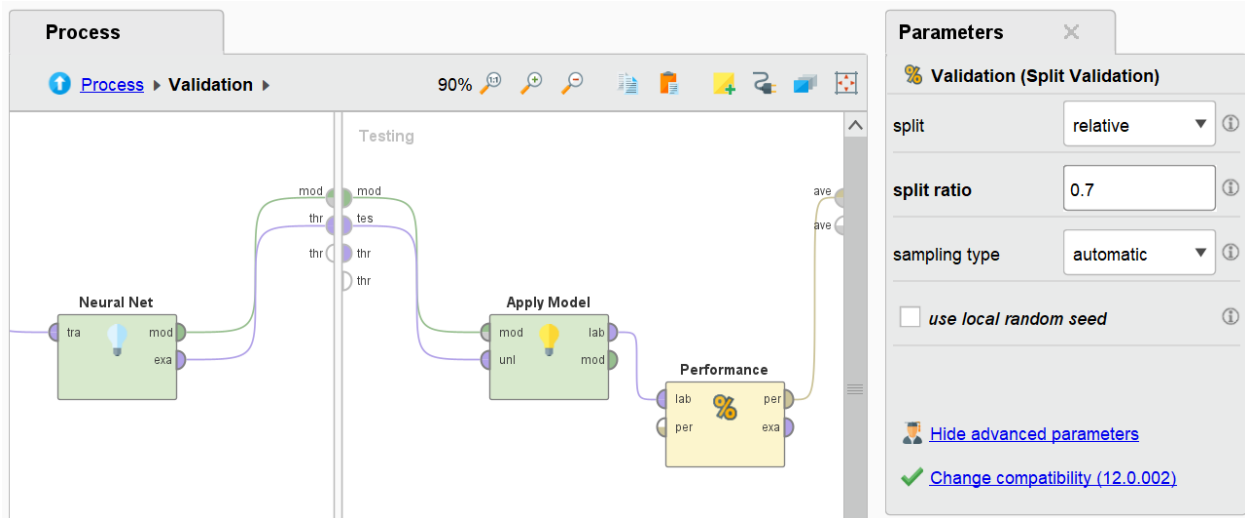
Split Validation: 60%



accuracy: 91.54%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	395	0	11	6	0	2	11	92.94%
pred. Cancer	1	227	0	0	4	0	1	97.42%
pred. Obesity	4	0	700	0	17	26	27	90.44%
pred. Healthy	5	0	0	632	0	1	0	99.06%
pred. Diab...	0	5	23	0	1298	1	12	96.94%
pred. Arthritis	6	0	32	3	6	266	77	68.21%
pred. Hype...	13	12	41	0	33	72	1371	88.91%
class recall	93.16%	93.03%	86.74%	98.60%	95.58%	72.28%	91.46%	

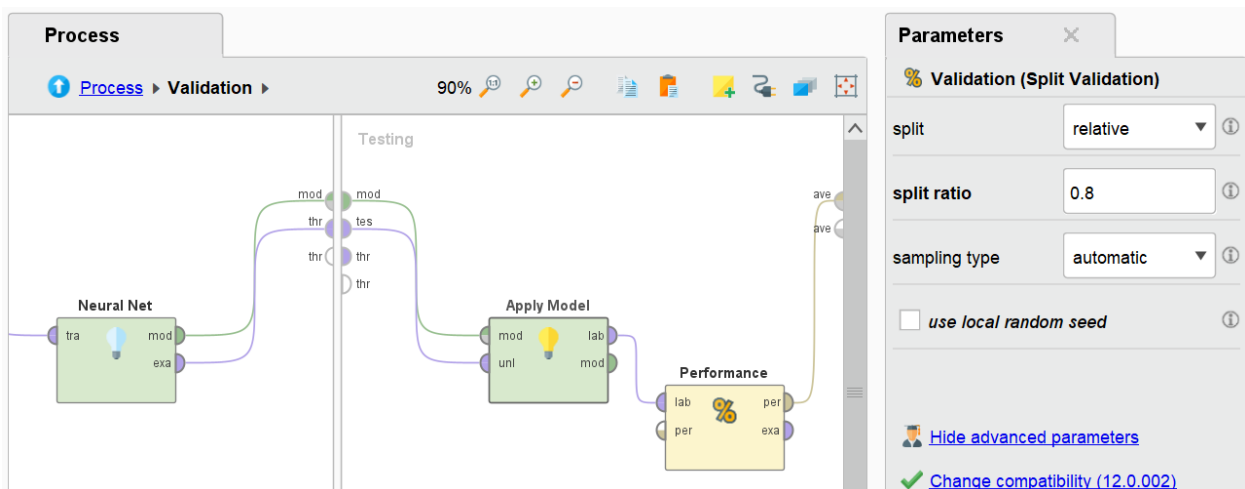
Split Validation: 70%



accuracy: 91.24%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	288	0	7	3	0	3	8	93.20%
pred. Cancer	0	176	0	0	5	0	0	97.24%
pred. Obesity	4	0	528	0	14	18	16	91.03%
pred. Healthy	4	0	0	474	0	0	1	98.96%
pred. Diab...	0	5	12	0	970	1	14	96.81%
pred. Arthritis	7	0	28	3	5	193	59	65.42%
pred. Hype...	15	2	30	1	25	61	1026	88.45%
class recall	90.57%	96.17%	87.27%	98.54%	95.19%	69.93%	91.28%	

Split Validation: 80%



accuracy: 91.24%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class preci...
pred. Asthma	184	0	3	1	0	2	1	96.34%
pred. Cancer	0	117	0	0	1	0	0	99.15%
pred. Obesity	4	0	356	0	9	16	14	89.22%
pred. Healthy	5	0	0	317	0	0	0	98.45%
pred. Diab...	0	5	7	0	650	1	13	96.15%
pred. Arthritis	7	0	11	1	2	125	35	69.06%
pred. Hype...	12	0	26	1	17	40	687	87.74%
class recall	86.79%	95.90%	88.34%	99.06%	95.73%	67.93%	91.60%	

Appendix D

Before fine tuning:

Weka:

Cross Validation: 5

The screenshot shows the Weka Explorer interface with the NaiveBayes classifier selected. The 'Test options' section is set to 'Cross-validation' with 'Folds' set to 5. The 'Classifier output' section displays the following statistics:

Statistic	Value
Correctly Classified Instances	12465
Incorrectly Classified Instances	890
Kappa statistic	0.9172
Mean absolute error	0.0281
Root mean squared error	0.1168
Relative absolute error	12.197 %
Root relative squared error	34.4202 %
Total Number of Instances	13355

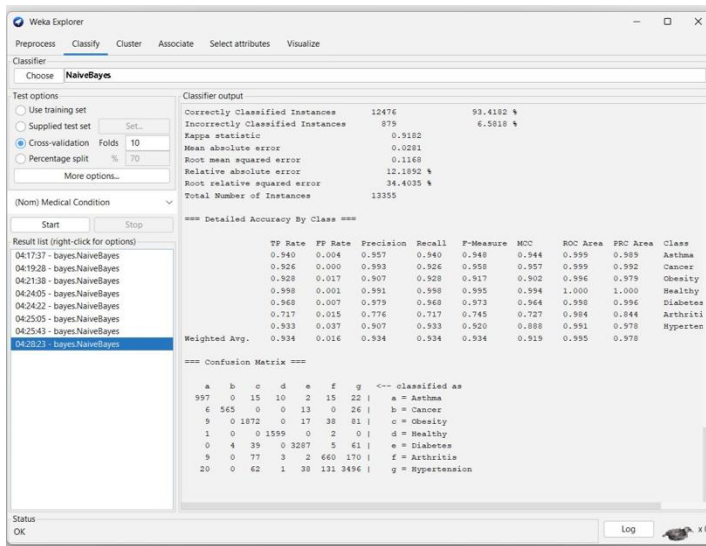
The 'Detailed Accuracy By Class' section shows the following metrics for each class:

Class	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area
Asthma	0.940	0.004	0.957	0.940	0.948	0.944	0.999	0.989
Cancer	0.926	0.000	0.993	0.926	0.950	0.957	0.999	0.992
Obesity	0.927	0.017	0.905	0.927	0.916	0.901	0.996	0.979
Healthy	0.998	0.001	0.991	0.998	0.995	0.994	1.000	1.000
Diabetes	0.968	0.007	0.979	0.968	0.973	0.964	0.998	0.996
Arthritis	0.711	0.015	0.773	0.711	0.741	0.723	0.984	0.844
Hypertension	0.932	0.038	0.905	0.932	0.918	0.886	0.991	0.978
Weighted Avg.	0.933	0.017	0.933	0.933	0.933	0.918	0.995	0.978

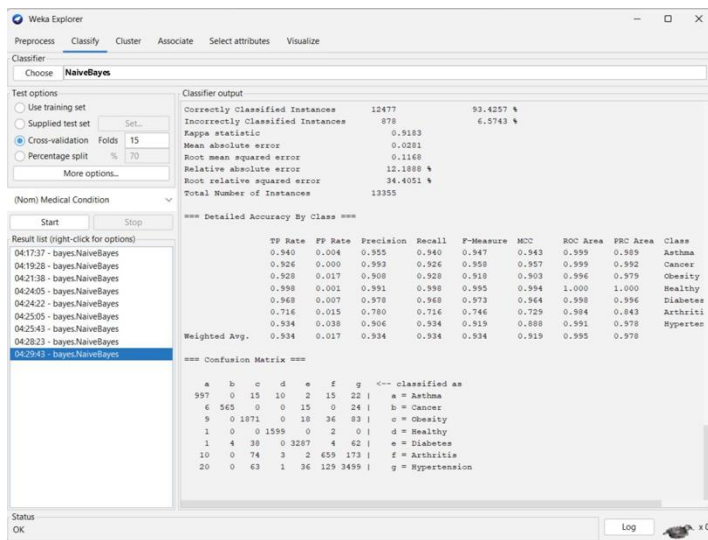
The 'Confusion Matrix' section shows the following data:

a \ b	a = Asthma	b = Cancer	c = Obesity	d = Healthy	e = Diabetes	f = Arthritis	g = Hypertension
a	184	0	3	1	0	2	1
b	0	117	0	0	1	0	0
c	4	0	356	0	9	16	14
d	5	0	0	317	0	0	0
e	0	5	7	0	650	1	13
f	7	0	11	1	2	125	35
g	12	0	26	1	17	40	687

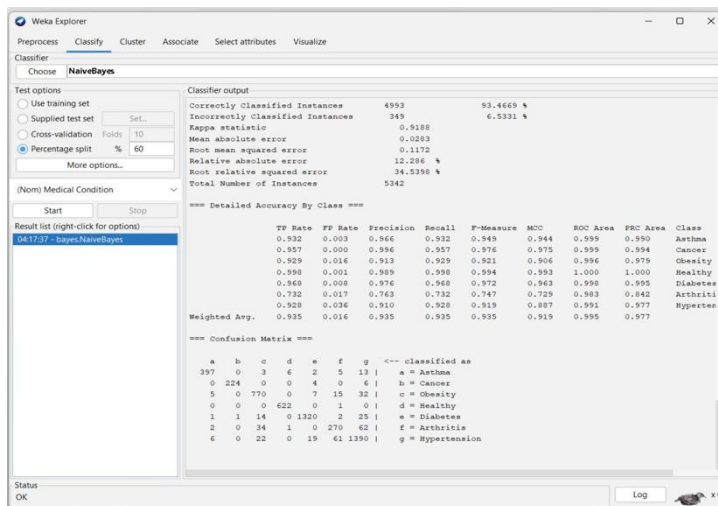
Cross Validation: 10



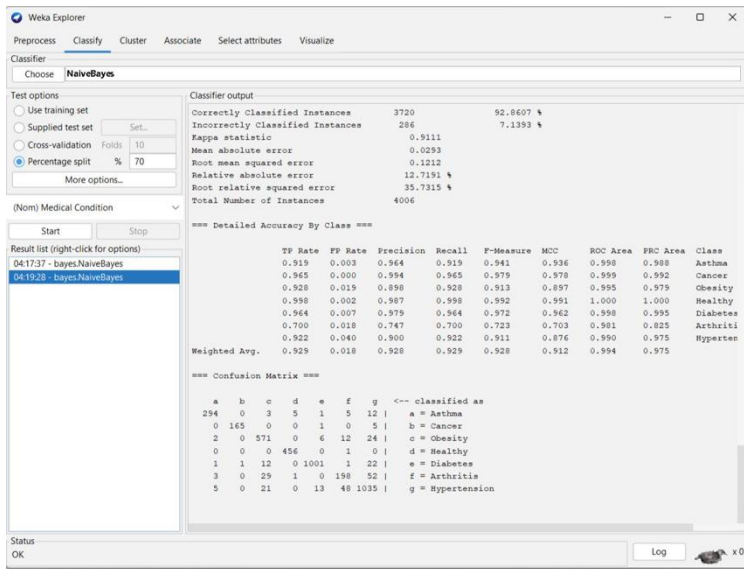
Cross Validation: 15



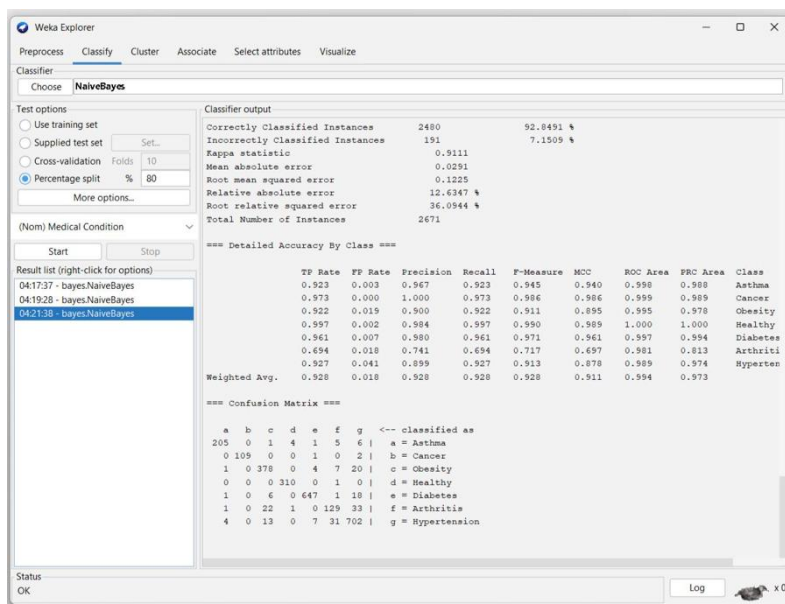
Split Validation: 60%



Split Validation: 70%



Split Validation: 80%



RapidMiner:

Cross Validation: 5

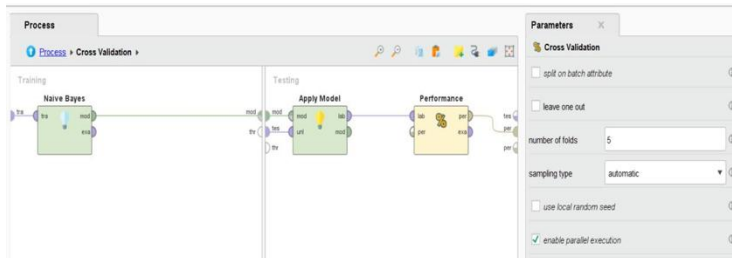
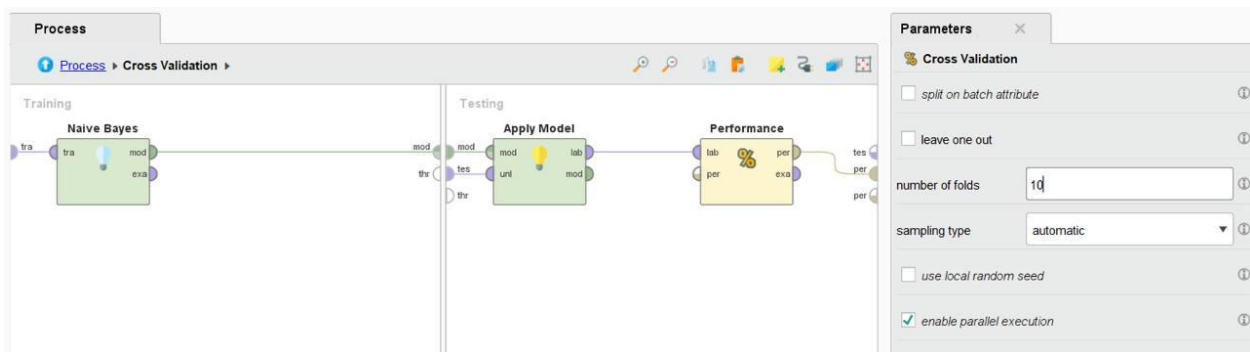


Table View Plot View

accuracy: 93.36% +/- 0.59% (micro average: 93.36%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	997	3	8	1	1	8	19	96.14%
pred. Cancer	0	571	0	0	7	0	1	98.62%
pred. Obesity	16	0	1874	0	40	78	65	90.40%
pred. Healthy	10	0	1	1598	0	3	1	99.07%
pred. Diabetes	2	11	17	0	3283	2	36	97.97%
pred. Arthritis	14	0	34	3	5	651	132	77.59%
pred. Hyperte...	22	25	83	0	60	179	3494	90.45%
class recall	93.97%	93.61%	92.91%	99.75%	96.67%	70.68%	93.22%	

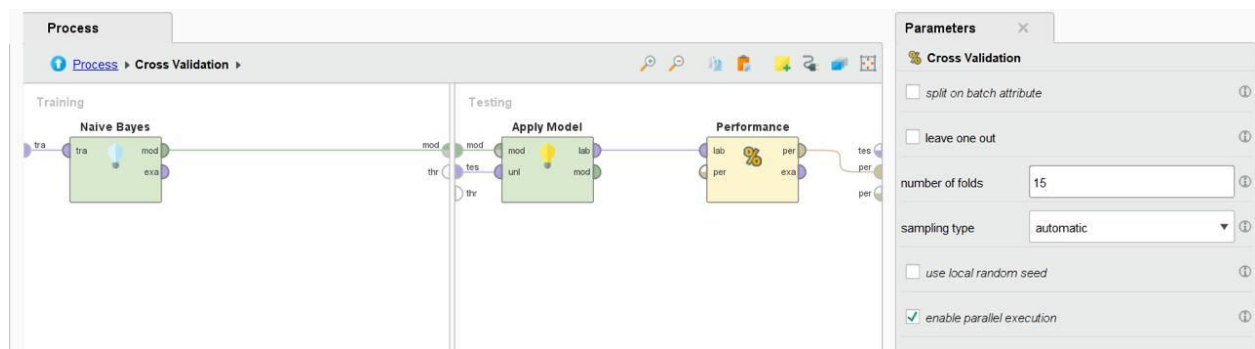
Cross Validation: 10



accuracy: 93.31% +/- 0.87% (micro average: 93.31%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	996	4	8	1	1	9	19	95.95%
pred. Cancer	0	570	0	0	8	0	1	98.45%
pred. Obesity	16	0	1871	0	41	77	65	90.39%
pred. Healthy	9	0	1	1598	0	3	1	99.13%
pred. Diabetes	2	12	17	0	3282	2	36	97.94%
pred. Arthritis	15	0	35	3	5	649	131	77.45%
pred. Hyperte...	23	24	85	0	59	181	3495	90.38%
class recall	93.87%	93.44%	92.76%	99.75%	96.64%	70.47%	93.25%	

Cross Validation: 15

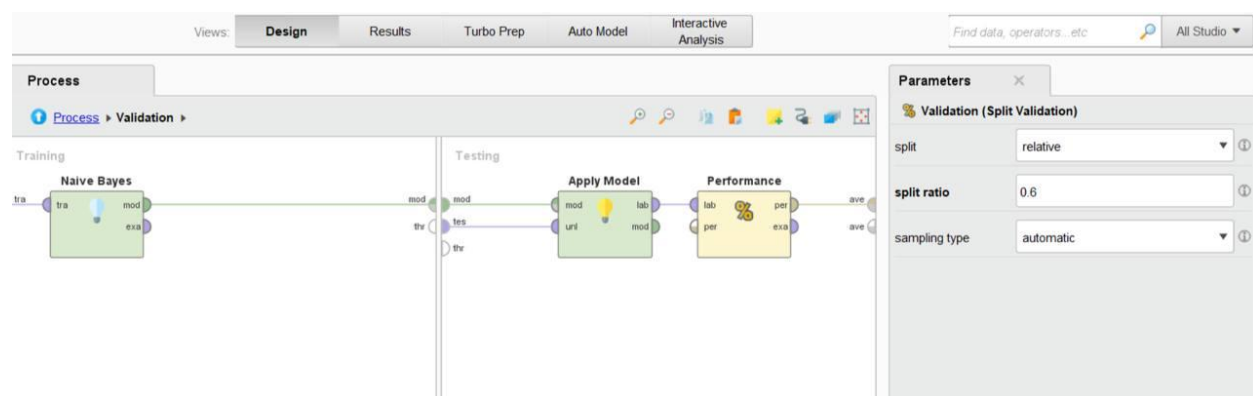


☒ Table View
 ☐ Plot View

accuracy: 93.34% +/- 0.87% (micro average: 93.34%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	996	5	8	1	0	9	20	95.86%
pred. Cancer	0	569	0	0	7	0	1	98.61%
pred. Obesity	15	0	1874	0	40	78	64	90.49%
pred. Healthy	9	0	1	1598	0	3	1	99.13%
pred. Diabetes	2	12	17	0	3283	2	36	97.94%
pred. Arthritis	15	0	32	3	6	650	131	77.66%
pred. Hyperte...	24	24	85	0	60	179	3495	90.38%
class recall	93.87%	93.28%	92.91%	99.75%	96.67%	70.58%	93.25%	

Split Validation: 60%

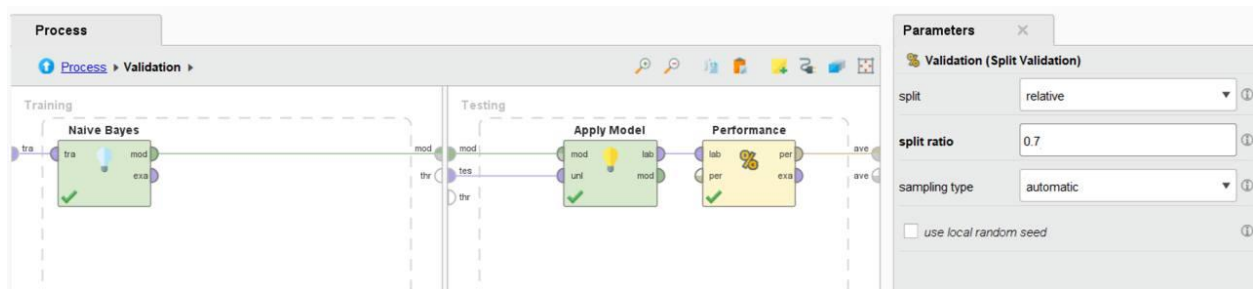


☒ Table View ☐ Plot View

accuracy: 93.39%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	403	1	2	1	1	3	9	95.95%
pred. Cancer	0	228	0	0	0	0	0	100.00%
pred. Obesity	4	0	748	0	18	38	26	89.69%
pred. Healthy	6	0	0	638	0	1	1	98.76%
pred. Diabetes	0	4	11	0	1308	0	11	98.05%
pred. Arthritis	5	0	17	2	4	266	55	76.22%
pred. Hyperte...	6	11	29	0	27	60	1397	91.31%
class recall	95.05%	93.44%	92.69%	99.53%	96.32%	72.28%	93.20%	

Split Validation: 70%

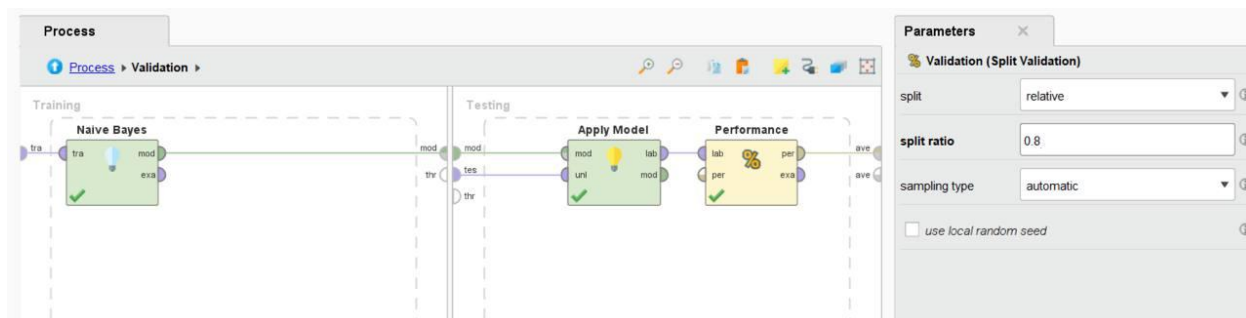


☒ Table View ☐ Plot View

accuracy: 93.38%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	300	1	2	1	0	3	6	95.85%
pred. Cancer	0	172	0	0	0	0	0	100.00%
pred. Obesity	2	0	559	0	11	34	20	89.30%
pred. Healthy	4	0	0	478	0	0	1	98.96%
pred. Diabetes	0	3	9	0	985	0	10	97.82%
pred. Arthritis	6	0	13	2	4	200	40	75.47%
pred. Hyperte...	6	7	22	0	19	39	1047	91.84%
class recall	94.34%	93.99%	92.40%	99.38%	96.66%	72.46%	93.15%	

Split Validation: 80%



☒ Table View ☐ Plot View

accuracy: 93.33%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	200	0	2	1	0	3	3	95.69%
pred. Cancer	0	118	0	0	0	0	0	100.00%
pred. Obesity	2	0	376	0	10	20	13	89.31%
pred. Healthy	2	0	0	318	0	0	1	99.07%
pred. Diabetes	0	2	6	0	653	0	8	97.61%
pred. Arthritis	4	0	8	1	2	132	30	74.58%
pred. Hyperte...	4	2	11	0	14	29	695	92.05%
class recall	94.34%	96.72%	93.30%	99.38%	96.17%	71.74%	92.67%	

After fine tuning:

Weka:

Cross Validation: 5

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes -K**

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation
 - Folds: **5**
 - Percentage split: **70**
- ☐ More options...

(Nom) Medical Condition

Start Stop

Result list (right-click for options):

- 04:17:37 - bayes.NaiveBayes
- 04:19:28 - bayes.NaiveBayes
- 04:21:38 - bayes.NaiveBayes
- 04:24:05 - bayes.NaiveBayes
- 04:24:22 - bayes.NaiveBayes
- 04:25:05 - bayes.NaiveBayes
- 04:25:43 - bayes.NaiveBayes
- 04:28:23 - bayes.NaiveBayes
- 04:29:43 - bayes.NaiveBayes
- 04:39:05 - bayes.NaiveBayes
- 04:39:56 - bayes.NaiveBayes
- 04:40:40 - bayes.NaiveBayes**

Classifier output:

Correctly Classified Instances 12623 94.5189 %

Incorrectly Classified Instances 732 5.4811 %

Kappa statistic 0.932

Mean absolute error 0.0233

Root mean squared error 0.1072

Relative absolute error 10.1362 %

Root relative squared error 31.5874 %

Total Number of Instances 13355

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.947	0.003	0.964	0.947	0.955	0.952	0.999	0.991	Asthma
	0.984	0.001	0.988	0.984	0.986	0.985	1.000	0.998	Cancer
	0.945	0.015	0.920	0.945	0.933	0.920	0.997	0.985	Obesity
	0.999	0.001	0.994	0.999	0.997	0.994	1.000	1.000	Healthy
	0.973	0.006	0.983	0.973	0.978	0.971	0.999	0.997	Diabetes
	0.747	0.014	0.801	0.747	0.773	0.757	0.987	0.867	Arthritis
	0.939	0.030	0.925	0.939	0.932	0.905	0.994	0.984	Hyperten
Weighted Avg.	0.945	0.013	0.945	0.945	0.945	0.932	0.996	0.982	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
1005	1	10	7	1	18	15	1	a = Asthma
0	600	0	0	10	0	0	1	b = Cancer
9	0	1907	0	10	32	59	1	c = Obesity
2	0	0	1600	0	0	0	1	d = Healthy
0	6	26	0	3305	4	55	1	e = Diabetes
8	0	70	2	2	688	151	1	f = Arthritis
19	0	60	0	34	117	3518	1	g = Hypertension

Status OK Log

Cross Validation: 10

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes** <-

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **10**
☐ Percentage split %: **70**
 More options...

(Nom) Medical Condition

Start Stop

Result list (right-click for options):

- 04:17:37 - bayes.NaiveBayes
- 04:19:28 - bayes.NaiveBayes
- 04:21:38 - bayes.NaiveBayes
- 04:24:05 - bayes.NaiveBayes
- 04:24:22 - bayes.NaiveBayes
- 04:25:05 - bayes.NaiveBayes
- 04:25:43 - bayes.NaiveBayes
- 04:28:23 - bayes.NaiveBayes
- 04:29:43 - bayes.NaiveBayes
- 04:39:05 - bayes.NaiveBayes
- 04:39:56 - bayes.NaiveBayes
- 04:40:49 - bayes.NaiveBayes
- 04:41:54 - bayes.NaiveBayes**

Status: OK

Classifier output:

Correctly Classified Instances	12626	94.5414 %
Incorrectly Classified Instances	729	5.4586 %
Kappa statistic	0.9322	
Mean absolute error	0.0232	
Root mean squared error	0.1069	
Relative absolute error	10.0685 %	
Root relative squared error	31.497 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.947	0.003	0.964	0.947	0.955	0.952	0.999	0.990	Asthma
	0.980	0.001	0.985	0.980	0.983	0.982	1.000	0.996	Cancer
	0.947	0.015	0.920	0.947	0.934	0.922	0.997	0.985	Obesity
	0.999	0.001	0.995	0.999	0.997	0.996	1.000	1.000	Healthy
	0.972	0.006	0.983	0.972	0.978	0.970	0.999	0.997	Diabetes
	0.747	0.013	0.807	0.747	0.776	0.760	0.987	0.868	Arthritis
	0.940	0.030	0.925	0.940	0.932	0.906	0.994	0.984	Hyperten
Weighted Avg.	0.945	0.013	0.945	0.945	0.945	0.932	0.996	0.982	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
1005	1	10	6	1	17	21		a = Asthma
0	598	0	0	12	0	0		b = Cancer
8	1	1910	0	8	32	58		c = Obesity
2	0	0	1600	0	0	0		d = Healthy
0	7	24	0	3302	4	59		e = Diabetes
8	0	72	2	2	688	149		f = Arthritis
20	0	59	0	34	112	3523		g = Hypertension

Status: OK

Log x0

Cross Validation: 15

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes** <-

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: **15**
☐ Percentage split %: **70**
 More options...

(Nom) Medical Condition

Start Stop

Result list (right-click for options):

- 04:17:37 - bayes.NaiveBayes
- 04:19:28 - bayes.NaiveBayes
- 04:21:38 - bayes.NaiveBayes
- 04:24:05 - bayes.NaiveBayes
- 04:24:22 - bayes.NaiveBayes
- 04:25:05 - bayes.NaiveBayes
- 04:25:43 - bayes.NaiveBayes
- 04:28:23 - bayes.NaiveBayes
- 04:29:43 - bayes.NaiveBayes
- 04:39:05 - bayes.NaiveBayes
- 04:39:56 - bayes.NaiveBayes
- 04:40:49 - bayes.NaiveBayes
- 04:41:54 - bayes.NaiveBayes
- 04:42:27 - bayes.NaiveBayes**

Status: OK

Classifier output:

Correctly Classified Instances	12631	94.5788 %
Incorrectly Classified Instances	724	5.4212 %
Kappa statistic	0.9327	
Mean absolute error	0.0232	
Root mean squared error	0.1069	
Relative absolute error	10.0503 %	
Root relative squared error	31.5028 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.947	0.003	0.964	0.947	0.957	0.953	0.999	0.990	Asthma
	0.985	0.001	0.984	0.985	0.984	0.984	1.000	0.996	Cancer
	0.946	0.014	0.921	0.946	0.934	0.922	0.997	0.985	Obesity
	0.999	0.001	0.995	0.999	0.997	0.997	1.000	1.000	Healthy
	0.972	0.005	0.984	0.972	0.978	0.971	0.998	0.997	Diabetes
	0.750	0.014	0.801	0.750	0.775	0.759	0.987	0.868	Arthritis
	0.940	0.029	0.926	0.940	0.933	0.906	0.994	0.984	Hyperten
Weighted Avg.	0.946	0.013	0.945	0.946	0.946	0.933	0.996	0.982	

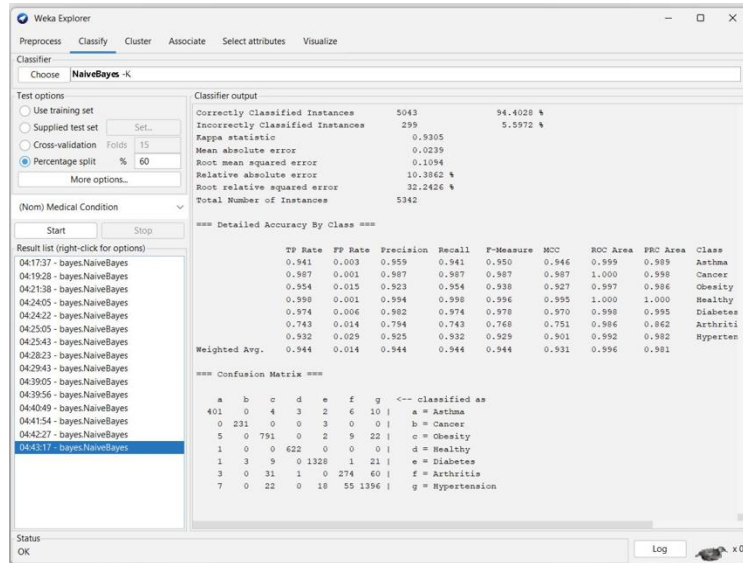
=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
1005	1	10	6	1	17	21		a = Asthma
0	601	0	0	9	0	0		b = Cancer
8	1	1909	0	9	33	57		c = Obesity
1	0	0	1601	0	0	0		d = Healthy
0	8	26	0	3302	4	56		e = Diabetes
7	0	70	2	2	691	149		f = Arthritis
19	0	57	0	32	118	3522		g = Hypertension

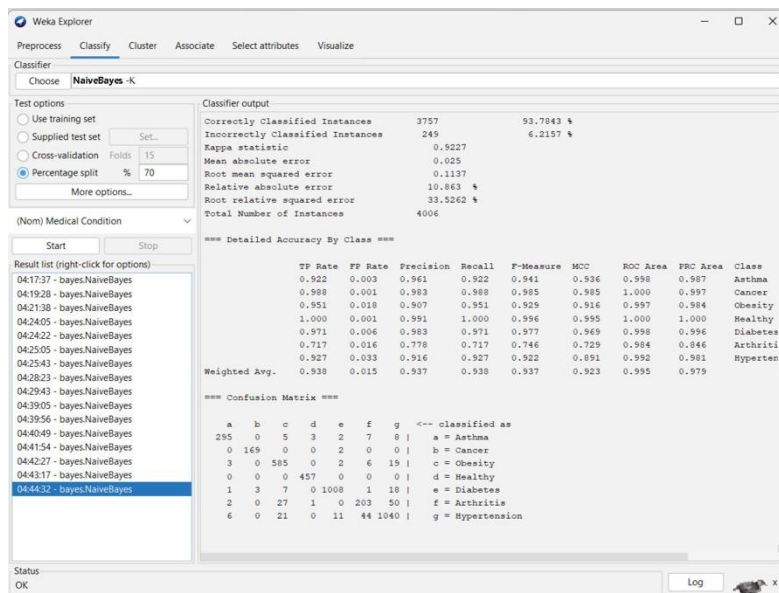
Status: OK

Log x0

Split Validation: 60%



Split Validation: 70%



Split Validation: 80%

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **NaiveBayes -K**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: 10
☒ Percentage split %: 80
 More options...

(Nom) Medical Condition

Start Stop

Result list (right-click for options):
 04-4653 - bayes2.NaiveBayes

Classifier output

Time taken to test model on test split: 4.46 seconds

=== Summary ===

Correctly Classified Instances	2505	93.7851 %
Incorrectly Classified Instances	166	6.2149 %
Kappa statistic	0.9228	
Mean absolute error	0.0248	
Root mean squared error	0.1142	
Relative absolute error	10.7491 %	
Root relative squared error	33.6516 %	
Total Number of Instances	2671	

=== Detailed Accuracy By Class ===

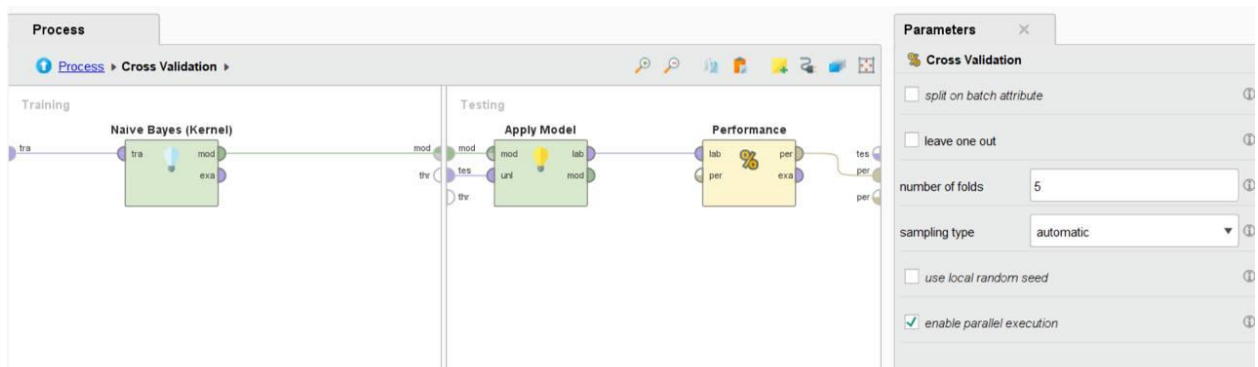
	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.928	0.003	0.963	0.928	0.945	0.940	0.997	0.985	Asthma
	0.982	0.000	0.991	0.982	0.987	0.986	1.000	0.997	Cancer
	0.939	0.018	0.904	0.939	0.921	0.907	0.997	0.983	Obesity
	1.000	0.002	0.987	1.000	0.994	0.993	1.000	1.000	Healthy
	0.973	0.006	0.983	0.973	0.978	0.971	0.998	0.995	Diabetes
	0.720	0.016	0.766	0.720	0.742	0.724	0.985	0.848	Arthritis
	0.930	0.031	0.921	0.930	0.926	0.896	0.993	0.982	Hypertension
Weighted Avg.	0.938	0.015	0.938	0.938	0.938	0.923	0.995	0.979	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	← classified as
a = Asthma	206	0	2	3	2	5	4	
b = Cancer	0	110	0	0	2	0	0	
c = Obesity	3	0	385	0	1	6	15	
d = Healthy	0	0	0	311	0	0	0	
e = Diabetes	1	1	5	0	635	1	10	
f = Arthritis	1	0	19	1	0	134	31	
g = Hypertension	3	0	15	0	6	29	704	

RapidMiner:

Cross Validation: 5



☒ Table View
 ☐ Plot View

accuracy: 94.17% +/- 0.76% (micro average: 94.17%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	999	0	7	1	1	9	22	96.15%
pred. Cancer	0	591	0	0	5	0	0	99.16%
pred. Obesity	13	0	1900	0	23	75	59	91.79%
pred. Healthy	9	0	1	1601	0	3	1	99.13%
pred. Diabetes	1	9	12	0	3305	2	30	98.39%
pred. Arthritis	16	0	34	0	4	666	122	79.10%
pred. Hyperte...	23	10	63	0	58	166	3514	91.65%
class recall	94.16%	96.89%	94.20%	99.94%	97.32%	72.31%	93.76%	

Cross Validation: 10

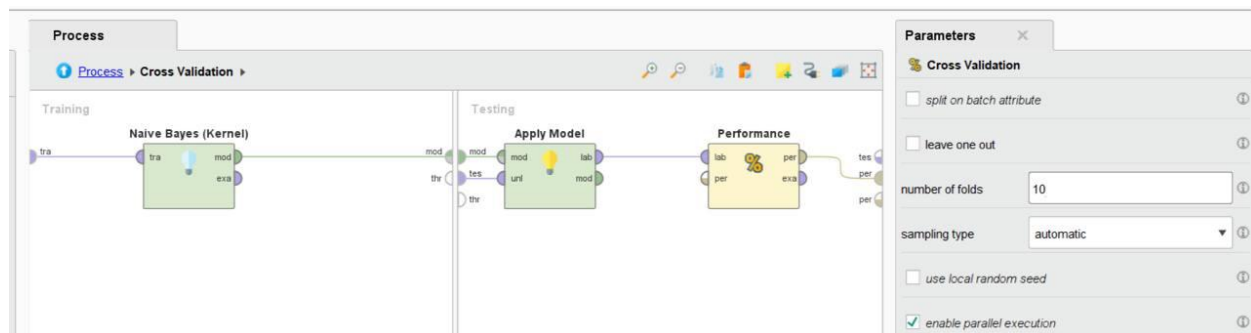


Table View Plot View

accuracy: 94.15% +/- 1.07% (micro average: 94.15%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	1002	0	7	0	1	9	21	96.35%
pred. Cancer	0	589	0	0	5	0	0	99.16%
pred. Obesity	12	0	1904	0	25	77	58	91.71%
pred. Healthy	9	0	1	1602	0	3	1	99.13%
pred. Diabetes	1	10	11	0	3302	2	31	98.36%
pred. Arthritis	15	0	36	0	3	658	120	79.09%
pred. Hyperte...	22	11	58	0	60	172	3517	91.59%
class recall	94.44%	96.56%	94.40%	100.00%	97.23%	71.44%	93.84%	

Cross Validation: 15

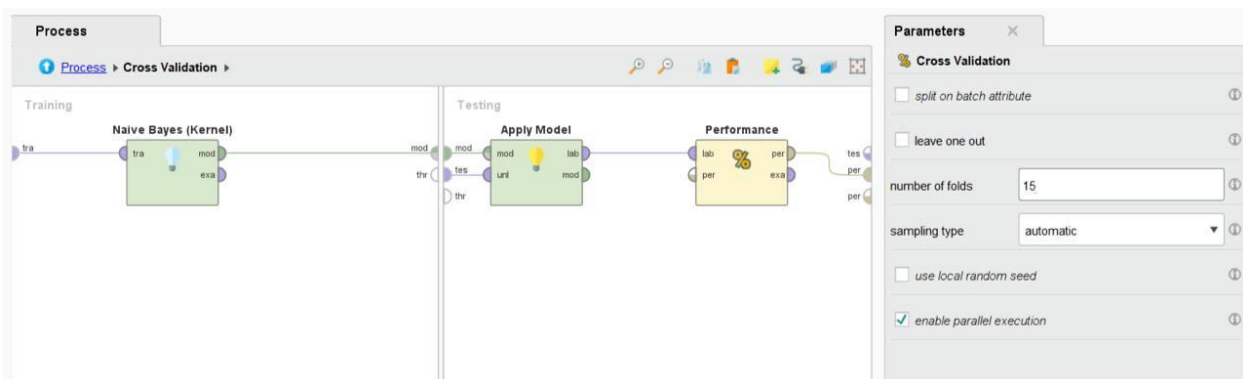


Table View Plot View

accuracy: 94.11% +/- 0.97% (micro average: 94.11%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	1001	1	8	1	1	10	22	95.88%
pred. Cancer	0	588	0	0	5	0	0	99.16%
pred. Obesity	12	0	1900	0	24	74	59	91.83%
pred. Healthy	9	0	1	1601	0	3	1	99.13%
pred. Diabetes	1	10	12	0	3303	2	32	98.30%
pred. Arthritis	16	0	35	0	4	662	120	79.09%
pred. Hyperte...	22	11	61	0	59	170	3514	91.58%
class recall	94.34%	96.39%	94.20%	99.94%	97.26%	71.88%	93.76%	

Split Validation: 60%

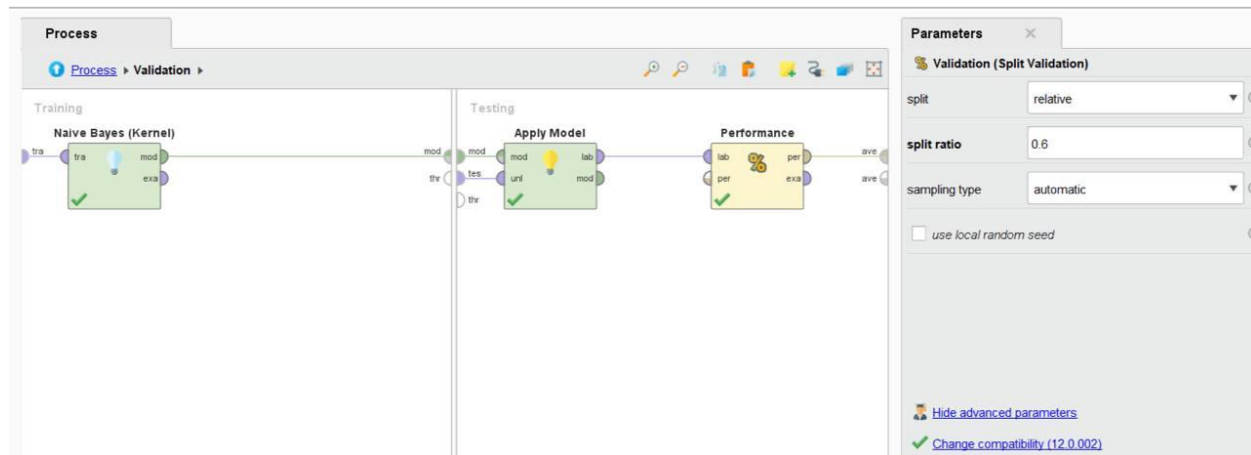


Table View Plot View

accuracy: 93.93%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	402	0	3	0	1	4	10	95.71%
pred. Cancer	0	233	0	0	0	0	0	100.00%
pred. Obesity	4	0	760	0	11	33	25	91.24%
pred. Healthy	6	0	1	641	0	0	1	98.77%
pred. Diabetes	0	7	6	0	1309	0	7	98.50%
pred. Arthritis	7	0	17	0	2	265	49	77.94%
pred. Hyperte...	5	4	20	0	35	66	1407	91.54%
class recall	94.81%	95.49%	94.18%	100.00%	96.39%	72.01%	93.86%	

Split Validation: 70%

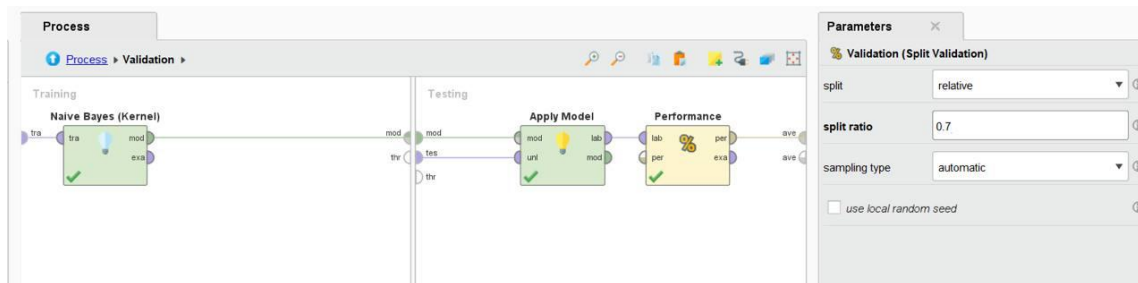
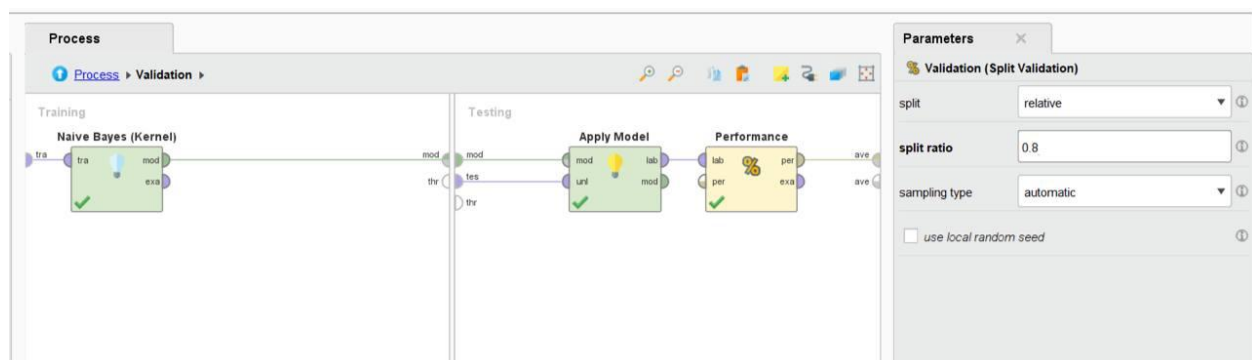


Table View Plot View

accuracy: 94.03%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	302	0	3	0	0	4	6	95.87%
pred. Cancer	0	178	0	0	0	0	0	100.00%
pred. Obesity	2	0	568	0	6	29	21	90.73%
pred. Healthy	5	0	0	481	0	0	1	98.77%
pred. Diabetes	0	3	4	0	985	0	7	98.60%
pred. Arthritis	5	0	13	0	1	202	38	77.99%
pred. Hyperte...	4	2	17	0	27	41	1051	92.03%
class recall	94.97%	97.27%	93.88%	100.00%	96.66%	73.19%	93.51%	

Split Validation: 80%



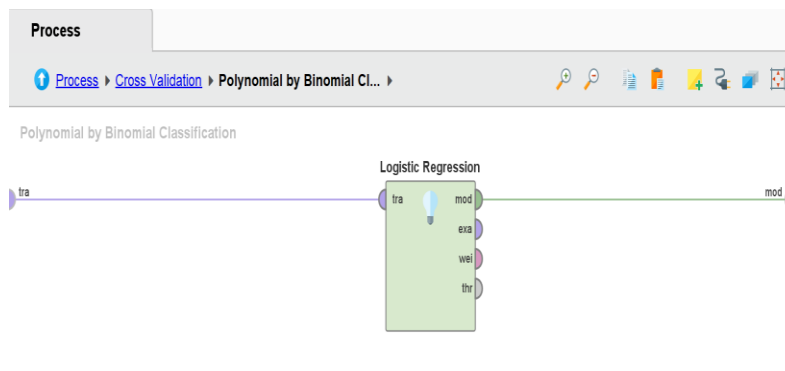
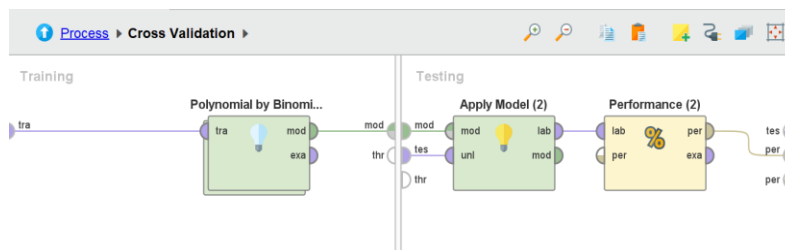
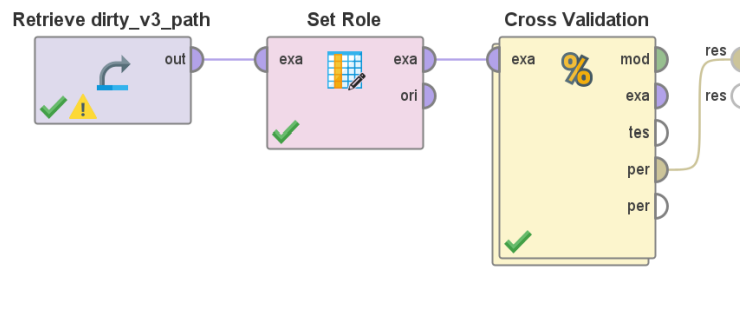
☒ Table View ☐ Plot View

accuracy: 93.82%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hyperten...	class precisio...
pred. Asthma	201	0	2	0	0	4	2	96.17%
pred. Cancer	0	120	0	0	0	0	0	100.00%
pred. Obesity	2	0	382	0	7	17	15	90.31%
pred. Healthy	2	0	0	320	0	0	1	99.07%
pred. Diabetes	0	2	3	0	653	0	8	98.05%
pred. Arthritis	3	0	9	0	0	133	28	76.88%
pred. Hyperte...	4	0	7	0	19	30	696	92.06%
class recall	94.81%	98.36%	94.79%	100.00%	96.17%	72.28%	92.80%	

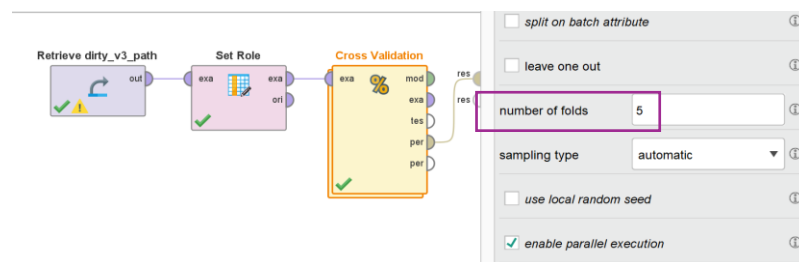
Appendix E

RapidMiner :



Before fine tuning:

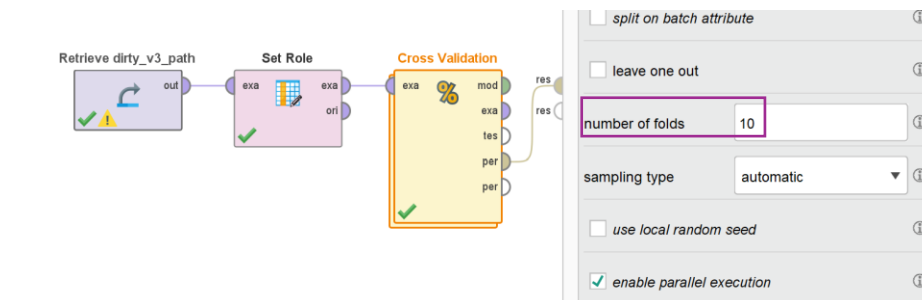
Cross validation : fold 5



accuracy: 90.36% +/- 1.02% (micro average: 90.36%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	886	12	17	3	2	12	33	91.81%
pred. Cancer	3	577	0	0	24	0	1	95.37%
pred. Obesity	45	0	1824	0	44	91	78	87.61%
pred. Healthy	16	0	1	1593	0	3	3	98.58%
pred. Diabe...	2	7	47	0	3241	10	70	95.97%
pred. Arthritis	20	1	28	5	13	489	106	73.87%
pred. Hyper...	89	13	100	1	72	316	3457	85.40%
class recall	83.51%	94.59%	90.43%	99.44%	95.44%	53.09%	92.24%	

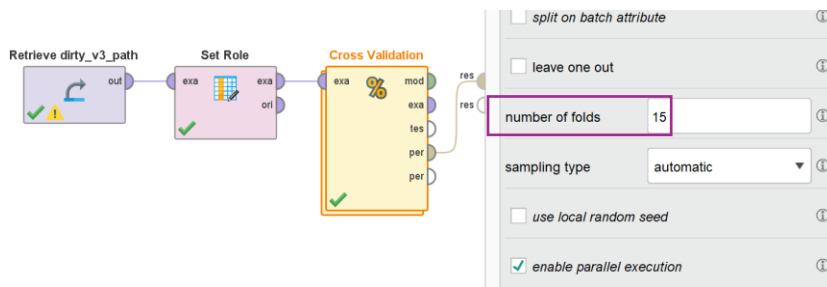
Cross validation : fold 10



accuracy: 90.37% +/- 1.38% (micro average: 90.37%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	891	12	15	3	2	10	33	92.24%
pred. Cancer	3	576	0	0	22	0	1	95.68%
pred. Obesity	44	0	1822	0	44	94	77	87.55%
pred. Healthy	13	0	1	1594	0	2	3	98.82%
pred. Diabe...	2	7	47	0	3240	10	71	95.94%
pred. Arthritis	18	1	31	4	14	489	106	73.76%
pred. Hyper...	90	14	101	1	74	316	3457	85.29%
class recall	83.98%	94.43%	90.33%	99.50%	95.41%	53.09%	92.24%	

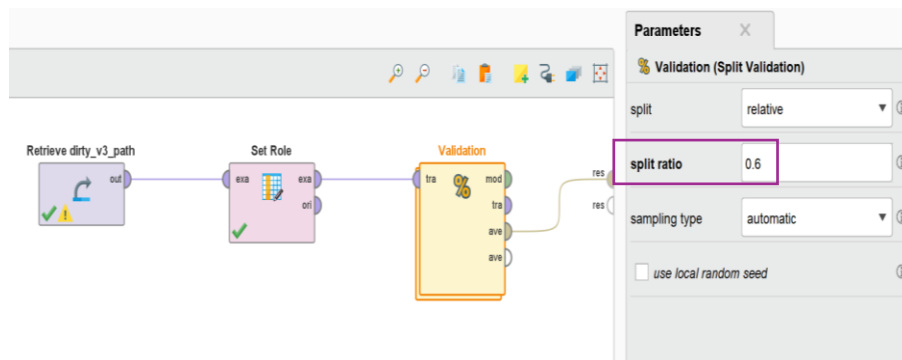
Cross validation : fold 15



accuracy: 90.38% +/- 1.07% (micro average: 90.38%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	891	13	16	2	2	11	32	92.14%
pred. Cancer	3	576	0	0	22	0	1	95.68%
pred. Obesity	42	0	1823	0	43	91	78	87.77%
pred. Healthy	15	0	1	1595	0	3	3	98.64%
pred. Diabe...	2	7	47	0	3240	11	72	95.89%
pred. Arthritis	17	1	30	4	14	487	104	74.12%
pred. Hyper...	91	13	100	1	75	318	3458	85.26%
class recall	83.98%	94.43%	90.38%	99.56%	95.41%	52.88%	92.26%	

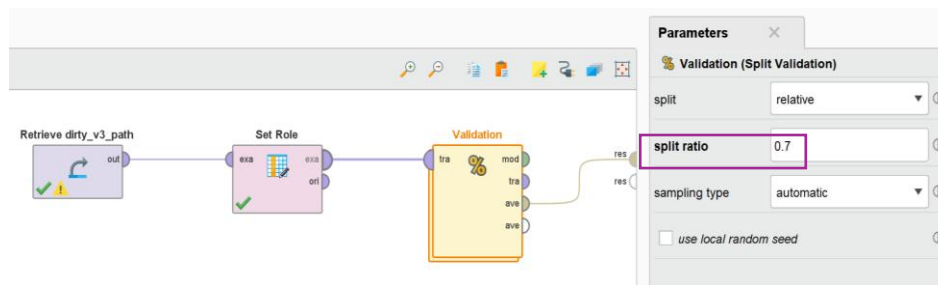
Split 60 :



accuracy: 90.54%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	358	5	12	1	2	5	9	91.33%
pred. Cancer	2	230	0	0	6	0	1	96.23%
pred. Obesity	16	0	720	0	21	41	25	87.48%
pred. Healthy	7	0	0	635	0	0	0	98.91%
pred. Diabe...	1	3	26	0	1292	5	23	95.70%
pred. Arthritis	10	1	14	4	5	205	45	72.18%
pred. Hyper...	30	5	35	1	32	112	1396	86.65%
class recall	84.43%	94.26%	89.22%	99.06%	95.14%	55.71%	93.13%	

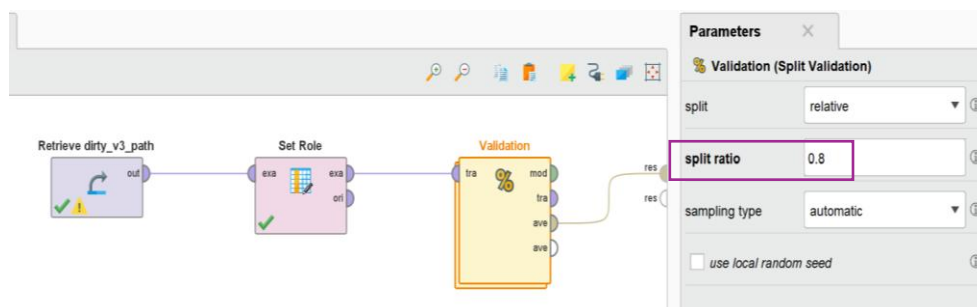
Split 70 :



accuracy: 90.39%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	269	4	12	1	1	3	6	90.88%
pred. Cancer	1	173	0	0	6	0	1	95.58%
pred. Obesity	8	0	535	0	15	37	18	87.28%
pred. Healthy	3	0	0	476	0	0	0	99.37%
pred. Diabe...	1	3	23	0	969	3	18	95.28%
pred. Arthritis	10	0	10	3	4	149	31	71.98%
pred. Hyper...	26	3	25	1	24	84	1050	86.56%
class recall	84.59%	94.54%	88.43%	98.96%	95.09%	53.99%	93.42%	

Split 80 :



accuracy: 90.00%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	176	2	8	1	0	3	3	91.19%
pred. Cancer	2	117	0	0	6	0	1	92.86%
pred. Obesity	6	0	360	0	11	22	12	87.59%
pred. Healthy	4	0	0	316	0	0	0	98.75%
pred. Diabe...	1	2	12	0	646	2	17	95.00%
pred. Arthritis	6	0	8	2	2	96	25	69.06%
pred. Hyper...	17	1	15	1	14	61	692	86.39%
class recall	83.02%	95.90%	89.33%	98.75%	95.14%	52.17%	92.27%	

Weka

Cross Validation:5

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **Logistic** R 1.0E-8 - M -1 - num decimal places 4

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds **5**

☐ Percentage split % 66

More options...

(Nom) Medical Condition

Start Stop

Result list (right-click for options)

20:13:49 - functions.Logistic

Classifier output

Stress Level	0.7681	1.6647	0.6091	0.3202	0.7508	0.6103
Sleep Hours	0.5579	0.666	1.477	17.5979	1.0713	1.4396

Time taken to build model: 3.55 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 12240 91.6511 %

Incorrectly Classified Instances 1115 8.3489 %

Kappa statistic 0.8962

Mean absolute error 0.0336

Root mean squared error 0.1312

Relative absolute error 14.5971 %

Root relative squared error 38.6526 %

Total Number of Instances 13355

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.918	0.006	0.928	0.918	0.923	0.916	0.997	0.978	Asthma	
0.954	0.002	0.962	0.954	0.958	0.956	1.000	0.993	Cancer	
0.908	0.019	0.894	0.908	0.901	0.883	0.993	0.970	Obesity	
0.992	0.001	0.991	0.992	0.992	0.990	1.000	1.000	Healthy	
0.959	0.011	0.968	0.959	0.963	0.951	0.997	0.993	Diabetes	
0.637	0.018	0.726	0.637	0.679	0.658	0.975	0.767	Arthritis	
0.912	0.047	0.883	0.912	0.897	0.857	0.986	0.966	Hypertension	
Weighted Avg.	0.917	0.021	0.915	0.917	0.916	0.896	0.992	0.966	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
974	1	21	10	2	13	40		a = Asthma
5	582	0	0	12	0	11		b = Cancer
13	0	1832	0	35	40	97		c = Obesity
7	0	0	1589	0	6	0		d = Healthy
4	20	43	0	3256	5	68		e = Diabetes
15	0	73	2	6	587	238		f = Arthritis
32	2	81	2	54	157	3420		g = Hypertension

Status OK

Log x 0

Cross Validation:10

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **Logistic** R 1.0E-8 - M -1 - num decimal places 4

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds **10**

☐ Percentage split % 66

More options...

(Nom) Medical Condition

Start Stop

Result list (right-click for options)

20:13:49 - functions.Logistic

20:15:46 - functions.Logistic

Classifier output

Stress Level	0.7681	1.6647	0.6091	0.3202	0.7508	0.6103
Sleep Hours	0.5579	0.666	1.477	17.5979	1.0713	1.4396

Time taken to build model: 2.55 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 12241 91.6586 %

Incorrectly Classified Instances 1114 8.3414 %

Kappa statistic 0.8963

Mean absolute error 0.0336

Root mean squared error 0.1308

Relative absolute error 14.5747 %

Root relative squared error 38.5481 %

Total Number of Instances 13355

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.920	0.006	0.929	0.920	0.924	0.918	0.998	0.979	Asthma	
0.959	0.002	0.964	0.959	0.961	0.960	1.000	0.993	Cancer	
0.908	0.019	0.892	0.908	0.900	0.882	0.994	0.970	Obesity	
0.993	0.001	0.993	0.993	0.993	0.992	1.000	1.000	Healthy	
0.958	0.011	0.968	0.958	0.963	0.950	0.997	0.993	Diabetes	
0.637	0.018	0.726	0.637	0.679	0.658	0.975	0.769	Arthritis	
0.912	0.047	0.883	0.912	0.897	0.856	0.986	0.966	Hypertension	
Weighted Avg.	0.917	0.021	0.916	0.917	0.916	0.896	0.992	0.966	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
976	1	20	8	2	14	40		a = Asthma
5	585	0	0	9	0	11		b = Cancer
13	0	1832	0	35	39	98		c = Obesity
5	0	0	1590	0	7	0		d = Healthy
4	19	45	0	3253	7	68		e = Diabetes
15	0	73	1	6	587	239		f = Arthritis
33	2	83	2	55	155	3418		g = Hypertension

Status OK

Log x 0

Cross Validation:15

The screenshot shows the Weka Explorer Classifier window with the 'Logistic - R 1.0E-8 - M -1 - num decimal places 4' model selected. The 'Test options' section has 'Cross-validation' selected with 'Folds' set to 15. The 'Classifier output' pane displays the following data:

Stress Level: 0.7681, 1.6647, 0.6091, 0.3202, 0.7508, 0.6103
Sleep Hours: 0.5579, 0.666, 1.477, 17.5979, 1.0713, 1.4396

Time taken to build model: 2.6 seconds

==== Stratified cross-validation ====

==== Summary ====

Metric	Value
Correctly Classified Instances	12245
Incorrectly Classified Instances	1110
Kappa statistic	0.8967
Mean absolute error	0.0336
Root mean squared error	0.1308
Relative absolute error	14.5749 %
Root relative squared error	38.532 %
Total Number of Instances	13355

==== Detailed Accuracy By Class ====

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.919	0.006	0.931	0.919	0.925	0.919	0.997	0.979	Asthma
0.959	0.002	0.957	0.959	0.958	0.956	1.000	0.994	Cancer
0.910	0.019	0.893	0.910	0.901	0.884	0.994	0.970	Obesity
0.992	0.001	0.991	0.992	0.992	0.990	1.000	1.000	Healthy
0.958	0.011	0.968	0.958	0.963	0.950	0.997	0.993	Diabetes
0.638	0.018	0.728	0.638	0.680	0.660	0.975	0.770	Arthritis
0.913	0.047	0.883	0.913	0.898	0.857	0.986	0.966	Hypertension
Weighted Avg.	0.917	0.021	0.916	0.917	0.916	0.897	0.992	

==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
975	1	19	11	2	12	41	1	a = Asthma
4	585	0	0	10	0	11	1	b = Cancer
12	0	1835	0	34	42	94	1	c = Obesity
7	0	0	1589	0	6	0	1	d = Healthy
3	23	45	0	3252	7	66	1	e = Diabetes
13	0	72	1	6	588	241	1	f = Arthritis
33	2	83	2	54	153	3421	1	g = Hypertension

Split 60

The screenshot shows the Weka Explorer Classifier window with the 'Logistic - R 1.0E-8 - M -1 - num decimal places 4' model selected. The 'Test options' section has 'Percentage split' selected with a percentage of 60. The 'Classifier output' pane displays the following data:

Time taken to build model: 2.58 seconds

==== Evaluation on test split ====

Time taken to test model on test split: 0.01 seconds

==== Summary ====

Metric	Value
Correctly Classified Instances	4877
Incorrectly Classified Instances	465
Kappa statistic	0.8917
Mean absolute error	0.0341
Root mean squared error	0.1338
Relative absolute error	14.8147 %
Root relative squared error	39.4367 %
Total Number of Instances	5342

==== Detailed Accuracy By Class ====

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
0.887	0.006	0.924	0.887	0.905	0.898	0.997	0.972	Asthma
0.966	0.001	0.970	0.966	0.968	0.966	1.000	0.995	Cancer
0.908	0.018	0.901	0.908	0.905	0.887	0.994	0.971	Obesity
0.994	0.002	0.984	0.994	0.989	0.987	1.000	1.000	Healthy
0.959	0.014	0.960	0.959	0.960	0.946	0.996	0.992	Diabetes
0.642	0.018	0.727	0.642	0.682	0.661	0.974	0.778	Arthritis
0.906	0.050	0.877	0.906	0.891	0.848	0.985	0.962	Hypertension
Weighted Avg.	0.913	0.022	0.912	0.913	0.912	0.891	0.992	

==== Confusion Matrix ====

	a	b	c	d	e	f	g	<-- classified as
378	0	6	8	3	6	25	1	a = Asthma
1	226	0	0	3	0	4	1	b = Cancer
7	0	753	0	17	15	37	1	c = Obesity
3	0	0	619	0	1	0	1	d = Healthy
1	6	14	0	1307	5	30	1	e = Diabetes
5	0	31	0	1	237	95	1	f = Arthritis
14	1	32	2	30	62	1357	1	g = Hypertension

Split 70

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier is 'Logistic - R 1.0E-8 - M -1 -num-decimal-places 4'. The test options are set to 'Percentage split' at 70%. The classifier output shows the following results:

Time taken to build model: 2.58 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.01 seconds

=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	3633	90.689 %
Incorrectly Classified Instances	373	9.311 %
Kappa statistic	0.8841	
Mean absolute error	0.0347	
Root mean squared error	0.137	
Relative absolute error	15.0796 %	
Root relative squared error	40.394 %	
Total Number of Instances	4006	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.875	0.008	0.909	0.875	0.892	0.883	0.996	0.968	Asthma
	0.977	0.001	0.971	0.977	0.974	0.973	1.000	0.994	Cancer
	0.899	0.021	0.885	0.899	0.892	0.872	0.993	0.969	Obesity
	0.993	0.003	0.978	0.993	0.986	0.984	1.000	1.000	Healthy
	0.956	0.012	0.965	0.956	0.960	0.947	0.996	0.992	Diabetes
	0.643	0.019	0.722	0.643	0.680	0.659	0.973	0.768	Arthritis
	0.896	0.053	0.869	0.896	0.882	0.835	0.983	0.959	Hypertension
Weighted Avg.	0.907	0.023	0.906	0.907	0.906	0.884	0.991	0.962	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
280	0	6	7	2	5	20	1	a = Asthma
1	167	0	0	2	0	1	1	b = Cancer
6	0	553	1	12	11	32	1	c = Obesity
2	0	0	454	0	1	0	1	d = Healthy
1	4	12	0	992	2	27	1	e = Diabetes
4	0	24	0	1	182	72	1	f = Arthritis
14	1	30	2	19	51	1005	1	g = Hypertension

Status: OK

Split 80

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The classifier is 'Logistic - R 1.0E-8 - M -1 -num-decimal-places 4'. The test options are set to 'Percentage split' at 80%. The classifier output shows the following results:

Time taken to build model: 2.42 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Metric	Value	Percentage
Correctly Classified Instances	2428	90.5023 %
Incorrectly Classified Instances	243	9.0977 %
Kappa statistic	0.8868	
Mean absolute error	0.0344	
Root mean squared error	0.1364	
Relative absolute error	14.9207 %	
Root relative squared error	40.214 %	
Total Number of Instances	2671	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.892	0.007	0.921	0.892	0.906	0.898	0.996	0.970	Asthma
	0.973	0.002	0.965	0.973	0.969	0.968	0.999	0.993	Cancer
	0.898	0.020	0.891	0.898	0.894	0.875	0.993	0.967	Obesity
	0.990	0.003	0.981	0.990	0.986	0.984	1.000	0.999	Healthy
	0.957	0.012	0.964	0.957	0.960	0.947	0.996	0.991	Diabetes
	0.651	0.018	0.729	0.651	0.687	0.667	0.974	0.775	Arthritis
	0.898	0.053	0.870	0.898	0.884	0.837	0.984	0.960	Hypertension
Weighted Avg.	0.909	0.023	0.908	0.909	0.908	0.886	0.991	0.963	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
198	0	3	5	2	3	11	1	a = Asthma
0	109	0	0	2	0	1	1	b = Cancer
3	0	368	0	7	7	25	1	c = Obesity
2	0	0	308	0	1	0	1	d = Healthy
0	3	4	0	644	2	20	1	e = Diabetes
2	0	18	0	0	121	45	1	f = Arthritis
10	1	20	1	13	32	680	1	g = Hypertension

Status: OK

After fine tuning :

RapidMiner :

Cross validation : fold 5

accuracy: 90.36% +/- 1.02% (micro average: 90.36%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	886	12	17	3	2	12	33	91.81%
pred. Cancer	3	577	0	0	24	0	1	95.37%
pred. Obesity	45	0	1824	0	44	91	78	87.61%
pred. Healthy	16	0	1	1593	0	3	3	98.58%
pred. Diabe...	2	7	47	0	3241	10	70	95.97%
pred. Arthritis	20	1	28	5	13	489	106	73.87%
pred. Hyper...	89	13	100	1	72	316	3457	85.40%
class recall	83.51%	94.59%	90.43%	99.44%	95.44%	53.09%	92.24%	

Cross validation : fold 10

accuracy: 90.37% +/- 1.38% (micro average: 90.37%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	891	12	15	3	2	10	33	92.24%
pred. Cancer	3	576	0	0	22	0	1	95.68%
pred. Obesity	44	0	1822	0	44	94	77	87.55%
pred. Healthy	13	0	1	1594	0	2	3	98.82%
pred. Diabe...	2	7	47	0	3240	10	71	95.94%
pred. Arthritis	18	1	31	4	14	489	106	73.76%
pred. Hyper...	90	14	101	1	74	316	3457	85.29%
class recall	83.98%	94.43%	90.33%	99.50%	95.41%	53.09%	92.24%	

Cross validation : fold 15

accuracy: 90.38% +/- 1.07% (micro average: 90.38%)

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	891	13	16	2	2	11	32	92.14%
pred. Cancer	3	576	0	0	22	0	1	95.68%
pred. Obesity	42	0	1823	0	43	91	78	87.77%
pred. Healthy	15	0	1	1595	0	3	3	98.64%
pred. Diabe...	2	7	47	0	3240	11	72	95.89%
pred. Arthritis	17	1	30	4	14	487	104	74.12%
pred. Hyper...	91	13	100	1	75	318	3458	85.26%
class recall	83.98%	94.43%	90.38%	99.56%	95.41%	52.88%	92.26%	

Split 60:

accuracy: 90.54%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	358	5	12	1	2	5	9	91.33%
pred. Cancer	2	230	0	0	6	0	1	96.23%
pred. Obesity	16	0	720	0	21	41	25	87.48%
pred. Healthy	7	0	0	635	0	0	0	98.91%
pred. Diabe...	1	3	26	0	1292	5	23	95.70%
pred. Arthritis	10	1	14	4	5	205	45	72.18%
pred. Hyper...	30	5	35	1	32	112	1396	86.65%
class recall	84.43%	94.26%	89.22%	99.06%	95.14%	55.71%	93.13%	

Split 70:

accuracy: 90.39%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	269	4	12	1	1	3	6	90.88%
pred. Cancer	1	173	0	0	6	0	1	95.58%
pred. Obesity	8	0	535	0	15	37	18	87.28%
pred. Healthy	3	0	0	476	0	0	0	99.37%
pred. Diabe...	1	3	23	0	969	3	18	95.28%
pred. Arthritis	10	0	10	3	4	149	31	71.98%
pred. Hyper...	26	3	25	1	24	84	1050	86.56%
class recall	84.59%	94.54%	88.43%	98.96%	95.09%	53.99%	93.42%	

Split 80:

accuracy: 90.00%

	true Asthma	true Cancer	true Obesity	true Healthy	true Diabetes	true Arthritis	true Hypert...	class precis...
pred. Asthma	176	2	8	1	0	3	3	91.19%
pred. Cancer	2	117	0	0	6	0	1	92.86%
pred. Obesity	6	0	360	0	11	22	12	87.59%
pred. Healthy	4	0	0	316	0	0	0	98.75%
pred. Diabe...	1	2	12	0	646	2	17	95.00%
pred. Arthritis	6	0	8	2	2	96	25	69.06%
pred. Hyper...	17	1	15	1	14	61	692	86.39%
class recall	83.02%	95.90%	89.33%	98.75%	95.14%	52.17%	92.27%	

Weka:

Cross Validation: 5

```
Time taken to build model: 4.22 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      12244           91.681 %
Incorrectly Classified Instances    1111           8.319 %
Kappa statistic                    0.8965
Mean absolute error                 0.0363
Root mean squared error             0.1315
Relative absolute error             15.776 %
Root relative squared error         38.7572 %
Total Number of Instances          13355

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      0.907    0.006    0.933     0.907    0.920     0.913    0.997    0.975    Asthma
      0.954    0.001    0.973     0.954    0.964     0.962    1.000    0.992    Cancer
      0.909    0.020    0.891     0.909    0.900     0.882    0.993    0.969    Obesity
      0.993    0.002    0.986     0.993    0.989     0.988    1.000    1.000    Healthy
      0.960    0.010    0.969     0.960    0.965     0.953    0.997    0.993    Diabetes
      0.634    0.017    0.737     0.634    0.682     0.662    0.975    0.772    Arthritis
      0.916    0.049    0.880     0.916    0.897     0.857    0.986    0.966    Hypertension
Weighted Avg.    0.917    0.021    0.916     0.917    0.916     0.897    0.992    0.966

=== Confusion Matrix ===

  a   b   c   d   e   f   g  <-- classified as
962  1  24  18   2  12  42 |  a = Asthma
  5 582   0   0  11   0  12 |  b = Cancer
13   0 1833   1  34  37  99 |  c = Obesity
  5   0   0 1590   0   7   0 |  d = Healthy
  4  12  44   0 3261   5  70 |  e = Diabetes
12   0  73   1   6  584 245 |  f = Arthritis
30   3   83   2  51 147 3432 |  g = Hypertension
```

Cross Validation: 10

```
Time taken to build model: 4.29 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      12237           91.6286 %
Incorrectly Classified Instances    1118           8.3714 %
Kappa statistic                    0.8959
Mean absolute error                 0.0361
Root mean squared error             0.1313
Relative absolute error             15.6717 %
Root relative squared error         38.6885 %
Total Number of Instances          13355

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      0.909    0.006    0.932     0.909    0.920     0.914    0.997    0.976    Asthma
      0.954    0.001    0.973     0.954    0.964     0.962    1.000    0.992    Cancer
      0.910    0.020    0.890     0.910    0.900     0.882    0.994    0.970    Obesity
      0.992    0.002    0.988     0.992    0.990     0.988    1.000    1.000    Healthy
      0.959    0.011    0.968     0.959    0.964     0.952    0.997    0.993    Diabetes
      0.632    0.017    0.735     0.632    0.680     0.660    0.975    0.773    Arthritis
      0.914    0.049    0.880     0.914    0.897     0.855    0.986    0.966    Hypertension
Weighted Avg.    0.916    0.021    0.915     0.916    0.915     0.896    0.992    0.966

=== Confusion Matrix ===

  a   b   c   d   e   f   g  <-- classified as
964  1  24  16   2  10  44 |  a = Asthma
  5 582   0   0  11   0  12 |  b = Cancer
13   0 1836   1  34  37  96 |  c = Obesity
  6   0   0 1589   0   7   0 |  d = Healthy
  4  13  45   0 3258   7  69 |  e = Diabetes
11   0  73   1   6  582 248 |  f = Arthritis
31   2   85   2  53 149 3426 |  g = Hypertension
```

Cross Validation: 15

Time taken to build model: 4.31 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	12243	91.6735 %
Incorrectly Classified Instances	1112	8.3265 %
Kappa statistic	0.8964	
Mean absolute error	0.036	
Root mean squared error	0.1312	
Relative absolute error	15.6378 %	
Root relative squared error	38.6523 %	
Total Number of Instances	13355	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.906	0.006	0.934	0.906	0.920	0.913	0.997	0.976	Asthma
	0.954	0.001	0.973	0.954	0.964	0.962	1.000	0.992	Cancer
	0.910	0.020	0.890	0.910	0.900	0.882	0.994	0.970	Obesity
	0.993	0.002	0.987	0.993	0.990	0.988	1.000	1.000	Healthy
	0.959	0.010	0.970	0.959	0.964	0.952	0.997	0.993	Diabetes
	0.633	0.017	0.735	0.633	0.680	0.661	0.975	0.772	Arthritis
	0.916	0.049	0.880	0.916	0.898	0.857	0.986	0.966	Hypertension
Weighted Avg.	0.917	0.021	0.916	0.917	0.916	0.896	0.992	0.966	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
961	1	24	17	2	10	46		a = Asthma
5	582	0	0	11	0	12		b = Cancer
12	0	1836	1	33	39	96		c = Obesity
5	0	0	1590	0	7	0		d = Healthy
4	12	48	0	3258	7	67		e = Diabetes
12	0	72	1	6	583	247		f = Arthritis
30	3	83	2	50	147	3433		g = Hypertension

Split 60

Classifier output

Time taken to build model: 8.62 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0.02 seconds

=== Summary ===

Correctly Classified Instances	4879	91.3328 %
Incorrectly Classified Instances	463	8.6672 %
Kappa statistic	0.8921	
Mean absolute error	0.0374	
Root mean squared error	0.1337	
Relative absolute error	16.2328 %	
Root relative squared error	39.4154 %	
Total Number of Instances	5342	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.876	0.005	0.933	0.876	0.903	0.896	0.996	0.971	Asthma
	0.974	0.001	0.979	0.974	0.976	0.975	0.999	0.994	Cancer
	0.911	0.019	0.896	0.911	0.903	0.885	0.994	0.971	Obesity
	0.994	0.003	0.979	0.994	0.986	0.985	1.000	0.999	Healthy
	0.959	0.012	0.964	0.959	0.961	0.948	0.996	0.992	Diabetes
	0.637	0.017	0.737	0.637	0.683	0.663	0.974	0.781	Arthritis
	0.909	0.051	0.874	0.909	0.891	0.848	0.985	0.962	Hypertension
Weighted Avg.	0.913	0.023	0.912	0.913	0.912	0.892	0.992	0.965	

=== Confusion Matrix ===

	a	b	c	d	e	f	g	<-- classified as
373	0	9	9	3	5	27		a = Asthma
0	228	0	0	4	0	2		b = Cancer
7	0	755	1	14	14	38		c = Obesity
3	0	0	619	0	1	0		d = Healthy
2	4	14	0	1307	3	33		e = Diabetes
3	0	32	1	1	235	97		f = Arthritis
12	1	33	2	27	61	1362		g = Hypertension

Log

x 0

Split 70

```
Time taken to test model on test split: 0.02 seconds

=== Summary ===

Correctly Classified Instances      3637          90.7888 %
Incorrectly Classified Instances    369           9.2112 %
Kappa statistic                    0.8852
Mean absolute error                 0.0377
Root mean squared error             0.137
Relative absolute error             16.3582 %
Root relative squared error         40.3918 %
Total Number of Instances          4006

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      0.863    0.007    0.920     0.863    0.890     0.882    0.996    0.966    Asthma
      0.977    0.001    0.971     0.977    0.974     0.973    0.999    0.993    Cancer
      0.904    0.022    0.881     0.904    0.892     0.873    0.993    0.969    Obesity
      0.996    0.004    0.972     0.996    0.984     0.982    1.000    0.999    Healthy
      0.955    0.010    0.970     0.955    0.962     0.949    0.996    0.992    Diabetes
      0.636    0.017    0.738     0.636    0.683     0.663    0.973    0.772    Arthritis
      0.902    0.054    0.866     0.902    0.883     0.837    0.983    0.959    Hypertension
Weighted Avg.   0.908    0.024    0.907     0.908    0.907     0.885    0.991    0.962

=== Confusion Matrix ===

  a   b   c   d   e   f   g  <-- classified as
276  0   8   9   2   4  21 |  a = Asthma
  0 167   0   0   2   0   2 |  b = Cancer
  6   0 556   1  10  11  31 |  c = Obesity
  2   0   0 455   0   0   0 |  d = Healthy
  2   4  11   0 991   1  29 |  e = Diabetes
  2   0  25   1   1 180  74 |  f = Arthritis
 12   1  31   2  16  48 1012 |  g = Hypertension
```

Log

x 0

Split 80:

```
Time taken to test model on test split: 0.01 seconds

=== Summary ===

Correctly Classified Instances      2432          91.052 %
Incorrectly Classified Instances    239           8.948 %
Kappa statistic                    0.8886
Mean absolute error                 0.037
Root mean squared error             0.1364
Relative absolute error             16.0719 %
Root relative squared error         40.2149 %
Total Number of Instances          2671

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
      0.874    0.007    0.924     0.874    0.898     0.890    0.995    0.967    Asthma
      0.973    0.001    0.982     0.973    0.978     0.977    0.999    0.990    Cancer
      0.900    0.022    0.881     0.900    0.890     0.870    0.993    0.967    Obesity
      0.994    0.004    0.972     0.994    0.983     0.980    1.000    0.998    Healthy
      0.960    0.011    0.969     0.960    0.964     0.952    0.996    0.991    Diabetes
      0.651    0.016    0.752     0.651    0.697     0.679    0.974    0.779    Arthritis
      0.904    0.053    0.871     0.904    0.887     0.842    0.984    0.961    Hypertension
Weighted Avg.   0.911    0.023    0.910     0.911    0.910     0.888    0.991    0.963

=== Confusion Matrix ===

  a   b   c   d   e   f   g  <-- classified as
194  0   5   8   2   2  11 |  a = Asthma
  0 109   0   0   2   0   1 |  b = Cancer
  4   0 369   0   5   7  25 |  c = Obesity
  2   0   0 309   0   0   0 |  d = Healthy
  1   1   5   0 646   1  19 |  e = Diabetes
  1   0  19   0   0 121  45 |  f = Arthritis
  8   1  21   1  12  30 684 |  g = Hypertension
```

Log

x 0